

# ImageNet Large Scale Visual Recognition Challenge

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**Abstract** The ImageNet Large Scale Visual Recognition Challenge is a benchmark in object category classification and detection on hundreds of object categories and millions of images. The challenge has been run annually from 2010 to present, attracting participation from more than fifty institutions.

This paper describes the creation of this benchmark dataset and the advances in object recognition that have been possible as a result. We discuss the chal-

lenges of collecting large-scale ground truth annotation, highlight key breakthroughs in categorical object recognition, provide a detailed analysis of the current state of the field of large-scale image classification and object detection, and compare the state-of-the-art computer vision accuracy with human accuracy. We conclude with lessons learned in the five years of the challenge, and propose future directions and improvements.

**Keywords** Dataset · Large-scale · Benchmark · Object recognition · Object detection

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## 1 Introduction

*Overview.* The ImageNet Large Scale Visual Recognition Challenge (ILSVRC) has been running annually for five years (since 2010) and has become the standard benchmark for large-scale object recognition.<sup>1</sup> ILSVRC follows in the footsteps of the PASCAL VOC challenge (Everingham et al., 2012), established in 2005, which set the precedent for standardized evaluation of recognition algorithms in the form of yearly competitions. As in PASCAL VOC, ILSVRC consists of two components: (1) a publically available *dataset*, and (2) an annual *competition* and corresponding workshop. The dataset allows for the development and comparison of categorical object recognition algorithms, and the competition and workshop provide a way to track the progress and discuss the lessons learned from the most successful and innovative entries each year.

<sup>1</sup> In this paper, we will be using the term *object recognition* broadly to encompass both *image classification* (a task requiring an algorithm to determine what object classes are present in the image) as well as *object detection* (a task requiring an algorithm to localize all objects present in the image).

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**Abstract** ImageNet大规模视觉识别挑战赛是一个在数百个物体类别和数百万张图像上进行物体分类与检测的基准测试。该挑战自2010年起每年举办，至今已吸引了超过五十家机构的参与。

本文介绍了该基准数据集的创建过程，以及由此推动的物体识别技术进展。我们探讨了挑

在收集大规模地面真实标注方面面临的挑战，突出分类物体识别的关键突破，详细分析当前大规模图像分类和物体检测领域的现状，并比较最先进的计算机视觉精度与人类精度。我们总结了挑战赛五年来的经验教训，并提出了未来的方向和改进建议。

**Keywords** 数据集 · 大规模 · 基准测试 · 物体识别 · 物体检测

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## 1 Introduction

*Overview.* ImageNet大规模视觉识别挑战赛 (ILSVRC) 已连续举办五年 (自2010年起)，成为大规模物体识别的标准基准。<sup>1</sup> ILSVRC沿袭了2005年设立的PASCAL VOC挑战赛 (Everingham等人, 2012) 的传统，后者以年度竞赛的形式为识别算法的标准化评估开创了先例。与PASCAL VOC类似，ILSVRC包含两个部分：(1) 公开可用的 *dataset*；(2) 年度 *competition* 及相应研讨会。该数据集支持分类物体识别算法的开发与比较，而竞赛和研讨会则为追踪年度进展、探讨每年最成功与最具创新性的参赛方案所获经验提供了平台。

<sup>1</sup> In this paper, we will be using the term *object recognition* broadly to encompass both *image classification* (a task requiring an algorithm to determine what object classes are present in the image) as well as *object detection* (a task requiring an algorithm to localize all objects present in the image).

The publically released dataset contains a set of manually annotated *training* images. A set of *test* images is also released, with the manual annotations withheld.<sup>2</sup> Participants train their algorithms using the training images and then automatically annotate the test images. These predicted annotations are submitted to the *evaluation server*. Results of the evaluation are revealed at the end of the competition period and authors are invited to share insights at the workshop held at the International Conference on Computer Vision (ICCV) or European Conference on Computer Vision (ECCV) in alternate years.

ILSVRC annotations fall into one of two categories: (1) *image-level annotation* of a binary label for the presence or absence of an object class in the image, e.g., “there are cars in this image” but “there are no tigers,” and (2) *object-level annotation* of a tight bounding box and class label around an object instance in the image, e.g., “there is a screwdriver centered at position (20,25) with width of 50 pixels and height of 30 pixels”.

*Large-scale challenges and innovations.* In creating the dataset, several challenges had to be addressed. Scaling up from 19,737 images in PASCAL VOC 2010 to 1,461,406 in ILSVRC 2010 and from 20 object classes to 1000 object classes brings with it several challenges. It is no longer feasible for a small group of annotators to annotate the data as is done for other datasets (Fei-Fei et al., 2004; Criminisi, 2004; Everingham et al., 2012; Xiao et al., 2010). Instead we turn to designing novel crowdsourcing approaches for collecting large-scale annotations (Su et al., 2012; Deng et al., 2009, 2014).

Some of the 1000 object classes may not be as easy to annotate as the 20 categories of PASCAL VOC: e.g., bananas which appear in bunches may not be as easy to delineate as the basic-level categories of aeroplanes or cars. Having more than a million images makes it infeasible to annotate the locations of all objects (much less with object segmentations, human body parts, and other detailed annotations that subsets of PASCAL VOC contain). New evaluation criteria have to be defined to take into account the facts that obtaining perfect manual annotations in this setting may be infeasible.

Once the challenge dataset was collected, its scale allowed for unprecedented opportunities both in evaluation of object recognition algorithms and in developing new techniques. Novel algorithmic innovations emerge with the availability of large-scale training data. The broad spectrum of object categories motivated the need for algorithms that are even able to distinguish classes which are visually very similar. We highlight the most

successful of these algorithms in this paper, and compare their performance with human-level accuracy.

Finally, the large variety of object classes in ILSVRC allows us to perform an analysis of statistical properties of objects and their impact on recognition algorithms. This type of analysis allows for a deeper understanding of object recognition, and for designing the next generation of general object recognition algorithms.

*Goals.* This paper has three key goals:

1. To discuss the challenges of creating this large-scale object recognition benchmark dataset,
2. To highlight the developments in object classification and detection that have resulted from this effort, and
3. To take a closer look at the current state of the field of categorical object recognition.

The paper may be of interest to researchers working on creating large-scale datasets, as well as to anybody interested in better understanding the history and the current state of large-scale object recognition.

The collected dataset and additional information about ILSVRC can be found at:

<http://image-net.org/challenges/LSVRC/>

## 1.1 Related work

We briefly discuss some prior work in constructing benchmark image datasets.

*Image classification datasets.* Caltech 101 (Fei-Fei et al., 2004) was among the first standardized datasets for multi-category image classification, with 101 object classes and commonly 15-30 training images per class. Caltech 256 (Griffin et al., 2007) increased the number of object classes to 256 and added images with greater scale and background variability. The TinyImages dataset (Torralba et al., 2008) contains 80 million 32x32 low resolution images collected from the internet using synsets in WordNet (Miller, 1995) as queries. However, since this data has not been manually verified, there are many errors, making it less suitable for algorithm evaluation. Datasets such as 15 Scenes (Oliva and Torralba, 2001; Fei-Fei and Perona, 2005; Lazebnik et al., 2006) or recent Places (Zhou et al., 2014) provide a single scene category label (as opposed to an object category).

The ImageNet dataset (Deng et al., 2009) is the backbone of ILSVRC. ImageNet is an image dataset organized according to the WordNet hierarchy (Miller, 1995). Each concept in WordNet, possibly described by multiple words or word phrases, is called a “synonym

<sup>2</sup> In 2010, the test annotations were later released publicly; since then the test annotation have been kept hidden.

公开发布的数据集包含一组手动标注的*training*图像。同时，还发布了一组*test*图像，但未提供其手动标注。<sup>2</sup>参与者使用训练图像训练其算法，然后自动标注测试图像。这些预测标注将提交至*evaluation server*。评估结果将在竞赛期结束时公布，并邀请作者在交替举办的国际计算机视觉大会（ICCV）或欧洲计算机视觉大会（ECCV）研讨会上分享见解。

ILSVRC的标注分为两类：(1)

*image-level annotation* 图像中是否存在某物体类别的二元标签，例如，“这张图中有汽车”但“没有老虎”；(2) *object-level annotation* 图像中物体实例的紧密边界框及类别标签，例如，“有一个螺丝刀，中心位于位置(20,25)，宽度为50像素，高度为30像素”。

*Large-scale challenges and innovations.* 在构建数据集的过程中，需要解决若干挑战。从PASCAL VOC 2010的19,737张图像扩展到ILSVRC 2010的1,461,406张图像，以及从20个物体类别增加到1000个物体类别，带来了诸多难题。像其他数据集那样由少数标注者完成标注工作已不再可行（Fei-Fei et al., 2004; Criminisi, 2004; Everingham et al., 2012; Xiao et al., 2010）。为此，我们转而设计新颖的众包方法以收集大规模标注数据（Sutcliffe et al., 2012; Deng et al., 2009, 2014）。

在1000个物体类别中，部分类别可能不如PASCAL VOC的20个类别那样易于标注：例如成串出现的香蕉，其轮廓标注难度可能高于飞机、汽车等基础类别。超过百万张的图像规模使得标注所有物体的位置变得不可行（更不用说PASCAL VOC子集中包含的物体分割、人体部位等精细标注）。需要制定新的评估标准，以考虑到在此规模下获得完美人工标注可能无法实现的客观情况。

一旦挑战数据集被收集完成，其规模为物体识别算法的评估和新技术的开发提供了前所未有的机遇。随着大规模训练数据的可用性，新颖的算法创新不断涌现。广泛的物体类别范围促使人们需要能够区分视觉上非常相似的类别的算法。我们重点介绍最

本文中这些算法的成功，并将其性能与人类水平的准确度进行比较。

最后，ILSVRC中大量的物体类别使我们能够分析物体的统计特性及其对识别算法的影响。这类分析有助于更深入地理解物体识别，并为设计下一代通用物体识别算法提供依据。

*Goals.* 本文有三个关键目标：

1. 探讨创建这一大规模物体识别基准数据集所面临的挑战，2. 强调由此努力带来的物体分类与检测领域的发展，3. 并深入审视当前类别物体识别领域的研究现状。

这篇论文可能会引起致力于创建大规模数据集的研究人员的兴趣，同时也适合任何希望更深入了解大规模物体识别的历史与现状的人士。

收集的数据集以及关于ILSVRC的更多信息可以在以下位置找到：

<http://image-net.org/challenges/LSVRC/>

## 1.1 相关工作

我们简要讨论一些在构建基准图像数据集方面的先前工作。

*Image classification datasets.* Caltech 101（Fei-Fei等人，2004年）是最早用于多类别图像分类的标准化数据集之一，包含101个物体类别，通常每个类别有15-30张训练图像。Caltech 256（Griffin等人，2007年）将物体类别数量增加到256个，并添加了更具尺度变化和背景多样性的图像。TinyImages数据集（Torralba等人，2008年）包含8000万张从互联网收集的32x32低分辨率图像，使用WordNet（Miller，1995年）中的同义词集作为查询词。然而，由于这些数据未经人工验证，存在大量错误，因此不太适合用于算法评估。诸如15 Scenes（Oliva和Torralba，2001年；Fei-Fei和Perona，2005年；Lazebnik等人，2006年）或近年提出的Places（Zhou等人，2014年）等数据集则提供单一场景类别标签（与物体类别标签相对应）。

ImageNet数据集（Deng等人，2009年）是ILSVRC的支柱。ImageNet是一个依据WordNet层次结构（Miller，1995年）组织的图像数据集。WordNet中的每个概念（可能由多个单词或词组描述）被称为一个“同义词集”

<sup>2</sup> In 2010, the test annotations were later released publicly; since then the test annotation have been kept hidden.



set” or “synset”. ImageNet populates 21,841 synsets of WordNet with an average of 650 manually verified and full resolution images. As a result, ImageNet contains 14,197,122 annotated images organized by the semantic hierarchy of WordNet (as of August 2014). ImageNet is larger in scale and diversity than the other image classification datasets. ILSVRC uses a subset of ImageNet images for training the algorithms and some of ImageNet’s image collection protocols for annotating additional images for testing the algorithms.

*Image parsing datasets.* Many datasets aim to provide richer image annotations beyond image-category labels. LabelMe (Russell et al., 2007) contains general photographs with multiple objects per image. It has bounding polygon annotations around objects, but the object names are not standardized: annotators are free to choose which objects to label and what to name each object. The SUN2012 (Xiao et al., 2010) dataset contains 16,873 manually cleaned up and fully annotated images more suitable for standard object detection training and evaluation. SIFT Flow (Liu et al., 2011) contains 2,688 images labeled using the LabelMe system. The LotusHill dataset (Yao et al., 2007) contains very detailed annotations of objects in 636,748 images and video frames, but it is not available for free. Several datasets provide pixel-level segmentations: for example, MSRC dataset (Criminisi, 2004) with 591 images and 23 object classes, Stanford Background Dataset (Gould et al., 2009) with 715 images and 8 classes, and the Berkeley Segmentation dataset (Arbelaez et al., 2011) with 500 images annotated with object boundaries. OpenSurfaces segments surfaces from consumer photographs and annotates them with surface properties, including material, texture, and contextual information (Bell et al., 2013).

The closest to ILSVRC is the PASCAL VOC dataset (Everingham et al., 2010, 2014), which provides a standardized test bed for object detection, image classification, object segmentation, person layout, and action classification. Much of the design choices in ILSVRC have been inspired by PASCAL VOC and the similarities and differences between the datasets are discussed at length throughout the paper. ILSVRC scales up PASCAL VOC’s goal of standardized training and evaluation of recognition algorithms by more than an order of magnitude in number of object classes and images: PASCAL VOC 2012 has 20 object classes and 21,738 images compared to ILSVRC2012 with 1000 object classes and 1,431,167 annotated images.

The recently released COCO dataset (Lin et al., 2014b) contains more than 328,000 images with 2.5 million object instances manually segmented. It has fewer

object categories than ILSVRC (91 in COCO versus 200 in ILSVRC object detection) but more instances per category (27K on average compared to about 1K in ILSVRC object detection). Further, it contains object segmentation annotations which are not currently available in ILSVRC. COCO is likely to become another important large-scale benchmark.

*Large-scale annotation.* ILSVRC makes extensive use of Amazon Mechanical Turk to obtain accurate annotations (Sorokin and Forsyth, 2008). Works such as (Welinder et al., 2010; Sheng et al., 2008; Vittayakorn and Hays, 2011) describe quality control mechanisms for this marketplace. (Vondrick et al., 2012) provides a detailed overview of crowdsourcing video annotation. A related line of work is to obtain annotations through well-designed games, e.g. (von Ahn and Dabbish, 2005). Our novel approaches to crowdsourcing accurate image annotations are in Sections 3.1.3, 3.2.1 and 3.3.3.

*Standardized challenges.* There are several datasets with standardized online evaluation similar to ILSVRC: the aforementioned PASCAL VOC (Everingham et al., 2012), Labeled Faces in the Wild (Huang et al., 2007) for unconstrained face recognition, Reconstruction meets Recognition (Urtasun et al., 2014) for 3D reconstruction and KITTI (Geiger et al., 2013) for computer vision in autonomous driving. These datasets along with ILSVRC help benchmark progress in different areas of computer vision. Works such as (Torralba and Efros, 2011) emphasize the importance of examining the bias inherent in any standardized dataset.

## 1.2 Paper layout

We begin with a brief overview of ILSVRC challenge tasks in Section 2. Dataset collection and annotation are described at length in Section 3. Section 4 discusses the evaluation criteria of algorithms in the large-scale recognition setting. Section 5 provides an overview of the methods developed by ILSVRC participants.

Section 6 contains an in-depth analysis of ILSVRC results: Section 6.1 documents the progress of large-scale recognition over the years, Section 6.2 concludes that ILSVRC results are statistically significant, Section 6.3 thoroughly analyzes the current state of the field of object recognition, and Section 6.4 compares state-of-the-art computer vision accuracy with human accuracy. We conclude and discuss lessons learned from ILSVRC in Section 7.

“集合”或“同义词集”。ImageNet 为 WordNet 的 21,841 个同义词集填充了平均 650 张经过人工验证的全分辨率图像。因此, ImageNet 包含了 14,197,122 张带标注的图像, 这些图像按照 WordNet 的语义层次结构进行组织(截至 2014 年 8 月)。ImageNet 在规模和多样性上都大于其他图像分类数据集。ILSVRC 使用 ImageNet 图像的一个子集来训练算法, 并采用 ImageNet 的部分图像收集协议来标注额外的图像, 用于测试算法。

*Image parsing datasets.* 许多数据集旨在提供比图像类别标签更丰富的图像标注。LabelMe (Russell 等人, 2007) 包含每张图像有多个物体的普通照片, 它提供了物体周围的边界多边形标注, 但物体名称未标准化: 标注者可自由选择标注哪些物体以及为每个物体命名。SUN 2012 (Xiao 等人, 2010) 数据集包含 16,873 张经过人工清理和完整标注的图像, 更适合标准的物体检测训练和评估。SIFT Flow (Liu 等人, 2011) 包含 2,688 张使用 LabelMe 系统标注的图像。LotusHill 数据集 (Yao 等人, 2007) 对 636,748 张图像和视频帧中的物体提供了非常详细的标注, 但该数据集并非免费提供。若干数据集提供了像素级分割标注: 例如包含 591 张图像和 23 个物体类别的 MSRC 数据集 (Criminisi, 2004)、包含 715 张图像和 8 个类别的斯坦福背景数据集 (Gould 等人, 2009), 以及包含 500 张标注了物体边界的图像的伯克利分割数据集 (Arbelaez 等人, 2011)。OpenSurfaces 从消费级照片中分割表面, 并用表面属性(包括材质、纹理和上下文信息)进行标注 (Bell 等人, 2013)。

与 ILSVRC 最接近的是 PASCAL VOC 数据集 (Everingham 等人, 2010 年, 2014 年), 它为物体检测、图像分类、物体分割、人物布局和动作分类提供了一个标准化的测试平台。ILSVRC 中的许多设计选择都受到了 PASCAL VOC 的启发, 并且本文详细讨论了这两个数据集之间的相似之处和差异。ILSVRC 在物体类别和图像数量上, 将 PASCAL VOC 标准化训练和评估识别算法的目标扩展了一个数量级以上: PASCAL VOC 2012 包含 20 个物体类别和 21,738 张图像, 而 ILSVRC 2012 则包含 100 个物体类别和 1,431,167 张标注图像。

最近发布的 COCO 数据集 (Lin 等人, 2014b) 包含超过 328,000 张图像, 其中 250 万个物体实例经过人工分割。它的数量较少

与 ILSVRC 相比, COCO 的物体类别更少 (COCO 为 91 类, 而 ILSVRC 物体检测为 200 类), 但每个类别的实例更多 (平均 27K, 而 ILSVRC 物体检测约为 1K)。此外, 它包含了 ILSVRC 目前尚未提供的物体分割标注。COCO 有望成为另一个重要的大规模基准数据集。

*Large-scale annotation.* ILSVRC 广泛使用亚马逊土耳其机器人 (Amazon Mechanical Turk) 来获取精确标注 (Soroikin 和 Forsyth, 2008 年)。诸如 (Welinder 等人, 2010 年; Sheng 等人, 2008 年; Vittayakorn 和 Hays, 2011 年) 等研究描述了该平台的质量控制机制。(Vondrick 等人, 2012 年) 对众包视频标注进行了详细综述。另一相关研究方向是通过精心设计的游戏获取标注, 例如 (von Ahn 和 Dabbish, 2005 年)。我们在第 3.1.3 节、3.2.1 节和 3.3.3 节中提出了众包精确图像标注的新方法。

*Standardized challenges.* 存在多个与 ILSVRC 类似、具备标准化在线评估的数据集: 前文提及的 PASCAL VOC (Everingham 等人, 2012)、用于无约束人脸识别的 Labeled Faces in the Wild (Huang 等人, 2007)、面向三维重建的 Reconstruction meets Recognition (Urtasun 等人, 2014), 以及用于自动驾驶计算机视觉的 KITTI (Geiger 等人, 2013)。这些数据集与 ILSVRC 共同为计算机视觉不同领域的进展提供了基准参照。诸如 (Torralba 和 Efros, 2011) 的研究工作, 则强调了审视任何标准化数据集固有偏差的重要性。

## 1.2 论文结构

我们从第 2 节对 ILSVRC 挑战赛任务的简要概述开始。第 3 节详细描述了数据集的收集与标注过程。第 4 节讨论了大规模识别场景下算法的评估标准。第 5 节概述了 ILSVRC 参赛者所开发的方法。

第 6 节对 ILSVRC 结果进行了深入分析: 6.1 节记录了大尺度识别技术多年来的进展, 6.2 节得出 ILSVRC 结果具有统计显著性的结论, 6.3 节全面分析了当前物体识别领域的发展现状, 6.4 节将最先进的计算机视觉准确率与人类准确率进行了对比。我们在第 7 节进行总结, 并讨论从 ILSVRC 中获得的经验教训。

## 2 Challenge tasks

The goal of ILSVRC is to estimate the content of photographs for the purpose of retrieval and automatic annotation. Test images are presented with no initial annotation, and algorithms have to produce labelings specifying what objects are present in the images. New test images are collected and labeled especially for this competition and are not part of the previously published ImageNet dataset (Deng et al., 2009).

ILSVRC over the years has consisted of one or more of the following tasks (years in parentheses):<sup>3</sup>

1. **Image classification** (2010-2014): Algorithms produce a list of object categories present in the image.
2. **Single-object localization** (2011-2014): Algorithms produce a list of object categories present in the image, along with an axis-aligned bounding box indicating the position and scale of *one* instance of each object category.
3. **Object detection** (2013-2014): Algorithms produce a list of object categories present in the image along with an axis-aligned bounding box indicating the position and scale of *every* instance of each object category.

This section provides an overview and history of each of the three tasks. Table 1 shows summary statistics.

### 2.1 Image classification task

Data for the image classification task consists of photographs collected from Flickr<sup>4</sup> and other search engines, manually labeled with the presence of one of 1000 object categories. Each image contains one ground truth label.

For each image, algorithms produce a list of object categories present in the image. The quality of a labeling is evaluated based on the label that best matches the ground truth label for the image (see Section 4.1).

Constructing ImageNet was an effort to scale up an image classification dataset to cover most nouns in English using tens of millions of manually verified photographs (Deng et al., 2009). The image classification task of ILSVRC came as a direct extension of this effort. A subset of categories and images was chosen and fixed to provide a standardized benchmark while the rest of ImageNet continued to grow.

<sup>3</sup> In addition, ILSVRC in 2012 also included a taster fine-grained classification task, where algorithms would classify dog photographs into one of 120 dog breeds (Khosla et al., 2011). Fine-grained classification has evolved into its own Fine-Grained classification challenge in 2013 (Berg et al., 2013), which is outside the scope of this paper.

<sup>4</sup> [www.flickr.com](http://www.flickr.com)

### 2.2 Single-object localization task

The single-object localization task, introduced in 2011, built off of the image classification task to evaluate the ability of algorithms to learn the appearance of the target object itself rather than its image context.

Data for the single-object localization task consists of the same photographs collected for the image classification task, hand labeled with the presence of one of 1000 object categories. Each image contains one ground truth label. Additionally, every instance of this category is annotated with an axis-aligned bounding box.

For each image, algorithms produce a list of object categories present in the image, along with a bounding box indicating the position and scale of one instance of each object category. The quality of a labeling is evaluated based on the object category label that best matches the ground truth label, with the additional requirement that the location of the predicted instance is also accurate (see Section 4.2).

### 2.3 Object detection task

The object detection task went a step beyond single-object localization and tackled the problem of localizing multiple object categories in the image. This task has been a part of the PASCAL VOC for many years on the scale of 20 object categories and tens of thousands of images, but scaling it up by an order of magnitude in object categories and in images proved to be very challenging from a dataset collection and annotation point of view (see Section 3.3).

Data for the detection tasks consists of new photographs collected from Flickr using scene-level queries. The images are annotated with axis-aligned bounding boxes indicating the position and scale of every instance of each target object category. The training set is additionally supplemented with (a) data from the single-object localization task, which contains annotations for all instances of just one object category, and (b) negative images known not to contain any instance of some object categories.

For each image, algorithms produce bounding boxes indicating the position and scale of all instances of all target object categories. The quality of labeling is evaluated by *recall*, or number of target object instances detected, and *precision*, or the number of spurious detections produced by the algorithm (see Section 4.3).

## 2 Challenge tasks

ILSVRC的目标是评估照片内容，以实现检索和自动标注。测试图像在无初始标注的情况下呈现，算法需生成标签以指明图像中存在哪些物体。专门为此竞赛收集并标注了新的测试图像，这些图像不属于先前发布的ImageNet数据集（Deng et al., 2009）。

ILSVRC 多年来包含以下一项或多项任务（年份在括号内）：<sup>3</sup>

1. **Image classification** (2010-2014): 算法生成图像中存在的物体类别列表。 2.

**Single-object localization** (2011-2014): 算法生成图像中存在的物体类别列表，并提供一个轴对齐的边界框，用于指示*one*每个物体类别实例的位置和尺度。

3. **Object detection** (2013-2014): 算法生成图像中存在的物体类别列表，并提供一个轴对齐的边界框，用于指示每个物体类别*every*实例的位置和尺度。

本节概述了三个任务各自的历史与概况。表1展示了汇总统计数据。

### 2.1 图像分类任务

图像分类任务的数据包括从Flickr<sup>4</sup>和其他搜索引擎收集的照片，这些照片经过人工标注，包含1000个对象类别中的一种。每张图像都有一个真实标签。

对于每张图像，算法会生成一个存在于图像中的物体类别列表。标注质量基于与图像真实标签最匹配的标注进行评估（参见第4.1节）。

构建ImageNet是为了扩展图像分类数据集，使其覆盖英语中的大多数名词，使用了数千万张经过人工验证的照片（Deng等人，2009年）。ILSVRC的图像分类任务正是这一努力的直接延伸。从中选取并固定了一部分类别和图像，以提供一个标准化的基准，而ImageNet的其余部分则持续增长。

<sup>3</sup> In addition, ILSVRC in 2012 also included a taster fine-grained classification task, where algorithms would classify dog photographs into one of 120 dog breeds (Khosla et al., 2011). Fine-grained classification has evolved into its own Fine-Grained classification challenge in 2013 (Berg et al., 2013), which is outside the scope of this paper.

<sup>4</sup> [www.flickr.com](http://www.flickr.com)

### 2.2 单目标定位任务

单目标定位任务于2011年被提出，它建立在图像分类任务的基础上，旨在评估算法学习目标对象本身外观而非其图像上下文的能力。

单目标定位任务的数据集与图像分类任务所收集的照片相同，这些照片已手工标注了1000个物体类别中某一类的存在。每张图像包含一个真实标签。此外，该类别的每个实例都用一个轴对齐的边界框进行了标注。

对于每张图像，算法会生成一个存在于图像中的物体类别列表，以及一个指示每个物体类别一个实例位置和尺度的边界框。标注质量基于与真实标签最匹配的物体类别标签进行评估，同时要求预测实例的位置也必须准确（参见第4.2节）。

### 2.3 目标检测任务

物体检测任务在单物体定位的基础上更进一步，致力于解决图像中多类别物体的定位问题。多年来，该任务一直是PASCAL VOC挑战赛的核心内容，其规模维持在20个物体类别和数万张图像。然而，若要将物体类别和图像数量提升一个数量级，从数据集收集与标注的角度来看极具挑战性（详见第3.3节）。

检测任务的数据包括通过场景级查询从Flickr收集的新照片。这些图像用轴对齐的边界框进行标注，指示每个目标对象类别每个实例的位置和尺度。训练集还额外补充了：(a) 来自单对象定位任务的数据，其中仅包含一个对象类别所有实例的标注，以及(b) 已知不包含某些对象类别任何实例的负样本图像。

对于每张图像，算法会生成边界框，标示出所有目标对象类别实例的位置和尺度。标注质量通过*recall*（即检测到的目标对象实例数量）和*precision*（即算法产生的误检数量）进行评估（详见第4.3节）。

| Task                                        |                                              | Image classification | Single-object localization   | Object detection            |
|---------------------------------------------|----------------------------------------------|----------------------|------------------------------|-----------------------------|
| Manual labeling on training set             | Number of object classes annotated per image | 1                    | 1                            | 1 or more                   |
|                                             | Locations of annotated classes               | —                    | all instances on some images | all instances on all images |
| Manual labeling on validation and test sets | Number of object classes annotated per image | 1                    | 1                            | all target classes          |
|                                             | Locations of annotated classes               | —                    | all instances on all images  | all instances on all images |

**Table 1** Overview of the provided annotations for each of the tasks in ILSVRC.

### 3 Dataset construction at large scale

Our process of constructing large-scale object recognition image datasets consists of three key steps.

The first step is defining the set of target object categories. To do this, we select from among the existing ImageNet (Deng et al., 2009) categories. By using WordNet as a backbone (Miller, 1995), ImageNet already takes care of disambiguating word meanings and of combining together synonyms into the same object category. Since the selection of object categories needs to be done only once per challenge task, we use a combination of automatic heuristics and manual post-processing to create the list of target categories appropriate for each task. For example, for image classification we may include broader scene categories such as a type of beach, but for single-object localization and object detection we want to focus only on object categories which can be unambiguously localized in images (Sections 3.1.1 and 3.3.1).

The second step is collecting a diverse set of candidate images to represent the selected categories. We use both automatic and manual strategies on multiple search engines to do the image collection. The process is modified for the different ILSVRC tasks. For example, for object detection we focus our efforts on collecting scene-like images using generic queries such as “African safari” to find pictures likely to contain multiple animals in one scene (Section 3.3.2).

The third (and most challenging) step is annotating the millions of collected images to obtain a clean dataset. We carefully design crowdsourcing strategies targeted to each individual ILSVRC task. For example, the bounding box annotation system used for localization and detection tasks consists of three distinct parts in order to include automatic crowdsourced quality control (Section 3.2.1). Annotating images fully with all target object categories (on a reasonable budget) for object detection requires an additional hierarchical image labeling system (Section 3.3.3).

We describe the data collection and annotation procedure for each of the ILSVRC tasks in order: image

classification (Section 3.1), single-object localization (Section 3.2), and object detection (Section 3.3), focusing on the three key steps for each dataset.

#### 3.1 Image classification dataset construction

The image classification task tests the ability of an algorithm to name the objects present in the image, without necessarily localizing them.

We describe the choices we made in constructing the ILSVRC image classification dataset: selecting the target object categories from ImageNet (Section 3.1.1), collecting a diverse set of candidate images by using multiple search engines and an expanded set of queries in multiple languages (Section 3.1.2), and finally filtering the millions of collected images using the carefully designed crowdsourcing strategy of ImageNet (Deng et al., 2009) (Section 3.1.3).

##### 3.1.1 Defining object categories for the image classification dataset

The 1000 categories used for the image classification task were selected from the ImageNet (Deng et al., 2009) categories. The 1000 synsets are selected such that there is no overlap between synsets: for any synsets  $i$  and  $j$ ,  $i$  is not an ancestor of  $j$  in the ImageNet hierarchy. These synsets are part of the larger hierarchy and may have children in ImageNet; however, for ILSVRC we do not consider their child subcategories. The synset hierarchy of ILSVRC can be thought of as a “trimmed” version of the complete ImageNet hierarchy. Figure 1 visualizes the diversity of the ILSVRC2012 object categories.

The exact 1000 synsets used for the image classification and single-object localization tasks have changed over the years. There are 639 synsets which have been used in all five ILSVRC challenges so far. In the first year of the challenge synsets were selected randomly from the available ImageNet synsets at the time, followed by manual filtering to make sure the object categories were not too obscure. With the introduction of

| Task                                        |                                              | Image classification | Single-object localization   | Object detection            |
|---------------------------------------------|----------------------------------------------|----------------------|------------------------------|-----------------------------|
| Manual labeling on training set             | Number of object classes annotated per image | 1                    | 1                            | 1 or more                   |
|                                             | Locations of annotated classes               | —                    | all instances on some images | all instances on all images |
| Manual labeling on validation and test sets | Number of object classes annotated per image | 1                    | 1                            | all target classes          |
|                                             | Locations of annotated classes               | —                    | all instances on all images  | all instances on all images |

**Table 1** ILSVRC各项任务所提供标注的概述。

### 3 Dataset construction at large scale

我们构建大规模物体识别图像数据集的过程包含三个关键步骤。

第一步是定义目标物体类别的集合。为此，我们从现有的ImageNet（Deng等人，2009年）类别中进行筛选。通过以WordNet为骨干（Miller，1995年），ImageNet已经负责消除词语歧义并将同义词合并到同一物体类别中。由于物体类别的选择在每个挑战任务中只需进行一次，我们结合自动启发式方法和人工后处理，为每个任务创建合适的目标类别列表。例如，对于图像分类，我们可能包含更广泛的场景类别，如某种海滩类型；但对于单物体定位和物体检测，我们希望仅关注那些能在图像中明确{v\*}定位的物体类别（第3.1.1节和第3.3.1节）。

第二步是收集一组多样化的候选图像来代表所选类别。我们利用多种搜索引擎，结合自动与手动策略进行图像采集。针对不同的ILSVRC任务，这一流程会相应调整。例如，在物体检测任务中，我们重点通过“非洲野生动物园”等通用查询来收集场景类图像，以获取可能包含多个动物同框的画面（详见第3.3.2节）。

第三步（也是最富挑战性的一步）是对收集的数百万张图像进行标注，以获取一个干净的数据集。我们针对每个ILSVRC任务精心设计了众包策略。例如，用于定位和检测任务的边界框标注系统包含三个独立部分，以实现自动化的众包质量控制（第3.2.1节）。若要在合理预算内为物体检测任务完整标注所有目标物体类别，则需要额外的分层图像标注系统（第3.3.3节）。

我们依次描述ILSVRC各项任务的数据收集和标注流程：图像

分类（第3.1节）、单目标定位（第3.2节）和目标检测（第3.3节），重点阐述每个数据集的三个关键步骤。

#### 3.1 图像分类数据集构建

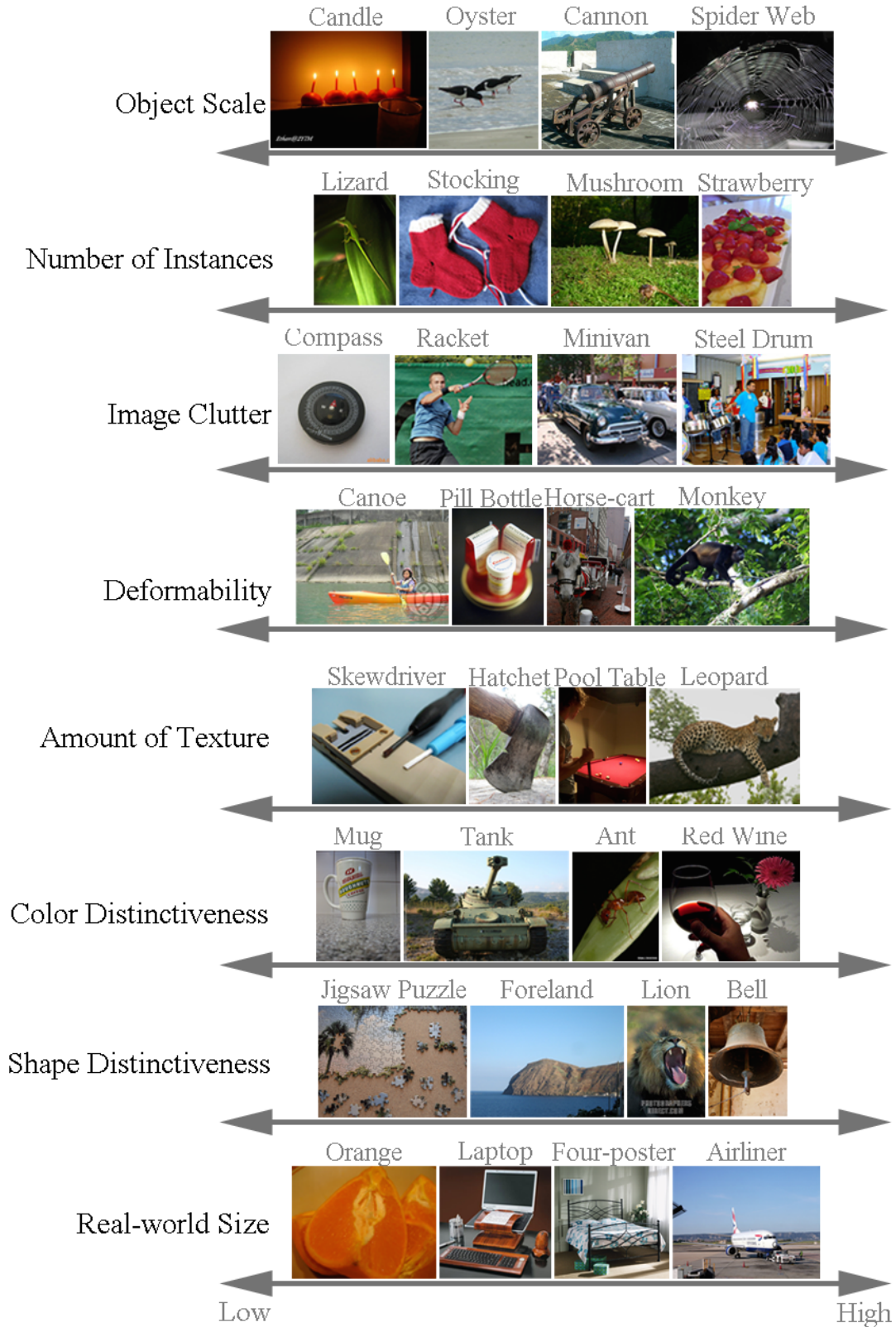
图像分类任务测试算法对图像中存在的物体进行命名的能力，无需精确定位它们。

我们阐述了构建ILSVRC图像分类数据集时所做的选择：从ImageNet中选定目标对象类别（第3.1.1节），通过使用多个搜索引擎和多种语言的扩展查询集来收集多样化的候选图像（第3.1.2节），最后借助ImageNet精心设计的众包策略（Deng et al., 2009）对数百万张收集到的图像进行筛选（第3.1.3节）。

##### 3.1.1 Defining object categories for the image classification dataset

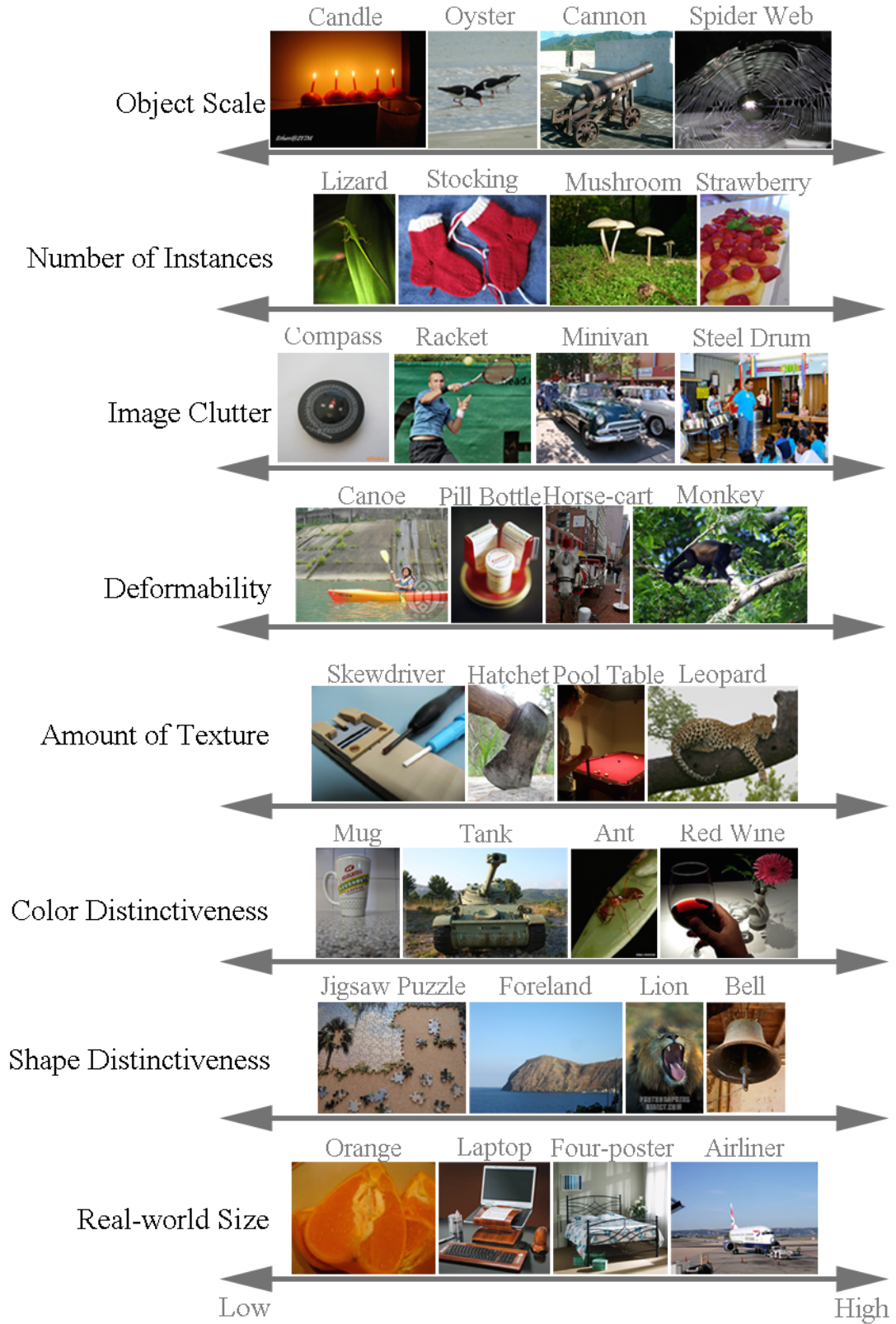
用于图像分类任务的1000个类别选自ImageNet（Deng等人，2009年）的类别。这1000个同义词集的选择确保了同义词集之间没有重叠：对于任意同义词集*i*和*j*，在ImageNet层次结构中*i*不是*j*的祖先。这些同义词集是更大层次结构的一部分，在ImageNet中可能拥有子类别；然而，对于ILSVRC，我们不考虑它们的子类别。ILSVRC的同义词集层次结构可以视为完整ImageNet层次结构的“修剪”版本。图1展示了ILSVRC2012对象类别的多样性。

用于图像分类和单目标定位任务的1000个精确同义词集（synsets）在过去几年中发生了变化。迄今为止，在所有五次ILSVRC挑战中，有639个同义词集被持续使用。在挑战赛的第一年，同义词集是从当时可用的ImageNet同义词集中随机选取的，随后经过人工筛选，以确保目标类别不会过于冷门。随着



**Fig. 1** The diversity of data in the ILSVRC image classification and single-object localization tasks. For each of the eight dimensions, we show example object categories along the range of that property. Object scale, number of instances and image clutter for each object category are computed using the metrics defined in Section 3.2.2 and in Appendix B. The other properties were computed by asking human subjects to annotate each of the 1000 object categories (Russakovsky et al., 2013).





**Fig. 1** ILSVRC图像分类与单目标定位任务中数据的多样性。针对八个维度中的每一个，我们沿该属性范围展示示例对象类别。每个对象类别的物体尺度、实例数量和图像杂乱度均使用第3.2.2节及附录B定义的度量标准进行计算。其余属性通过邀请受试者对1000个对象类别进行标注得出（Russakovsky等人，2013年）。



the object localization challenge in 2011 there were 321 synsets that changed: categories such as “New Zealand beach” which were inherently difficult to localize were removed, and some new categories from ImageNet containing object localization annotations were added. In ILSVRC2012, 90 synsets were replaced with categories corresponding to dog breeds to allow for evaluation of more fine-grained object classification, as shown in Figure 2. The synsets have remained consistent since year 2012. Appendix A provides the complete list of object categories used in ILSVRC2012-2014.

### *3.1.2 Collecting candidate images for the image classification dataset*

Image collection for ILSVRC classification task is the same as the strategy employed for constructing ImageNet (Deng et al., 2009). Training images are taken directly from ImageNet. Additional images are collected for the ILSVRC using this strategy and randomly partitioned into the validation and test sets.

We briefly summarize the process; (Deng et al., 2009) contains further details. Candidate images are collected from the Internet by querying several image search engines. For each synset, the queries are the set of WordNet synonyms. Search engines typically limit the number of retrievable images (on the order of a few hundred to a thousand). To obtain as many images as possible, we expand the query set by appending the queries with the word from parent synsets, if the same word appears in the glossary of the target synset. For example, when querying “whippet”, according to WordNet’s glossary a “small slender dog of greyhound type developed in England”, we also use “whippet dog” and “whippet greyhound.” To further enlarge and diversify the candidate pool, we translate the queries into other languages, including Chinese, Spanish, Dutch and Italian. We obtain accurate translations using WordNets in those languages.

### *3.1.3 Image classification dataset annotation*

Annotating images with corresponding object classes follows the strategy employed by ImageNet (Deng et al., 2009). We summarize it briefly here.

To collect a highly accurate dataset, we rely on humans to verify each candidate image collected in the previous step for a given synset. This is achieved by using Amazon Mechanical Turk (AMT), an online platform on which one can put up tasks for users for a monetary reward. With a global user base, AMT is particularly suitable for large scale labeling. In each of our

labeling tasks, we present the users with a set of candidate images and the definition of the target synset (including a link to Wikipedia). We then ask the users to verify whether each image contains objects of the synset. We encourage users to select images regardless of occlusions, number of objects and clutter in the scene to ensure diversity.

While users are instructed to make accurate judgment, we need to set up a quality control system to ensure this accuracy. There are two issues to consider. First, human users make mistakes and not all users follow the instructions. Second, users do not always agree with each other, especially for more subtle or confusing synsets, typically at the deeper levels of the tree. The solution to these issues is to have multiple users independently label the same image. An image is considered positive only if it gets a convincing majority of the votes. We observe, however, that different categories require different levels of consensus among users. For example, while five users might be necessary for obtaining a good consensus on Burmese cat images, a much smaller number is needed for cat images. We develop a simple algorithm to dynamically determine the number of agreements needed for different categories of images. For each synset, we first randomly sample an initial subset of images. At least 10 users are asked to vote on each of these images. We then obtain a confidence score table, indicating the probability of an image being a good image given the consensus among user votes. For each of the remaining candidate images in this synset, we proceed with the AMT user labeling until a pre-determined confidence score threshold is reached.

*Empirical evaluation.* Evaluation of the accuracy of the large-scale crowdsourced image annotation system was done on the entire ImageNet (Deng et al., 2009). A total of 80 synsets were randomly sampled at every tree depth of the mammal and vehicle subtrees. An independent group of subjects verified the correctness of each of the images. An average of 99.7% precision is achieved across the synsets. We expect similar accuracy on ILSVRC image classification dataset since the image annotation pipeline has remained the same. To verify, we manually checked 1500 ILSVRC2012-2014 image classification test set images (the test set has remained unchanged in these three years). We found 5 annotation errors, corresponding as expected to 99.7% precision.

### *3.1.4 Image classification dataset statistics*

Using the image collection and annotation procedure described in previous sections, we collected a large-scale dataset used for ILSVRC classification task. There

2011年的物体定位挑战中，有321个同义词集发生了变化：诸如“新西兰海滩”这类本质上难以定位的类别被移除，同时新增了来自ImageNet的、包含物体定位标注的部分新类别。在ILSVRC2012中，90个同义词集被替换为对应犬种细分类别的标签，以便评估更精细的物体分类能力，如图2所示。自2012年起，同义词集始终保持一致。附录A提供了ILSVRC2012-2014使用的完整物体类别列表。

### 3.1.2 Collecting candidate images for the image classification dataset

ILSVRC分类任务的图像收集采用了与构建ImageNet（Deng等人，2009年）相同的策略。训练图像直接取自ImageNet。使用该策略为ILSVRC收集的额外图像被随机划分为验证集和测试集。

我们简要概述这一过程；更多细节可参见(Deng et al., 2009)。候选图像通过向多个图像搜索引擎发起查询从互联网收集。对于每个同义词集，查询词为WordNet同义词集合。搜索引擎通常限制可检索图像数量（约数百至上千张）。为获取尽可能多的图像，我们通过将查询词与父同义词集中的词汇组合来扩展查询集——若该词汇同时出现在目标同义词集的词表中。例如查询“whippet”（根据WordNet释义为“在英国培育的一种灰猎犬型细长小型犬”）时，我们同时使用“whippet dog”和“whippet greyhound”。为进一步扩大并丰富候选图像池，我们将查询词翻译为其他语言，包括中文、西班牙语、荷兰语和意大利语，并借助对应语言的WordNet确保翻译准确性。

### 3.1.3 Image classification dataset annotation

为图像标注对应的物体类别遵循了ImageNet（Deng等人，2009年）采用的策略。我们在此简要概述。

为了收集高精度的数据集，我们依靠人工来验证前一步为给定同义词集收集的每个候选图像。这是通过使用亚马逊机械土耳其人（AMT）实现的，这是一个在线平台，用户可以在上面发布任务以获取金钱报酬。凭借全球用户基础，AMT特别适合大规模标注。在我们的每一个

在标注任务中，我们向用户展示一组候选图像以及目标同义词集的定义（包含维基百科链接）。随后请用用户验证每张图像是否包含该同义词集所指代的物体。我们鼓励用户选择图像时无需考虑遮挡情况、物体数量或场景杂乱程度，以确保数据多样性。

尽管用户被要求做出准确判断，我们仍需建立质量控制体系以确保这种准确性。这涉及两个需要考虑的问题。首先，人类用户会犯错，且并非所有用户都遵循指示。其次，用户之间并不总能达成一致，尤其是对于更微妙或易混淆的同义词集，通常位于树状结构的更深层级。解决这些问题的方法是让多位用户独立标注同一张图像。只有当图像获得令人信服的多数票时，才被视为正例。然而我们观察到，不同类别需要用户之间不同水平的共识。例如，虽然缅甸猫图像可能需要五位用户才能达成良好共识，但普通猫图像所需的用户数量则少得多。我们开发了一种简单算法，用于动态确定不同图像类别所需的赞同票数。对于每个同义词集，我们首先随机抽样一个初始图像子集。每张图像至少由10位用户进行投票。随后我们得到一个置信度评分表，该表显示了在用户投票共识下图像属于优质图像的概率。对于该同义词集中剩余的候选图像，我们会持续进行AMT用户标注，直至达到预设的置信度评分阈值。

*Empirical evaluation.* 对大规模众包图像标注系统准确性的评估是在整个ImageNet数据集上进行的（Deng等人，2009年）。在哺乳动物和车辆子树的每个树深度上，随机抽取了共计80个同义词集。一个独立的受试者小组验证了每张图像的正确性。在所有同义词集上，平均达到了99.7%的精确度。由于图像标注流程保持不变，我们预期在ILSVRC图像分类数据集上会有相似的准确性。为验证这一点，我们手动检查了1500张ILSVRC2012-2014图像分类测试集图像（该测试集在这三年中保持不变）。我们发现了5个标注错误，对应的精确度如预期为99.7%。

### 3.1.4 Image classification dataset statistics

利用前几节描述的图像收集和标注流程，我们收集了一个用于ILSVRC分类任务的大规模数据集。



**Fig. 2** The ILSVRC dataset contains many more fine-grained classes compared to the standard PASCAL VOC benchmark; for example, instead of the PASCAL “dog” category there are 120 different breeds of dogs in ILSVRC2012-2014 classification and single-object localization tasks.

are 1000 object classes and approximately 1.2 million training images, 50 thousand validation images and 100 thousand test images. Table 2 (top) documents the size of the dataset over the years of the challenge.

### 3.2 Single-object localization dataset construction

The single-object localization task evaluates the ability of an algorithm to localize one instance of an object category. It was introduced as a taster task in ILSVRC 2011, and became an official part of ILSVRC in 2012.

The key challenge was developing a scalable crowdsourcing method for object bounding box annotation. Our three-step self-verifying pipeline is described in Section 3.2.1. Having the dataset collected, we perform detailed analysis in Section 3.2.2 to ensure that the dataset is sufficiently varied to be suitable for evaluation of object localization algorithms.

*Object classes and candidate images.* The object classes for single-object localization task are the same as the object classes for image classification task described above in Section 3.1. The training images for localization task are a subset of the training images used for image classification task, and the validation and test images are the same between both tasks.

*Bounding box annotation.* Recall that for the image classification task every image was annotated with one

object class label, corresponding to one object that is present in an image. For the single-object localization task, every validation and test image and a subset of the training images are annotated with axis-aligned bounding boxes around every instance of this object.

Every bounding box is required to be as small as possible while including all visible parts of the object instance. An alternate annotation procedure could be to annotate the *full (estimated) extent* of the object: e.g., if a person’s legs are occluded and only the torso is visible, the bounding box could be drawn to include the likely location of the legs. However, this alternative procedure is inherently ambiguous and ill-defined, leading to disagreement among annotators and among researchers (what is the true “most likely” extent of this object?). We follow the standard protocol of only annotating visible object parts (Russell et al., 2007; Everingham et al., 2010).<sup>5</sup>

#### 3.2.1 Bounding box object annotation system

We summarize the crowdsourced bounding box annotation system described in detail in (Su et al., 2012). The goal is to build a system that is fully automated,

<sup>5</sup> Some datasets such as PASCAL VOC (Everingham et al., 2010) and LabelMe (Russell et al., 2007) are able to provide more detailed annotations: for example, marking individual object instances as being *truncated*. We chose not to provide this level of detail in favor of annotating more images and more object instances.



**Fig. 2** 与标准的PASCAL VOC基准相比，ILSVRC数据集包含了更多细粒度类别；例如，在ILSVRC2012-2014分类和单目标定位任务中，取代PASCAL的“狗”类别的是120种不同的犬种。

包含1000个对象类别，约120万张训练图像、5万张验证图像和10万张测试图像。表2（顶部）记录了挑战赛历年数据集的规模。

### 3.2 单目标定位数据集构建

单目标定位任务评估算法对某一物体类别单个实例的定位能力。该任务于2011年ILSVRC中作为体验性任务首次亮相，并于2012年成为ILSVRC的正式竞赛项目。

关键挑战在于开发一种可扩展的众包方法用于物体边界框标注。我们三步自验证流程在3.2.1节中详述。完成数据集收集后，我们在3.2.2节进行详细分析，以确保数据集具有充分的多样性，适用于物体定位算法的评估。

*Object classes and candidate images.* 单目标定位任务的对象类别与上文第3.1节描述的图像分类任务对象类别相同。定位任务的训练图像是图像分类任务所用训练图像的一个子集，而验证图像和测试图像在两项任务中保持一致。

*Bounding box annotation.* 回想一下，在图像分类任务中，每张图像都被标注了一个

物体类别标签，对应图像中存在的一个物体。对于单物体定位任务，每个验证和测试图像以及部分训练图像都标注了该物体每个实例的轴对齐边界框。

每个边界框都要求尽可能小，同时包含物体实例的所有可见部分。另一种标注方法可能是标注物体的 *full (estimated) extent*：例如，如果一个人的腿部被遮挡，只有躯干可见，边界框可以绘制为包含腿部可能的位置。然而，这种替代方法本质上具有模糊性和不明确性，会导致标注者和研究人员之间的分歧（这个物体的真实“最可能”范围是什么？）。我们遵循仅标注可见物体部分的标准协议（Russell等人，2007年；Everingham等人，2010年）。<sup>5</sup>

#### 3.2.1 Bounding box object annotation system

我们总结了（Su等人，2012）中详细描述众包边界框标注系统。目标是构建一个完全自动化的系统，

<sup>5</sup> Some datasets such as PASCAL VOC (Everingham et al., 2010) and LabelMe (Russell et al., 2007) are able to provide more detailed annotations: for example, marking individual object instances as being *truncated*. We chose not to provide this level of detail in favor of annotating more images and more object instances.



**Image classification annotations (1000 object classes)**

| Year          | Train images (per class) | Val images (per class) | Test images (per class) |
|---------------|--------------------------|------------------------|-------------------------|
| ILSVRC2010    | 1,261,406 (668-3047)     | 50,000 (50)            | 150,000 (150)           |
| ILSVRC2011    | 1,229,413 (384-1300)     | 50,000 (50)            | 100,000 (100)           |
| ILSVRC2012-14 | 1,281,167 (732-1300)     | 50,000 (50)            | 100,000 (100)           |

**Additional annotations for single-object localization (1000 object classes)**

| Year          | Train images with<br>bbox annotations<br>(per class) | Train bboxes<br>annotated<br>(per class) | Val images with<br>bbox annotations<br>(per class) | Val bboxes<br>annotated<br>(per class) | Test images with<br>bbox annotations |
|---------------|------------------------------------------------------|------------------------------------------|----------------------------------------------------|----------------------------------------|--------------------------------------|
| ILSVRC2011    | 315,525 (104-1256)                                   | 344,233 (114-1502)                       | 50,000 (50)                                        | 55,388 (50-118)                        | 100,000                              |
| ILSVRC2012-14 | 523,966 (91-1268)                                    | 593,173 (92-1418)                        | 50,000 (50)                                        | 64,058 (50-189)                        | 100,000                              |

**Table 2** Scale of ILSVRC image classification task (top) and single-object localization task (bottom). The numbers in parentheses correspond to (minimum per class - maximum per class). The 1000 classes change from year to year but are consistent between image classification and single-object localization tasks in the same year. All images from the image classification task may be used for single-object localization.

highly accurate, and cost-effective. Given a collection of images where the object of interest has been verified to exist, for each image the system collects a tight bounding box for every instance of the object.

There are two requirements:

- **Quality** Each bounding box needs to be tight, i.e. the smallest among all bounding boxes that contains all visible parts of the object. This facilitates the object detection learning algorithms by providing the precise location of each object instance;
- **Coverage** Every object instance needs to have a bounding box. This is important for training localization algorithms because it tells the learning algorithms with certainty what is not the object.

The core challenge of building such a system is effectively controlling the data quality with minimal cost. Our key observation is that drawing a bounding box is significantly more difficult and time consuming than giving answers to multiple choice questions. Thus quality control through additional verification tasks is more cost-effective than consensus-based algorithms. This leads to the following workflow with simple basic subtasks:

1. **Drawing** A worker draws one bounding box around one instance of an object on the given image.
2. **Quality verification** A second worker checks if the bounding box is correctly drawn.
3. **Coverage verification** A third worker checks if all object instances have bounding boxes.

The sub-tasks are designed following two principles. First, the tasks are made as simple as possible. For example, instead of asking the worker to draw all bounding boxes on the same image, we ask the worker to draw only one. This reduces the complexity of the task. Second, each task has a fixed and predictable amount of work. For example, assuming that the input images are

clean (object presence is correctly verified) and the coverage verification tasks give correct results, the amount of work of the drawing task is always that of providing exactly one bounding box.

Quality control on Tasks 2 and 3 is implemented by embedding “gold standard” images where the correct answer is known. Worker training for each of these subtasks is described in detail in (Su et al., 2012).

*Empirical evaluation.* The system is evaluated on 10 categories with ImageNet (Deng et al., 2009): balloon, bear, bed, bench, beach, bird, bookshelf, basketball hoop, bottle, and people. A subset of 200 images are randomly sampled from each category. On the image level, our evaluation shows that 97.9% images are completely covered with bounding boxes. For the remaining 2.1%, some bounding boxes are missing. However, these are all difficult cases: the size is too small, the boundary is blurry, or there is strong shadow.

On the bounding box level, 99.2% of all bounding boxes are accurate (the bounding boxes are visibly tight). The remaining 0.8% are somewhat off. No bounding boxes are found to have less than 50% intersection over union overlap with ground truth.

Additional evaluation of the overall cost and an analysis of quality control can be found in (Su et al., 2012).

### 3.2.2 Single-object localization dataset statistics

Using the annotation procedure described above, we collect a large set of bounding box annotations for the ILSVRC single-object classification task. All 50 thousand images in the validation set and 100 thousand images in the test set are annotated with bounding boxes around all instances of the ground truth object class (one object class per image). In addition, in ILSVRC2011 25% of training images are annotated with bounding

Image classification annotations (1000 object classes)

| Year          | Train images (per class) | Val images (per class) | Test images (per class) |
|---------------|--------------------------|------------------------|-------------------------|
| ILSVRC2010    | 1,261,406 (668-3047)     | 50,000 (50)            | 150,000 (150)           |
| ILSVRC2011    | 1,229,413 (384-1300)     | 50,000 (50)            | 100,000 (100)           |
| ILSVRC2012-14 | 1,281,167 (732-1300)     | 50,000 (50)            | 100,000 (100)           |

Additional annotations for single-object localization (1000 object classes)

| Year          | Train images with<br>bbox annotations<br>(per class) | Train bboxes<br>annotated<br>(per class) | Val images with<br>bbox annotations<br>(per class) | Val bboxes<br>annotated<br>(per class) | Test images with<br>bbox annotations |
|---------------|------------------------------------------------------|------------------------------------------|----------------------------------------------------|----------------------------------------|--------------------------------------|
| ILSVRC2011    | 315,525 (104-1256)                                   | 344,233 (114-1502)                       | 50,000 (50)                                        | 55,388 (50-118)                        | 100,000                              |
| ILSVRC2012-14 | 523,966 (91-1268)                                    | 593,173 (92-1418)                        | 50,000 (50)                                        | 64,058 (50-189)                        | 100,000                              |

**Table 2** ILSVRC图像分类任务（上）与单目标定位任务（下）的规模。括号中的数字对应（每类最少数量 - 每类最多数量）。1000个类别每年有所变动，但在同一年份中图像分类与单目标定位任务的类别保持一致。图像分类任务中的所有图像均可用于单目标定位任务。

高精度且成本效益显著。给定一组已验证存在感兴趣对象的图像，系统会为每张图像中该对象的每个实例收集一个紧密的边界框。

有两个要求：

- **Quality** 每个边界框都需要是紧凑的，即包含对象所有可见部分的最小边界框。这有助于物体检测学习算法，通过提供每个物体实例的精确位置；
- **Coverage** 每个物体实例都需要有一个边界框。这对于训练定位算法很重要，因为它明确告知学习算法哪些部分不属于该物体。

构建此类系统的核心挑战在于以最低成本有效控制数据质量。我们的关键观察是：绘制边界框远比回答多项选择题更为困难且耗时。因此，通过附加验证任务进行质量控制，比基于共识的算法更具成本效益。这催生了以下采用简单基础子任务的工作流程：

**1. Drawing** 一名工作人员在给定图像中为一个物体实例绘制一个边界框。**2. Quality verification** 第二名工作人员检查边界框是否正确绘制。**3.**

**Coverage verification** 第三名工作人员检查是否所有物体实例都已标注边界框。

子任务的设计遵循两个原则。首先，任务尽可能简化。例如，我们不要求工人在同一张图像上绘制所有边界框，而只要求绘制一个。这降低了任务的复杂度。其次，每个任务都有固定且可预估的工作量。例如，假设输入图像是

当清洁（物体存在性被正确验证）且覆盖验证任务给出正确结果时，绘图任务的工作量始终是提供恰好一个边界框。

任务2和3的质量控制通过嵌入已知正确答案的“黄金标准”图像来实现。这些子任务的工作人员培训详情见(Su et al., 2012)。

*Empirical evaluation.* 该系统在ImageNet (Deng等人, 2009)的10个类别上进行评估：气球、熊、床、长椅、海滩、鸟、书架、篮球框、瓶子和人物。每个类别随机抽取200张图像。在图像层面，我们的评估显示97.9%的图像被边界框完全覆盖。对于剩余的2.1%，部分边界框存在缺失。然而，这些均为困难案例：尺寸过小、边界模糊或存在强烈阴影。

在边界框层面，99.2%的边界框是精确的（边界框明显紧凑）。剩下的0.8%略有偏差。未发现任何边界框与真实标注的交并比重叠低于50%。

关于总体成本的额外评估以及质量控制的分析可在(Su et al., 2012)中查阅。

### 3.2.2 Single-object localization dataset statistics

利用上述标注流程，我们为ILSVRC单物体分类任务收集了大量边界框标注数据。验证集中的全部5万张图像及测试集中的10万张图像均已标注，所有真实物体类别的实例均被边界框标出（每张图像仅含一个物体类别）。此外，ILSVRC2011中25%的训练图像也完成了边界框标注。

boxes the same way, yielding more than 310 thousand annotated images with more than 340 thousand annotated object instances. In ILSVRC2012 40% of training images are annotated, yielding more than 520 thousand annotated images with more than 590 thousand annotated object instances. Table 2 (bottom) documents the size of this dataset.

In addition to the size of the dataset, we also analyze the level of difficulty of object localization in these images compared to the PASCAL VOC benchmark. We compute statistics on the ILSVRC2012 single-object localization validation set images compared to PASCAL VOC 2012 validation images.

Real-world scenes are likely to contain multiple instances of some objects, and nearby object instances are particularly difficult to delineate. The average object category in ILSVRC has 1.61 target object instances on average per positive image, with each instance having on average 0.47 neighbors (adjacent instances of the same object category). This is comparable to 1.69 instances per positive image and 0.52 neighbors per instance for an average object class in PASCAL.

As described in (Hoiem et al., 2012), smaller objects tend to be significantly more difficult to localize. In the average object category in PASCAL the object occupies 24.1% of the image area, and in ILSVRC 35.8%. However, PASCAL has only 20 object categories while ILSVRC has 1000. The 537 object categories of ILSVRC with the smallest objects on average occupy the same fraction of the image as PASCAL objects: 24.1%. Thus even though on average the object instances tend to be bigger in ILSVRC images, there are more than 25 times more object categories than in PASCAL VOC with the same average object scale.

Appendix B and (Russakovsky et al., 2013) have additional comparisons.

### 3.3 Object detection dataset construction

The ILSVRC task of object detection evaluates the ability of an algorithm to name and localize *all* instances of *all* target objects present in an image. It is much more challenging than object localization because some object instances may be small/occluded/difficult to accurately localize, and the algorithm is expected to locate them all, not just the one it finds easiest.

There are three key challenges in collecting the object detection dataset. The first challenge is selecting the set of common objects which tend to appear in cluttered photographs and are well-suited for benchmarking object detection performance. Our approach relies on statistics of the object localization dataset and the tradition of the PASCAL VOC challenge (Section 3.3.1).

| Class name in PASCAL VOC (20 classes) | Closest class in ILSVRC-DET (200 classes) | Avg object scale (%) |            |
|---------------------------------------|-------------------------------------------|----------------------|------------|
|                                       |                                           | PASCAL VOC           | ILSVRC-DET |
| aeroplane                             | airplane                                  | 29.7                 | 22.4       |
| bicycle                               | bicycle                                   | 29.3                 | 14.3       |
| bird                                  | bird                                      | 15.9                 | 20.1       |
| boat                                  | watercraft                                | 15.2                 | 16.5       |
| bottle                                | wine bottle                               | 7.3                  | 10.4       |
| bus                                   | bus                                       | 29.9                 | 22.1       |
| car                                   | car                                       | 14.0                 | 13.4       |
| cat                                   | domestic cat                              | 46.8                 | 29.8       |
| chair                                 | chair                                     | 12.8                 | 10.1       |
| cow                                   | cattle                                    | 19.3                 | 13.5       |
| dining table                          | table                                     | 29.1                 | 30.3       |
| dog                                   | dog                                       | 37.0                 | 28.9       |
| horse                                 | horse                                     | 29.5                 | 18.5       |
| motorbike                             | motorcycle                                | 32.0                 | 20.7       |
| person                                | person                                    | 17.5                 | 19.3       |
| potted plant                          | flower pot                                | 12.3                 | 8.1        |
| sheep                                 | sheep                                     | 12.2                 | 17.3       |
| sofa                                  | sofa                                      | 41.7                 | 44.4       |
| train                                 | train                                     | 35.4                 | 35.1       |
| tv/monitor                            | tv or monitor                             | 14.6                 | 11.2       |

**Table 3** Correspondences between the object classes in the PASCAL VOC (Everingham et al., 2010) and the ILSVRC detection task. Object scale is the fraction of image area (reported in percent) occupied by an object instance. It is computed on the validation sets of PASCAL VOC 2012 and of ILSVRC-DET. The average object scale is 24.1% across the 20 PASCAL VOC categories and 20.3% across the 20 corresponding ILSVRC-DET categories. Section 3.3.4 reports additional dataset statistics.

The second challenge is obtaining a much more varied set of scene images than those used for the image classification and single-object localization datasets. Section 3.3.2 describes the procedure for utilizing as much data from the single-object localization dataset as possible and supplementing it with Flickr images queried using hundreds of manually designed high-level queries.

The third, and biggest, challenge is completely annotating this dataset with all the objects. This is done in two parts. Section 3.3.3 describes the first part: our hierarchical strategy for obtaining the list of all target objects which occur within every image. This is necessary since annotating in a straight-forward way by creating a task for every (image, object class) pair is no longer feasible at this scale. Appendix E describes the second part: annotating the bounding boxes around these objects, using the single-object localization bounding box annotation pipeline of Section 3.2.1 along with extra verification to ensure that *every* instance of the object is annotated with exactly *one* bounding box.



以相同的方式标注边界框，产生了超过31万张标注图像，包含超过34万个标注对象实例。在ILSVRC2012中，40%的训练图像被标注，产生了超过52万张标注图像，包含超过59万个标注对象实例。表2（底部）记录了该数据集的规模。

除了数据集的大小，我们还分析了这些图像中物体定位的难度水平，并与PASCAL VOC基准进行了比较。我们计算了ILSVRC2012单物体定位验证集图像与PASCAL VOC 2012验证图像的统计信息。

现实场景中可能包含多个同类物体的实例，而相邻的物体实例尤其难以精确分割。在ILSVRC数据集中，平均每个正样本图像包含1.61个目标物体实例，每个实例平均有0.47个相邻实例（即同一物体类别中相邻的实例）。这一数据与PASCAL数据集中平均每个正样本图像包含1.69个实例、每个实例平均有0.52个相邻实例的情况相当。

如（Hoiem等人，2012年）所述，较小的物体往往明显更难定位。在PASCAL数据集的平均物体类别中，物体占据图像面积的24.1%，而在ILSVRC中这一比例为35.8%。然而，PASCAL仅包含20个物体类别，而ILSVRC有1000个。ILSVRC中平均物体尺寸最小的537个类别，其物体占图像面积的比例与PASCAL相同：24.1%。因此，尽管ILSVRC图像中的物体实例平均尺寸更大，但其物体类别数量是PASCAL VOC的25倍以上，且这些类别具有相同的平均物体尺度。

附录B和（Russakovsky等人，2013年）提供了更多比较。

### 3.3 目标检测数据集构建

ILSVRC的目标检测任务评估算法对图像中存在的*all*个目标对象的*all*个实例进行命名和定位的能力。这比目标定位更具挑战性，因为某些对象实例可能较小/被遮挡/难以准确定位，而算法需要定位所有实例，而不仅仅是它认为最容易找到的那一个。

收集目标检测数据集面临三个关键挑战。第一个挑战是选择那些在杂乱照片中常见且适合用于目标检测性能基准测试的物体类别。我们的方法依赖于目标定位数据集的统计数据以及PASCAL VOC挑战赛的傳統做法（第3.3.1节）。

| Class name in PASCAL VOC (20 classes) | Closest class in ILSVRC-DET (200 classes) | Avg object scale (%) |            |
|---------------------------------------|-------------------------------------------|----------------------|------------|
|                                       |                                           | PASCAL VOC           | ILSVRC-DET |
| aeroplane                             | airplane                                  | 29.7                 | 22.4       |
| bicycle                               | bicycle                                   | 29.3                 | 14.3       |
| bird                                  | bird                                      | 15.9                 | 20.1       |
| boat                                  | watercraft                                | 15.2                 | 16.5       |
| bottle                                | wine bottle                               | 7.3                  | 10.4       |
| bus                                   | bus                                       | 29.9                 | 22.1       |
| car                                   | car                                       | 14.0                 | 13.4       |
| cat                                   | domestic cat                              | 46.8                 | 29.8       |
| chair                                 | chair                                     | 12.8                 | 10.1       |
| cow                                   | cattle                                    | 19.3                 | 13.5       |
| dining table                          | table                                     | 29.1                 | 30.3       |
| dog                                   | dog                                       | 37.0                 | 28.9       |
| horse                                 | horse                                     | 29.5                 | 18.5       |
| motorbike                             | motorcycle                                | 32.0                 | 20.7       |
| person                                | person                                    | 17.5                 | 19.3       |
| potted plant                          | flower pot                                | 12.3                 | 8.1        |
| sheep                                 | sheep                                     | 12.2                 | 17.3       |
| sofa                                  | sofa                                      | 41.7                 | 44.4       |
| train                                 | train                                     | 35.4                 | 35.1       |
| tv/monitor                            | tv or monitor                             | 14.6                 | 11.2       |

**Table 3** PASCAL VOC（Everingham等人，2010）与ILSVRC检测任务中物体类别的对应关系。物体尺度指物体实例所占图像面积的比例（以百分比报告）。该数据基于PASCAL VOC 2012和ILSVRC-DET的验证集计算得出。在20个PASCAL VOC类别中，平均物体尺度为24.1%；在20个对应的ILSVRC-DET类别中，平均物体尺度为20.3%。第3.3.4节将报告更多数据集统计信息。

第二个挑战是获取比用于图像分类和单目标定位数据集的场景图像更为多样化的图像集。第3.3.2节描述了如何尽可能利用单目标定位数据集中的数据，并通过使用数百个手动设计的高级查询从Flickr获取的图像进行补充的过程。

第三个也是最大的挑战是完整标注数据集中所有对象。这项工作分为两部分：3.3.3节描述了第一部分——我们通过分层策略获取每张图像中所有目标对象的列表。由于在此规模下，为每个（图像，对象类别）组合单独创建标注任务已不可行，这一步骤显得尤为必要。附录E阐述了第二部分：利用3.2.1节中的单对象定位边界框标注流程，并辅以额外验证来确保每个对象的*every*实例都严格对应*one*个边界框，从而完成对这些对象的边界框标注。

### 3.3.1 Defining object categories for the object detection dataset

There are 200 object classes hand-selected for the detection task, each corresponding to a synset within ImageNet. These were chosen to be mostly basic-level object categories that would be easy for people to identify and label. The rationale is that the object detection system developed for this task can later be combined with a fine-grained classification model to further classify the objects if a finer subdivision is desired.<sup>6</sup> As with the 1000 classification classes, the synsets are selected such that there is no overlap: for any synsets  $i$  and  $j$ ,  $i$  is not an ancestor of  $j$  in the ImageNet hierarchy.

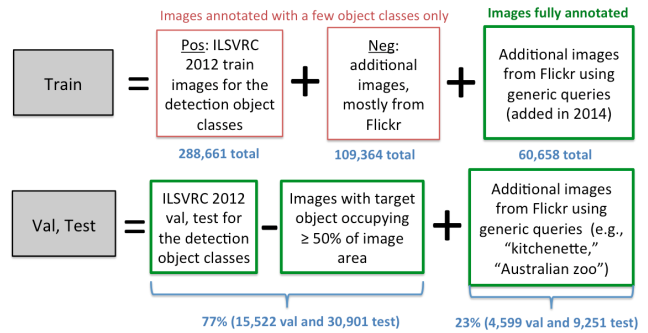
The selection of the 200 object detection classes in 2013 was guided by the ILSVRC 2012 classification and localization dataset. Starting with 1000 object classes and their bounding box annotations we first eliminated all object classes which tended to be too “big” in the image (on average the object area was greater than 50% of the image area). These were classes such as T-shirt, spiderweb, or manhole cover. We then manually eliminated all classes which we did not feel were well-suited for detection, such as hay, barbershop, or poncho. This left 494 object classes which were merged into basic-level categories: for example, different species of birds were merged into just the “bird” class. The classes remained the same in ILSVRC2014. Appendix D contains the complete list of object categories used in ILSVRC2013-2014 (in the context of the hierarchy described in Section 3.3.3).

Staying mindful of the tradition of the PASCAL VOC dataset we also tried to ensure that the set of 200 classes contains as many of the 20 PASCAL VOC classes as possible. Table 3 shows the correspondences. The changes that were done were to ensure more accurate and consistent crowdsourced annotations. The object class with the weakest correspondence is “potted plant” in PASCAL VOC, corresponding to “flower pot” in ILSVRC. “Potted plant” was one of the most challenging object classes to annotate consistently among the PASCAL VOC classes, and in order to obtain accurate annotations using crowdsourcing we had to restrict the definition to a more concrete object.

### 3.3.2 Collecting images for the object detection dataset

Many images for the detection task were collected differently than the images in ImageNet and the classifica-

<sup>6</sup> Some of the training objects are actually annotated with more detailed classes: for example, one of the 200 object classes is the category “dog,” and some training instances are annotated with the specific dog breed.



**Fig. 3** Summary of images collected for the detection task. Images in green (bold) boxes have all instances of all 200 detection object classes fully annotated. Table 4 lists the complete statistics.

tion and single-object localization tasks. Figure 3 summarizes the types of images that were collected. Ideally all of these images would be scene images fully annotated with all target categories. However, given budget constraints our goal was to provide as much suitable detection data as possible, even if the images were drawn from a few different sources and distributions.

The validation and test detection set images come from two sources (percent of images from each source in parentheses). The first source (77%) is images from ILSVRC2012 single-object localization validation and test sets corresponding to the 200 detection classes (or their children in the ImageNet hierarchy). Images where the target object occupied more than 50% of the image area were discarded, since they were unlikely to contain other objects of interest. The second source (23%) is images from Flickr collected specifically for detection task. We queried Flickr using a large set of manually defined queries, such as “kitchenette” or “Australian zoo” to retrieve images of scenes likely to contain several objects of interest. Appendix C contains the full list. We also added pairwise queries, or queries with two target object names such as “tiger lion,” which also often returned cluttered scenes.

Figure 4 shows a random set of both types of validation images. Images were randomly split, with 33% going into the validation set and 67% into the test set.<sup>7</sup>

The training set for the detection task comes from three sources of images (percent of images from each source in parentheses). The first source (63%) is all training images from ILSVRC2012 single-object localization task corresponding to the 200 detection classes (or their children in the ImageNet hierarchy). We did not filter by object size, allowing teams to take advan-

<sup>7</sup> The validation/test split is consistent with ILSVRC2012: validation images of ILSVRC2012 remained in the validation set of ILSVRC2013, and ILSVRC2012 test images remained in ILSVRC2013 test set.

### 3.3.1 Defining object categories for the object detection dataset

为检测任务手工选取了200个物体类别，每个类别对应ImageNet中的一个同义词集。这些类别大多选择为基础层级的物体分类，便于人们识别和标注。其设计理念在于：若需要更精细的划分，可在此任务开发的物体检测系统基础上，结合细粒度分类模型对物体进行进一步归类。<sup>6</sup> 与1000个分类类别类似，同义词集的选取遵循非重叠原则：对于任意同义词集 $i$ 和 $j$ ，在ImageNet层级结构中 $i$ 不是 $j$ 的祖先节点。

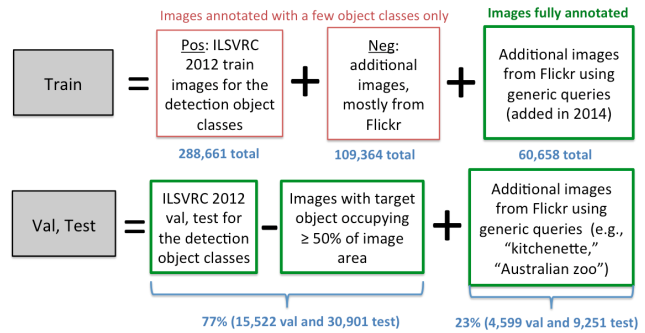
2013年选定的200个目标检测类别是基于ILSVRC 2012分类与定位数据集确定的。我们从1000个目标类别及其边界框标注出发，首先剔除了所有在图像中通常显得过于“庞大”的类别（即目标平均面积超过图像面积的50%），例如T恤衫、蜘蛛网或井盖等类别。随后，我们人工排除了所有我们认为不适合用于检测的类别，例如干草、理发店或披风。经过筛选后剩余494个目标类别，这些类别被合并为基础层级类别：例如，不同种类的鸟类被统一归为“鸟类”。ILSVRC2014中沿用了相同的类别设置。附录D完整列出了ILSVRC2013-2014使用的目标类别清单（其层级结构遵循第3.3.3节所述的分类体系）。

在铭记PASCAL VOC数据集传统的同时，我们也尽力确保这200个类别尽可能涵盖20个PASCAL VOC类别。表3展示了对应关系。所做的调整是为了保证更精确、更一致的众包标注。对应关系最弱的物体类别是PASCAL VOC中的“盆栽植物”，对应ILSVRC中的“花盆”。“盆栽植物”原是PASCAL VOC所有类别中最难保持标注一致性的对象之一，为通过众包获得精确标注，我们不得不将定义限制为更具体的物体。

### 3.3.2 Collecting images for the object detection dataset

检测任务中的许多图像采集方式与ImageNet中的图像不同，且分类任务中的图像采集方式也有所差异。

<sup>6</sup> Some of the training objects are actually annotated with more detailed classes: for example, one of the 200 object classes is the category “dog,” and some training instances are annotated with the specific dog breed.



**Fig. 3** 为检测任务收集的图像总结。绿色（粗体）框中的图像已完整标注了所有200个检测对象类别的全部实例。表4列出了完整的统计数据。

检测和单目标定位任务。图3总结了所收集图像的类型。理想情况下，所有这些图像都应是完全标注了所有目标类别的场景图像。然而，考虑到预算限制，我们的目标是提供尽可能多的合适检测数据，即使这些图像来自少数不同的来源和分布。

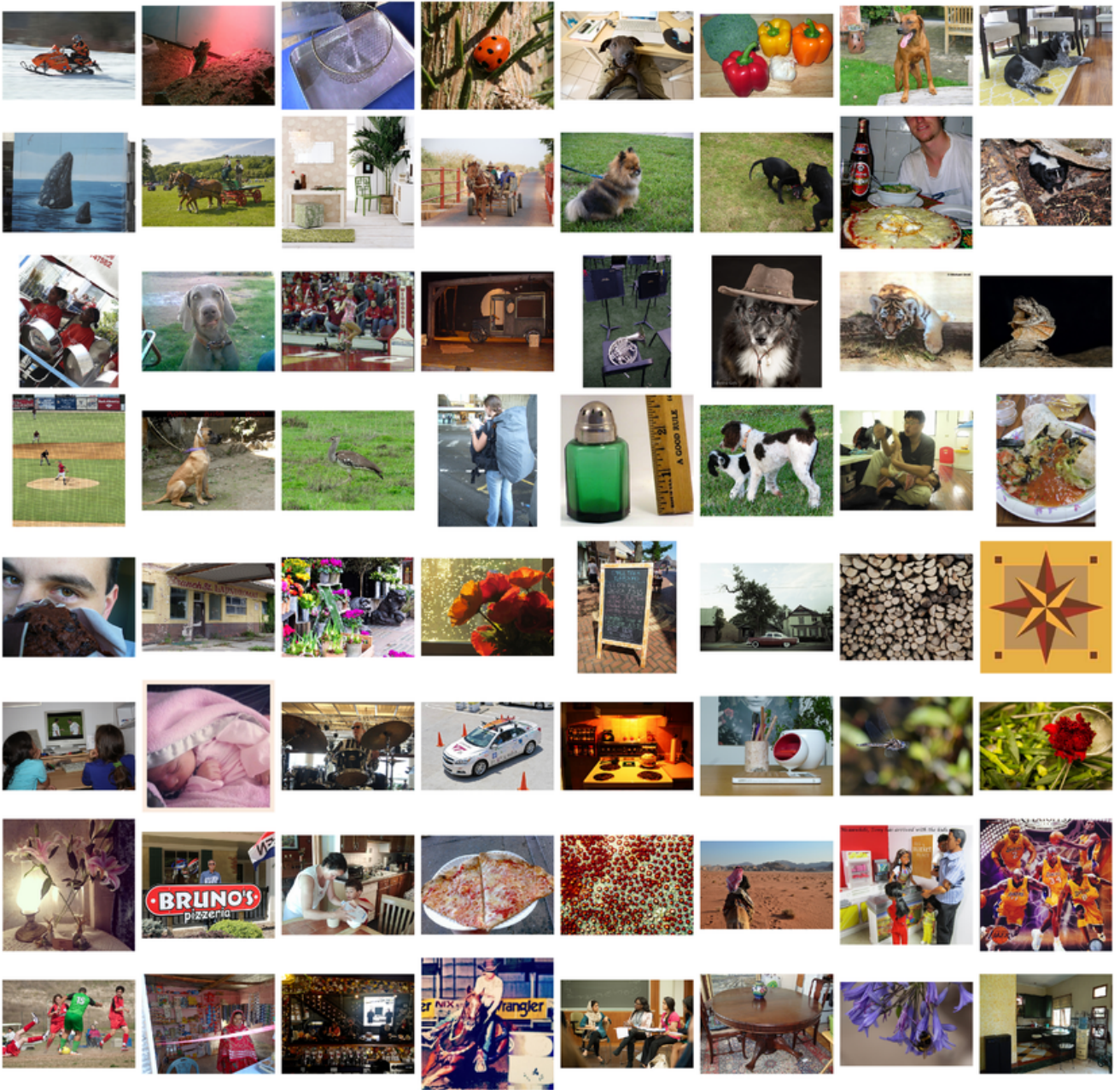
验证和测试检测集的图像来自两个来源（每个来源的图像百分比在括号中）。第一个来源（77%）是来自ILSVRC2012单目标定位验证集和测试集的图像，对应200个检测类别（或它们在ImageNet层级结构中的子类）。目标对象占据图像面积超过50%的图像被剔除，因为它们不太可能包含其他感兴趣的对象。第二个来源（23%）是专门为检测任务从Flickr收集的图像。我们使用大量手动定义的查询（如“小厨房”或“澳大利亚动物园”）在Flickr上检索可能包含多个感兴趣对象的场景图像。附录C包含完整列表。我们还添加了成对查询，或包含两个目标对象名称的查询（如“老虎狮子”），这类查询也经常返回杂乱场景的图像。

图4展示了两类验证图像的随机集合。图像被随机分割，其中33%进入验证集，67%进入测试集。 $\{v^*\}$

检测任务的训练集来自三个图像来源（每个来源的图像百分比在括号中）。第一个来源（63%）是ILSVRC2012单目标定位任务中所有与200个检测类别（或它们在ImageNet层级中的子类别）对应的训练图像。我们没有按物体大小进行筛选，以便让团队能够充分利用—

<sup>7</sup> The validation/test split is consistent with ILSVRC2012: validation images of ILSVRC2012 remained in the validation set of ILSVRC2013, and ILSVRC2012 test images remained in ILSVRC2013 test set.



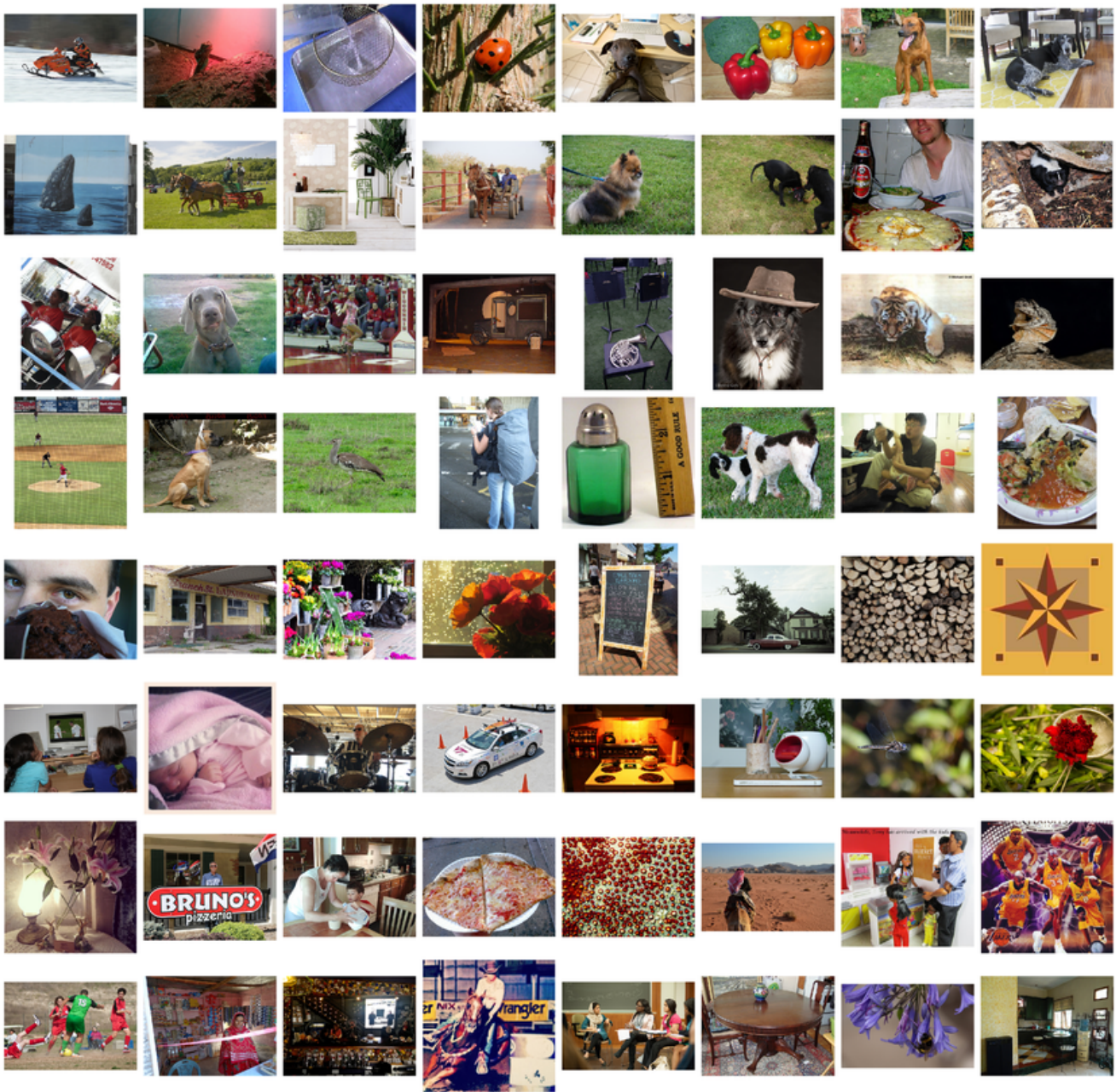


**Fig. 4** Random selection of images in ILSVRC detection validation set. The images in the top 4 rows were taken from ILSVRC2012 single-object localization validation set, and the images in the bottom 4 rows were collected from Flickr using scene-level queries.

tage of all the positive examples available. The second source (24%) is negative images which were part of the original ImageNet collection process but voted as negative: for example, some of the images were collected from Flickr and search engines for the ImageNet synset “animals” but during the manual verification step did not collect enough votes to be considered as containing an “animal.” These images were manually re-verified for the detection task to ensure that they did not in fact contain the target objects. The third source (13%)

is images collected from Flickr specifically for the detection task. These images were added for ILSVRC2014 following the same protocol as the second type of images in the validation and test set. This was done to bring the training and testing distributions closer together.

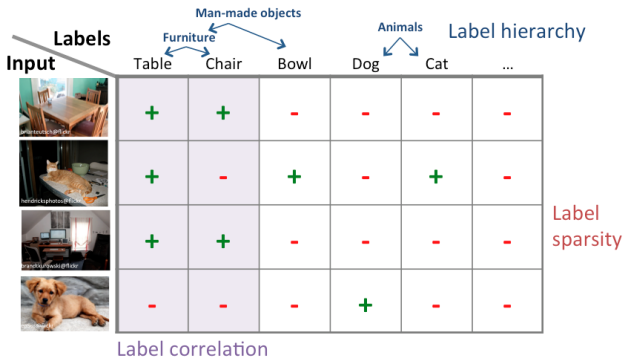




**Fig. 4** ILSVRC检测验证集中随机选取的图像。前4行的图像来自ILSVRC2012单目标定位验证集，后4行的图像是通过场景级查询从Flickr收集的。

所有可用正例的标签。第二个来源（24%）是负面图像，这些图像原本是ImageNet收集过程的一部分，但被投票判定为负面：例如，部分图像是从Flickr和搜索引擎中为ImageNet同义词集“动物”收集的，但在人工验证步骤中未获得足够票数被认定为包含“动物”。这些图像为检测任务进行了人工重新验证，以确保它们实际上不包含目标物体。第三个来源（13%）

这些图像是从Flickr专门为检测任务收集的。这些图像按照与验证集和测试集中第二类图像相同的协议，被添加至ILSVRC2014。此举旨在使训练和测试的数据分布更为接近。



**Fig. 5** Consider the problem of binary multi-label annotation. For each input (e.g., image) and each label (e.g., object), the goal is to determine the presence or absence (+ or -) of the label (e.g., decide if the object is present in the image). Multi-label annotation becomes much more efficient when considering real-world structure of data: correlation between labels, hierarchical organization of concepts, and sparsity of labels.

### 3.3.3 Complete image-object annotation for the object detection dataset

The key challenge in annotating images for the object detection task is that all objects in all images need to be labeled. Suppose there are  $N$  inputs (images) which need to be annotated with the presence or absence of  $K$  labels (objects). A naïve approach would query humans for each combination of input and label, requiring  $NK$  queries. However,  $N$  and  $K$  can be very large and the cost of this exhaustive approach quickly becomes prohibitive. For example, annotating 60,000 validation and test images with the presence or absence of 200 object classes for the detection task naïvely would take 80 times more effort than annotating 150,000 validation and test images with 1 object each for the classification task – and this is not even counting the additional cost of collecting bounding box annotations around each object instance. This quickly becomes infeasible.

In (Deng et al., 2014) we study strategies for scalable multilabel annotation, or for efficiently acquiring multiple labels from humans for a collection of items. We exploit three key observations for labels in real world applications (illustrated in Figure 5):

1. **Correlation.** Subsets of labels are often highly correlated. Objects such as a computer keyboard, mouse and monitor frequently co-occur in images. Similarly, some labels tend to all be absent at the same time. For example, all objects that require electricity are usually absent in pictures taken outdoors. This suggests that we could potentially fill in the values of multiple labels by grouping them into only one query for humans. Instead of checking if dog, cat, rabbit etc. are present in the photo, we just check

about the “animal” group. If the answer is no, then this implies a no for all categories in the group.

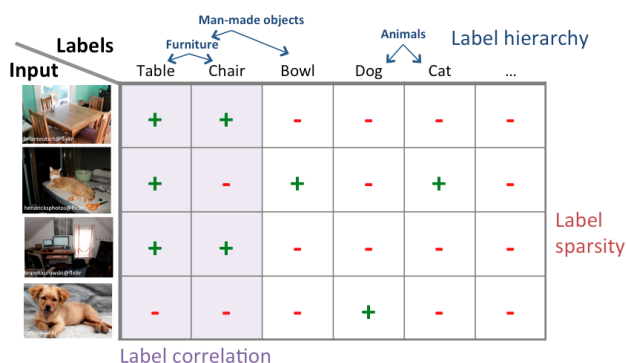
2. **Hierarchy.** The above example of grouping dog, cat, rabbit etc. into animal has implicitly assumed that labels can be grouped together and humans can efficiently answer queries about the group as a whole. This brings up our second key observation: humans organize semantic concepts into hierarchies and are able to efficiently categorize at higher semantic levels (Thorpe et al., 1996), e.g. humans can determine the presence of an animal in an image as fast as every type of animal individually. This leads to substantial cost savings.
3. **Sparsity.** The values of labels for each image tend to be sparse, i.e. an image is unlikely to contain more than a dozen types of objects, a small fraction of the hundreds of object categories. This enables rapid elimination of many objects by quickly filling in no. With a high degree of sparsity, an efficient algorithm can have a cost which grows logarithmically with the number of objects instead of linearly.

We propose algorithmic strategies that exploit the above intuitions. The key is to select a sequence of queries for humans such that we achieve the same labeling results with only a fraction of the cost of the naïve approach. The main challenges include how to measure cost and utility of queries, how to construct good queries, and how to dynamically order them. A detailed description of the generic algorithm, along with theoretical analysis and empirical evaluation, is presented in (Deng et al., 2014).

*Application of the generic multi-class labeling algorithm to our setting.* The generic algorithm automatically selects the most informative queries to ask based on object label statistics learned from the training set. In our case of 200 object classes, since obtaining the training set was by itself challenging we chose to design the queries by hand. We created a hierarchy of queries of the type “is there a... in the image?” For example, one of the high-level questions was “is there an animal in the image?” We ask the crowd workers this question about every image we want to label. The children of the “animal” question would correspond to specific examples of animals: for example, “is there a mammal in the image?” or “is there an animal with no legs?” To annotate images efficiently, these questions are asked only on images determined to contain an animal. The 200 leaf node questions correspond to the 200 target objects, e.g., “is there a cat in the image?”. A few sample iterations of the algorithm are shown in Figure 6.

Algorithm 1 is the formal algorithm for labeling an image with the presence or absence of each target object





**Fig. 5** 考虑二元多标签标注问题。对于每个输入（如图像）和每个标签（如物体），目标是确定标签的存在或缺失（+ 或 -）（例如判断图像中是否存在该物体）。当考虑到数据的真实世界结构时——标签间的相关性、概念的层次化组织以及标签的稀疏性——多标签标注的效率将大幅提升。

### 3.3.3 Complete image-object annotation for the object detection dataset

为物体检测任务标注图像的关键挑战在于，所有图像中的所有物体都需要被标记。假设有 $N$ 个输入（图像）需要标注 $K$ 个标签（物体）的存在与否。一种天真的方法会为每个输入和标签的组合向人类提问，需要进行 $NK$ 次查询。然而， $N$ 和 $K$ 可能非常大，这种穷举方法的成本很快就会变得难以承受。例如，为检测任务标注60,000张验证和测试图像中200个物体类别的存在与否，其工作量将比分类任务中标注150,000张验证和测试图像（每张图像仅含1个物体）多出80倍——这甚至还未计入为每个物体实例收集边界框标注的额外成本。这种方法很快就会变得不可行。

在（Deng等人，2014）中，我们研究了可扩展的多标签标注策略，即如何高效地从人类标注者处为项目集合获取多个标签。我们利用了现实应用场景中标签的三个关键观察（如图5所示）：

**1. Correlation.** 标签子集通常高度相关。例如计算机键盘、鼠标和显示器等物体在图像中频繁同时出现。类似地，某些标签往往同时全部缺失。例如所有需要电力的物体在户外拍摄的照片中通常都不存在。这表明我们可以通过将多个标签分组为单一人工查询项，来推测多个标签的取值。与其逐一检查照片中是否存在狗、猫、兔子等，我们只需确认

关于“动物”组，如果答案为否，则意味着该组中所有类别的答案均为否。

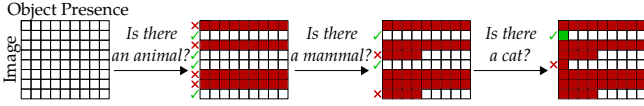
**2. Hierarchy.** 上述将狗、猫、兔子等归类为动物的例子隐含了一个前提：标签可以被分组，且人类能够高效地回答关于整个组别的查询。这引出了我们的第二个关键发现：人类将语义概念组织成层次结构，并能够在更高的语义层次上进行高效分类（Thorpe等人，1996），例如，人类判断图像中是否存在动物的速度，与判断每种具体动物个体的速度同样快。这能带来显著的成本节约。

**3. Sparsity.** 每张图像的标签值往往具有稀疏性，即一张图像不太可能包含超过十几种物体类型，这仅占数百种物体类别中的一小部分。这种特性使得我们能够通过快速填充“否”来迅速排除大量物体。在高度稀疏的情况下，高效算法的成本可以随物体数量呈对数增长，而非线性增长。

我们提出了利用上述直觉的算法策略。关键在于为人类选择一系列查询，使得我们仅需朴素方法成本的一小部分就能达到相同的标注效果。主要挑战包括如何衡量查询的成本与效用、如何构建有效的查询，以及如何动态排序。关于通用算法的详细描述，连同理论分析和实证评估，已在（Deng et al., 2014）中给出。

*Application of the generic multi-class labeling algorithm to our setting.* 通用算法基于从训练集中学习到的物体标签统计信息，自动选择最具信息量的查询进行提问。在我们涉及200个物体类别的情况下，由于获取训练集本身具有挑战性，我们选择手动设计查询问题。我们创建了一个层次化的查询体系，问题类型为“图像中是否有……？”。例如，一个高层级的问题是“图像中是否有动物？”。我们针对每张待标注图像向众包工作者提出这个问题。“动物”问题的子问题则对应具体的动物类别：例如“图像中是否有哺乳动物？”或“图像中是否有无腿动物？”。为高效标注图像，这类问题仅在对已确定包含动物的图像进行提问。200个叶节点问题对应200个目标物体，例如“图像中是否有猫？”。图6展示了该算法的几次示例迭代过程。

算法1是用于标注图像中每个目标对象存在与否的正式算法



**Fig. 6** Our algorithm dynamically selects the next query to efficiently determine the presence or absence of every object in every image. Green denotes a positive annotation and red denotes a negative annotation. This toy example illustrates a sample progression of the algorithm for one label (cat) on a set of images.

category. With this algorithm in mind, the hierarchy of questions was constructed following the principle that false positives only add extra cost whereas false negatives can significantly affect the quality of the labeling. Thus, it is always better to stick with more general but less ambiguous questions, such as “is there a mammal in the image?” as opposed to asking overly specific but potentially ambiguous questions, such as “is there an animal that can climb trees?” Constructing this hierarchy was a surprisingly time-consuming process, involving multiple iterations to ensure high accuracy of labeling and avoid question ambiguity. Appendix D shows the constructed hierarchy.

*Bounding box annotation.* Once all images are labeled with the presence or absence of all object categories we use the bounding box system described in Section 3.2.1 along with some additional modifications of Appendix E to annotate the location of every instance of every present object category.

### 3.3.4 Object detection dataset statistics

Using the procedure described above, we collect a large-scale dataset for ILSVRC object detection task. There are 200 object classes and approximately 450K training images, 20K validation images and 40K test images. Table 4 documents the size of the dataset over the years of the challenge. The major change between ILSVRC2013 and ILSVRC2014 was the addition of 60,658 fully annotated training images.

Prior to ILSVRC, the object detection benchmark was the PASCAL VOC challenge (Everingham et al., 2010). ILSVRC has 10 times more object classes than PASCAL VOC (200 vs 20), 10.6 times more fully annotated training images (60,658 vs 5,717), 35.2 times more training objects (478,807 vs 13,609), 3.5 times more validation images (20,121 vs 5823) and 3.5 times more validation objects (55,501 vs 15,787). ILSVRC has 2.8 annotated objects per image on the validation set, compared to 2.7 in PASCAL VOC. The average object in ILSVRC takes up 17.0% of the image area and in PASCAL VOC takes up 20.7%; Table 3 contains

**Input:** Image  $i$ , queries  $\mathcal{Q}$ , directed graph  $\mathcal{G}$  over  $\mathcal{Q}$

**Output:** Labels  $L : \mathcal{Q} \rightarrow \{\text{“yes”}, \text{“no”}\}$

Initialize labels  $L(q) = \emptyset \forall q \in \mathcal{Q}$ ;

Initialize candidates  $C = \{q : q \in \text{Root}(\mathcal{G})\}$ ;

**while**  $C$  not empty **do**

    Obtain answer  $A$  to query  $q^* \in C$ ;

$L(q^*) = A$ ;  $C = C \setminus \{q^*\}$ ;

**if**  $A$  is “yes” **then**

$\text{Chldr} = \{q \in \text{Children}(q^*, \mathcal{G}) : L(q) = \emptyset\}$ ;

$C = C \cup \text{Chldr}$ ;

**else**

$\text{Des} = \{q \in \text{Descendants}(q^*, \mathcal{G}) : L(q) = \emptyset\}$ ;

$L(q) = \text{“no”} \forall q \in \text{Des}$ ;

$C = C \setminus \text{Des}$ ;

**end**

**end**

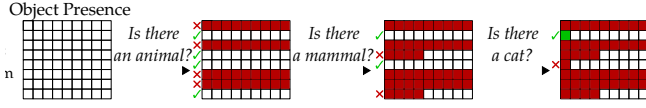
**Algorithm 1:** The algorithm for complete multi-class annotation. This is a special case of the algorithm described in (Deng et al., 2014). A hierarchy of questions  $\mathcal{G}$  is manually constructed. All root questions are asked on every image. If the answer to query  $q^*$  on image  $i$  is “no” then the answer is assumed to be “no” for all queries  $q$  such that  $q$  is a descendant of  $q^*$  in the hierarchy. We continue asking the queries until all queries are answered. For images taken from the single-object localization task we used the known object label to initialize  $L$ .

per-class comparisons. Additionally, ILSVRC contains a wide variety of objects, including tiny objects such as sunglasses (1.3% of image area on average), ping-pong balls (1.5% of image area on average) and basketballs (2.0% of image area on average).

## 4 Evaluation at large scale

Once the dataset has been collected, we need to define a standardized evaluation procedure for algorithms. Some measures have already been established by datasets such as the Caltech 101 (Fei-Fei et al., 2004) for image classification and PASCAL VOC (Everingham et al., 2012) for both image classification and object detection. To adapt these procedures to the large-scale setting we had to address three key challenges. First, for the image classification and single-object localization tasks only one object category could be labeled in each image due to the scale of the dataset. This created potential ambiguity during evaluation (addressed in Section 4.1). Second, evaluating localization of object instances is inherently difficult in some images which contain a cluster of objects (addressed in Section 4.2). Third, evaluating localization of object instances which occupy few pixels in the image is challenging (addressed in Section 4.3).

In this section we describe the standardized evaluation criteria for each of the three ILSVRC tasks.



**Fig. 6** 我们的算法动态选择下一个查询，以高效确定每个图像中每个对象的存在与否。绿色表示肯定标注，红色表示否定标注。这个简化示例展示了算法在针对一组图像中某个标签（猫）的典型执行过程。

类别。考虑到这一算法，问题的层级结构遵循以下原则构建：误判只会增加额外成本，而漏判则会显著影响标注质量。因此，始终应选择更通用、歧义更少的问题，例如“图像中有哺乳动物吗？”，而非提出过于具体但可能产生歧义的问题，例如“图像中有能爬树的动物吗？”。构建这一层级结构的过程出乎意料地耗时，需要多次迭代以确保标注的高准确性并避免问题歧义。附录D展示了构建的层级结构。

*Bounding box annotation.* 在所有图像都被标记了所有对象类别的存在与否后，我们使用第3.2.1节中描述的边界框系统，并结合附录E中的一些额外修改，来标注每个存在的对象类别中每个实例的位置。

### 3.3.4 Object detection dataset statistics

使用上述流程，我们为ILSVRC目标检测任务收集了一个大规模数据集。该数据集包含200个对象类别，约45万张训练图像、2万张验证图像和4万张测试图像。表4记录了历年挑战赛中数据集的规模变化。ILSVRC2013与ILSVRC2014之间的主要差异在于新增了60,658张完整标注的训练图像。

在ILSVRC之前，目标检测的基准是PASCAL VOC挑战赛（Everingham等人，2010年）。ILSVRC的目标类别数量是PASCAL VOC的10倍（200 vs 20），完全标注的训练图像数量是10.6倍（60,658 vs 5,717），训练目标数量是35.2倍（478,807 vs 13,609），验证图像数量是3.5倍（20,121 vs 5823），验证目标数量是3.5倍（55,501 vs 15,787）。在验证集上，ILSVRC每张图像有2.8个标注目标，而PASCAL VOC为2.7个。ILSVRC中的平均目标占据图像面积的17.0%，而PASCAL VOC中为20.7%；表3包含

**Input** 图像  $i$ , 查询  $Q$ , 有向图  $G$  覆盖  $Q$  **Output**: 标签  $L: Q \rightarrow \{ \text{“是”}, \text{“否”} \}$  初始化标签  
 $L(q) = \emptyset \forall q \in Q$ ; 初始化候选  $C = \{q: q \in \text{Root}(G)\}$   
**while**  $C$  not empty **do** 获取查询  $q^* \in C$  的答案  $A$   
**if**  $L(q^*) \leftarrow A$ ; **then**  $C \setminus \{q^*\}$ ;  
 $\text{Chldr} = \{q \in \text{Children}(q^*, G): L(q) = \emptyset\}$ ;  
**else**  $C \cup \text{Chldr}$ ;  
 $\text{Des} = \{q \in \text{Descendants}(q^*, G): L(q) = \emptyset\}$ ;  
 $L(q) = \text{“no”} \forall q \in \text{Des}; C = C \setminus \text{Des}$ ; **end**  
**end**

**Algorithm 1:** 完整多类别标注算法。这是(Deng et al., 2014)中所述算法的一个特例。我们手动构建了一个问题层次结构 $G$ 。所有根问题都会在每个图像上提问。若图像 $i$ 对查询 $q^*$ 的回答为“否”，则假定所有满足“ $q$ 是层次结构中 $q^*$ 的后代节点”条件的查询 $q$ 答案均为“否”。我们将持续提问直至所有查询获得回答。对于来自单物体定位任务的图像，我们使用已知物体标签来初始化 $L$ 。

每类比较。此外，ILSVRC包含了各种各样的物体，包括微小物体，如太阳镜（平均占图像面积的1.3%）、乒乓球（平均占图像面积的1.5%）和篮球（平均占图像面积的2.0%）。

## 4 Evaluation at large scale

数据集收集完成后，我们需要为算法定义标准化的评估流程。部分衡量标准已由现有数据集确立，例如用于图像分类的Caltech 101（Fei-Fei等，2004）以及同时涵盖图像分类与目标检测的PASCAL VOC（Everingham等，2012）。为将这些评估流程适配于大规模场景，我们必须解决三个关键挑战：首先，在图像分类与单目标定位任务中，由于数据集规模限制，每张图像仅能标注一个物体类别，这可能导致评估过程中的歧义（解决方案见第4.1节）；其次，对包含密集物体群的图像进行实例定位评估存在固有困难（解决方案见第4.2节）；最后，评估图像中仅占少量像素的物体实例定位极具挑战性（解决方案见第4.3节）。

在本节中，我们描述了针对ILSVRC三项任务中每一项的标准化评估标准。

**Object detection annotations (200 object classes)**

| Year       | Train images<br>(per class)                             | Train bboxes annotated<br>(per class) | Val images<br>(per class)              | Val bboxes annotated<br>(per class) | Test<br>images |
|------------|---------------------------------------------------------|---------------------------------------|----------------------------------------|-------------------------------------|----------------|
| ILSVRC2013 | 395909<br>(417-561-66911 pos,<br>185-4130-10073 neg)    | 345854<br>(438-660-73799)             | 21121<br>(23-58-5791 pos,<br>rest neg) | 55501<br>(31-111-12824)             | 40152          |
| ILSVRC2014 | 456567<br>(461-823-67513 pos,<br>42945-64614-70626 neg) | 478807<br>(502-1008-74517)            | 21121<br>(23-58-5791 pos,<br>rest neg) | 55501<br>(31-111-12824)             | 40152          |

**Table 4** Scale of ILSVRC object detection task. Numbers in parentheses correspond to (minimum per class - median per class - maximum per class).

We elaborate further on these and other more minor challenges with large-scale evaluation. Appendix F describes the submission protocol and other details of running the competition itself.

#### 4.1 Image classification

The scale of ILSVRC classification task (1000 categories and more than a million of images) makes it very expensive to label every instance of every object in every image. Therefore, on this dataset only one object category is labeled in each image. This creates ambiguity in evaluation. For example, an image might be labeled as a “strawberry” but contain both a strawberry and an apple. Then an algorithm would not know which one of the two objects to name. For the image classification task we allowed an algorithm to identify multiple (up to 5) objects in an image and not be penalized as long as one of the objects indeed corresponded to the ground truth label. Figure 7(top row) shows some examples.

Concretely, each image  $i$  has a single class label  $C_i$ . An algorithm is allowed to return 5 labels  $c_{i1}, \dots, c_{i5}$ , and is considered correct if  $c_{ij} = C_i$  for some  $j$ .

Let the error of a prediction  $d_{ij} = d(c_{ij}, C_i)$  be 1 if  $c_{ij} \neq C_i$  and 0 otherwise. The error of an algorithm is the fraction of test images on which the algorithm makes a mistake:

$$\text{error} = \frac{1}{N} \sum_{i=1}^N \min_j d_{ij} \quad (1)$$

We used two additional measures of error. First, we evaluated top-1 error. In this case algorithms were penalized if their highest-confidence output label  $c_{i1}$  did not match ground truth class  $C_i$ . Second, we evaluated hierarchical error. The intuition is that confusing two nearby classes (such as two different breeds of dogs) is not as harmful as confusing a dog for a container ship. For the hierarchical criteria, the cost of one misclassification,  $d(c_{ij}, C_i)$ , is defined as the height of the lowest

common ancestor of  $c_{ij}$  and  $C_i$  in the ImageNet hierarchy. The height of a node is the length of the longest path to a leaf node (leaf nodes have height zero).

However, in practice we found that all three measures of error (top-5, top-1, and hierarchical) produced the same ordering of results. Thus, since ILSVRC2012 we have been exclusively using the top-5 metric which is the simplest and most suitable to the dataset.

#### 4.2 Single-object localization

The evaluation for single-object localization is similar to object classification, again using a top-5 criteria to allow the algorithm to return unannotated object classes without penalty. However, now the algorithm is considered correct only if it both correctly identifies the target class  $C_i$  and accurately localizes one of its instances. Figure 7(middle row) shows some examples.

Concretely, an image is associated with object class  $C_i$ , with all instances of this object class annotated with bounding boxes  $B_{ik}$ . An algorithm returns  $\{(c_{ij}, b_{ij})\}_{j=1}^5$  of class labels  $c_{ij}$  and associated locations  $b_{ij}$ . The error of a prediction  $j$  is:

$$d_{ij} = \max(d(c_{ij}, C_i), \min_k d(b_{ij}, B_{ik})) \quad (2)$$

Here  $d(b_{ij}, B_{ik})$  is the error of localization, defined as 0 if the area of intersection of boxes  $b_{ij}$  and  $B_{ik}$  divided by the areas of their union is greater than 0.5, and 1 otherwise. (Everingham et al., 2010) The error of an algorithm is computed as in Eq. 1.

Evaluating localization is inherently difficult in some images. Consider a picture of a bunch of bananas or a carton of apples. It is easy to classify these images as containing bananas or apples, and even possible to localize a few instances of each fruit. However, in order for evaluation to be accurate *every* instance of banana or apple needs to be annotated, and that may be impossible. To handle the images where localizing individual object instances is inherently ambiguous we manually discarded 3.5% of images since ILSVRC2012. Some examples of discarded images are shown in Figure 8.

Object detection annotations (200 object classes)

| Year       | Train images<br>(per class)                             | Train bboxes annotated<br>(per class) | Val images<br>(per class)              | Val bboxes annotated<br>(per class) | Test<br>images |
|------------|---------------------------------------------------------|---------------------------------------|----------------------------------------|-------------------------------------|----------------|
| ILSVRC2013 | 395909<br>(417-561-66911 pos,<br>185-4130-10073 neg)    | 345854<br>(438-660-73799)             | 21121<br>(23-58-5791 pos,<br>rest neg) | 55501<br>(31-111-12824)             | 40152          |
| ILSVRC2014 | 456567<br>(461-823-67513 pos,<br>42945-64614-70626 neg) | 478807<br>(502-1008-74517)            | 21121<br>(23-58-5791 pos,<br>rest neg) | 55501<br>(31-111-12824)             | 40152          |

**Table 4** ILSVRC目标检测任务的规模。括号中的数字对应（每类最小数量 - 每类中位数数量）  
c类 - 每类最大数量）。

我们进一步详细阐述了这些以及其他在大规模评估中较为次要的挑战。附录F描述了提交协议以及举办竞赛本身的其他细节。

#### 4.1 图像分类

ILSVRC分类任务的规模（1000个类别及超过百万张图像）使得为每张图像中的每个物体实例进行标注成本极高。因此，在该数据集中，每张图像仅标注一个物体类别。这给评估带来了模糊性。例如，一张图像可能被标注为“草莓”，但其中同时包含草莓和苹果。此时算法将无法确定应识别两个物体中的哪一个。针对图像分类任务，我们允许算法识别图像中的多个物体（最多5个），只要其中一个物体与真实标签相符就不会受到惩罚。图7（顶部行）展示了一些示例。

具体来说，每张图像  $i$  都有一个单独的类别标签  $C_i$ 。算法可以返回 5 个标签  $c_{i1}, \dots, c_{i5}$ ，若存在某个  $j$  使得  $c_{ij} = C_i$  成立，则视为正确。

让预测的误差  $d_{ij} = d(c_{ij}, C_i)$  在  $c_{ij} \neq C_i$  时为 1，否则为 0。算法的误差是算法在测试图像上犯错的比率：

$$\text{error} = \frac{1}{N} \sum_{i=1}^N \min_j d_{ij} \quad (1)$$

我们使用了两种额外的误差度量方法。首先，我们评估了top-1误差。在这种情况下，如果算法最高置信度的输出标签  $c_{i1}$  与真实类别  $C_i$  不匹配，则算法会受到惩罚。其次，我们评估了层次误差。其直观思想是：混淆两个相近的类别（例如两种不同品种的狗）不如将狗误认为集装箱船那样有害。对于层次标准，一次错误分类的代价  $d(c_{ij}, C_i)$  被定义为最低共同祖先的高度。

$c_{ij}$  和  $C_i$  在ImageNet层次结构中的最近公共祖先。节点的高度是到叶节点的最长路径长度（叶节点的高度为零）。

然而，在实践中我们发现，所有三种误差度量（top-5、top-1和分层误差）得出的结果排序完全一致。因此，自ILSVRC2012起，我们便统一采用最简单且最适配该数据集的top-5评估指标。

#### 4.2 单目标定位

单目标定位的评估与物体分类类似，同样采用前5名的标准，允许算法返回未标注的物体类别而不受惩罚。然而，现在算法只有在同时正确识别目标类别  $C_i$  并准确定位其一个实例时，才被视为正确。图7（中间行）展示了一些示例。

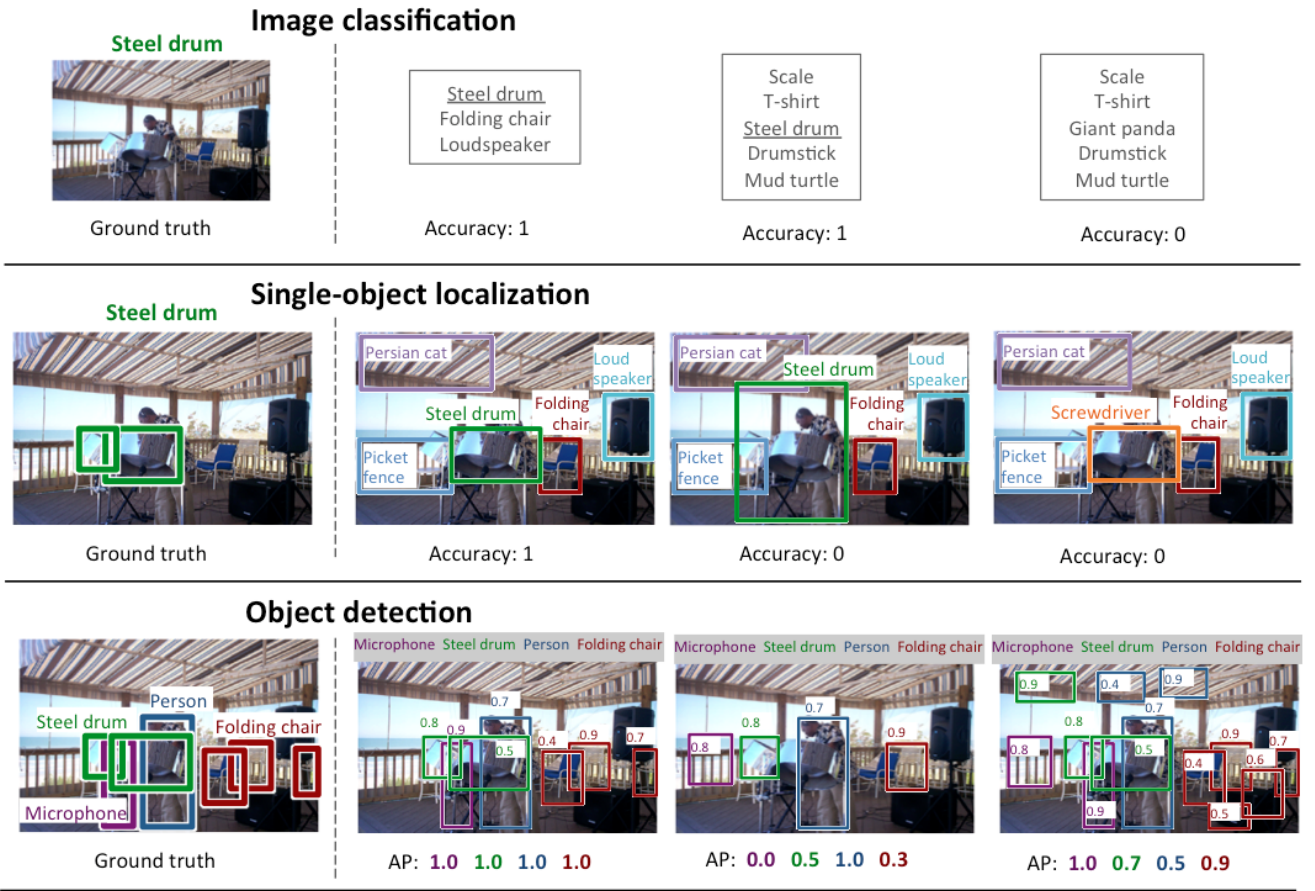
具体而言，一幅图像与物体类别  $C_i$  相关联，该物体类别的所有实例均用边界框  $B_{ik}$  进行标注。算法返回类别标签  $c_{ij}$  的  $\{(c_{ij}, b_{ij})\}_{j=1}^5$  及其对应位置  $b_{ij}$ 。预测  $j$  的误差为：

$$d_{ij} = \max(d(c_{ij}, C_i), \min_k d(b_{ij}, B_{ik})) \quad (2)$$

这里  $d(b_{ij}, B_{ik})$  是定位误差，定义为：若边界框  $b_{ij}$  与  $B_{ik}$  的交集面积除以其并集面积大于 0.5，则为 0，否则为 1。（Everingham 等人，2010）算法的误差按公式 1 计算。

在某些图像中评估定位本身就很困难。以一堆香蕉或一箱苹果的照片为例。将这些图像分类为包含香蕉或苹果很容易，甚至可能定位到每种水果的几个实例。然而，为了评估的准确性，每个香蕉或苹果的实例都需要被标注，而这可能无法实现。为了处理那些定位单个物体实例本身存在歧义的图像，自ILSVRC2012以来，我们手动剔除了3%至5%的图像。图8展示了一些被剔除图像的示例。





**Fig. 7** Tasks in ILSVRC. The first column shows the ground truth labeling on an example image, and the next three show three sample outputs with the corresponding evaluation score.



**Fig. 8** Images marked as “difficult” in the ILSVRC2012 single-object localization validation set. Please refer to Section 4.2 for details.

#### 4.3 Object detection

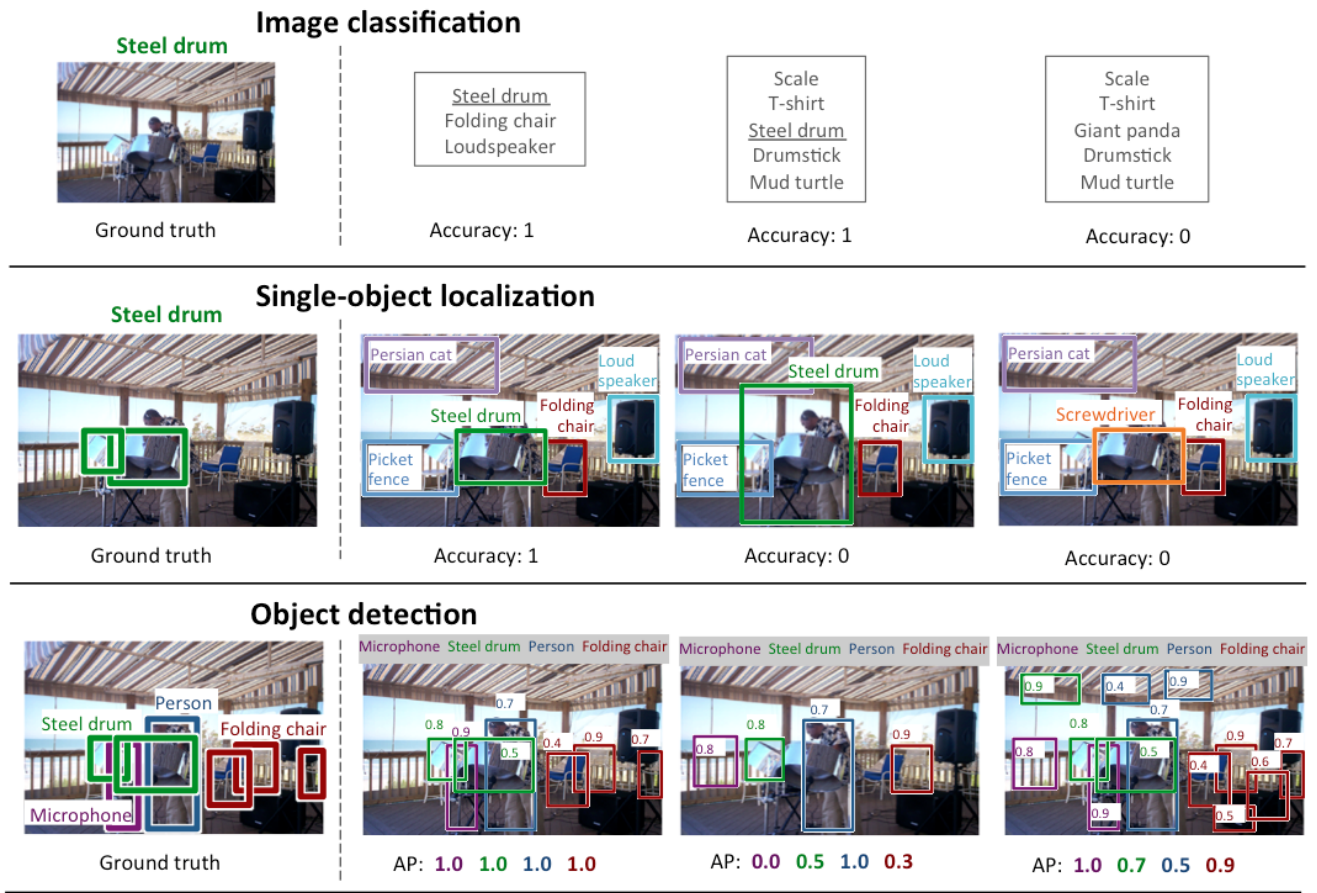
The criteria for object detection was adopted from PASCAL VOC (Everingham et al., 2010). It is designed to penalize the algorithm for missing object instances, for duplicate detections of one instance, and for false positive detections. Figure 7(bottom row) shows examples.

For each object class and each image  $I_i$ , an algorithm returns predicted detections  $(b_{ij}, s_{ij})$  of predicted locations  $b_{ij}$  with confidence scores  $s_{ij}$ . These detections are greedily matched to the ground truth boxes  $\{B_{ik}\}$  using Algorithm 2. For every detection  $j$  on image  $i$  the algorithm returns  $z_{ij} = 1$  if the detection is

matched to a ground truth box according to the threshold criteria, and 0 otherwise. For a given object class, let  $N$  be the total number of ground truth instances across all images. Given a threshold  $t$ , define *recall* as the fraction of the  $N$  objects detected by the algorithm, and *precision* as the fraction of correct detections out of the total detections returned by the algorithm. Concretely,

$$Recall(t) = \frac{\sum_{ij} 1[s_{ij} \geq t] z_{ij}}{N} \quad (3)$$

$$Precision(t) = \frac{\sum_{ij} 1[s_{ij} \geq t] z_{ij}}{\sum_{ij} 1[s_{ij} \geq t]} \quad (4)$$



**Fig. 7** ILSVRC中的任务。第一列展示了示例图像上的真实标注，接下来的三列则显示了三个样本输出及其对应的评估分数。



**Fig. 8** 在ILSVRC2012单目标定位验证集中标记为“困难”的图像。详情请参阅第4.2节。

### 4.3 目标检测

物体检测的标准采用了PASCAL VOC（Everingham等人，2010年）的方法。该标准旨在对算法因遗漏物体实例、对同一实例进行重复检测以及产生误报检测而进行惩罚。图7（底行）展示了相关示例。

对于每个物体类别和每张图像 $I_i$ ，算法会返回预测检测结果 $(b_{ij}, s_{ij})$ ，包含预测位置 $b_{ij}$ 和置信度分数 $s_{ij}$ 。这些检测结果通过算法2与真实标注框 $\{B_{ik}\}$ 进行贪心匹配。对于图像 $i$ 上的每个检测结果 $j$ ，若检测成立，算法将返回 $z_{ij} = 1$ 。

根据阈值标准与真实边界框匹配时，该值为1，否则为0。对于给定的物体类别，设 $N$ 为所有图像中真实实例的总数。给定阈值 $t$ ，定义 $recall$ 为算法检测出的 $N$ 个物体所占比例， $precision$ 为算法返回的总检测结果中正确检测所占的比例。具体而言，

$$Recall(t) = \frac{\sum_{ij} 1[s_{ij} \geq t] z_{ij}}{N} \quad (3)$$

$$Precision(t) = \frac{\sum_{ij} 1[s_{ij} \geq t] z_{ij}}{\sum_{ij} 1[s_{ij} \geq t]} \quad (4)$$

**Input:** Bounding box predictions with confidence scores  $\{(b_j, s_j)\}_{j=1}^M$  and ground truth boxes  $\mathcal{B}$  on image  $I$  for a given object class.  
**Output:** Binary results  $\{z_j\}_{j=1}^M$  of whether or not prediction  $j$  is a true positive detection  
Let  $\mathcal{U} = \mathcal{B}$  be the set of unmatched objects;  
Order  $\{(b_j, s_j)\}_{j=1}^M$  in descending order of  $s_j$ ;  
**for**  $j=1 \dots M$  **do**  
    Let  $\mathcal{C} = \{B_k \in \mathcal{U} : \text{IOU}(B_k, b_j) \geq \text{thr}(B_k)\}$ ;  
    **if**  $\mathcal{C} \neq \emptyset$  **then**  
        Let  $k^* = \arg \max_{\{k : B_k \in \mathcal{C}\}} \text{IOU}(B_k, b_j)$ ;  
        Set  $\mathcal{U} = \mathcal{U} \setminus B_{k^*}$ ;  
        Set  $z_j = 1$  since true positive detection;  
    **else**  
        Set  $z_j = 0$  since false positive detection;  
    **end**  
**end**

**Algorithm 2:** The algorithm for greedily matching object detection outputs to ground truth labels. The standard  $\text{thr}(B_k) = 0.5$  (Everingham et al., 2010). ILSVRC computes  $\text{thr}(B_k)$  using Eq. 5 to better handle low-resolution objects.

The final metric for evaluating an algorithm on a given object class is *average precision* over the different levels of recall achieved by varying the threshold  $t$ . The winner of each object class is then the team with the highest average precision, and then winner of the challenge is the team that wins on the most object classes.<sup>8</sup>

*Difference with PASCAL VOC.* Evaluating localization of object instances which occupy very few pixels in the image is challenging. The PASCAL VOC approach was to label such instances as “difficult” and ignore them during evaluation. However, since ILSVRC contains a more diverse set of object classes including, for example, “nail” and “ping pong ball” which have many very small instances, it is important to include even very small object instances in evaluation.

In Algorithm 2, a predicted bounding box  $b$  is considered to have properly localized by a ground truth bounding box  $B$  if  $\text{IOU}(b, B) \geq \text{thr}(B)$ . The PASCAL VOC metric uses the threshold  $\text{thr}(B) = 0.5$ . However, for small objects even deviations of a few pixels would be unacceptable according to this threshold. For example, consider an object  $B$  of size  $10 \times 10$  pixels, with a detection window of  $20 \times 20$  pixels which fully contains that object. This would be an error of approximately 5 pixels on each dimension, which is average human annotation error. However, the IOU in this case would be  $100/400 = 0.25$ , far below the threshold of 0.5. Thus

<sup>8</sup> In this paper we focus on the mean average precision across all categories as the measure of a team’s performance. This is done for simplicity and is justified since the ordering of teams by mean average precision was always the same as the ordering by object categories won.

for smaller objects we loosen the threshold in ILSVRC to allow for the annotation to extend up to 5 pixels on average in each direction around the object. Concretely, if the ground truth box  $B$  is of dimensions  $w \times h$  then

$$\text{thr}(B) = \min \left( 0.5, \frac{wh}{(w+10)(h+10)} \right) \quad (5)$$

In practice, this changes the threshold only on objects which are smaller than approximately  $25 \times 25$  pixels, and affects 5.5% of objects in the detection validation set.

*Practical consideration.* One additional practical consideration for ILSVRC detection evaluation is subtle and comes directly as a result of the scale of ILSVRC. In PASCAL, algorithms would often return many detections per class on the test set, including ones with low confidence scores. This allowed the algorithms to reach the level of high recall at least in the realm of very low precision. On ILSVRC detection test set if an algorithm returns 10 bounding boxes per object per image this would result in  $10 \times 200 \times 40K = 80M$  detections. Each detection contains an image index, a class index, 4 bounding box coordinates, and the confidence score, so it takes on the order of 28 bytes. The full set of detections would then require 2.24Gb to store and submit to the evaluation server, which is impractical. This means that algorithms are implicitly required to limit their predictions to only the most confident locations.

## 5 Methods

The ILSVRC dataset and the competition has allowed significant algorithmic advances in large-scale image recognition and retrieval.

### 5.1 Challenge entries

This section is organized chronologically, highlighting the particularly innovative and successful methods which participated in the ILSVRC each year. Tables 5, 6 and 7 list all the participating teams. We see a turning point in 2012 with the development of large-scale convolutional neural networks.

*ILSVRC2010.* The first year the challenge consisted of just the classification task. The winning entry from NEC team (Lin et al., 2011) used SIFT (Lowe, 2004) and LBP (Ahonen et al., 2006) features with two non-linear coding representations (Zhou et al., 2010; Wang

**Input** 对于给定物体类别，图像  $I$  上带有置信度分数  $\{(b_j, s_j)\}_{j=1}^M$  的边界框预测与真实框  $B$ 。 **Output**: 预测  $j$  是否为真正检测的二元结果  $\{z_j\}_{j=1}^M$ 。令  $\mathcal{U} = \mathcal{B}$  为未匹配物体的集合；将  $\{(b_j, s_j)\}_{j=1}^M$  按  $s_j$  降序排列；  
**for**  $j=1 \dots M$  **do** 令  $\mathcal{C} = \{B_k \in \mathcal{U} : \text{IOU}(B_k, b_j) \geq \text{阈值}(B_k)\}$ ; **if**  $\mathcal{C} \neq \emptyset$  **then** 令  $k^* = \arg \max_{\{k : B_k \in \mathcal{C}\}} \text{IOU}(B_k)$ ; **else** 设置  $\mathcal{U} = \mathcal{U} \setminus B_{k^*}$ ; 因是真正检测，设置  $z_j = 1$ ;  
 ; 因是假正检测，设置  $z_j = 0$ ; **end**

**Algorithm 2:** 贪婪匹配物体检测输出与真实标签的算法。标准阈值  $(B_k) = 0.5$  (Everingham等人, 2010年)。ILSVRC使用公式5计算阈值  $(B_k)$ ，以更好地处理低分辨率物体。

评估算法在给定对象类别上的最终指标是通过改变阈值  $t$  在不同召回率水平上获得的 *average precision*。每个对象类别的获胜者是平均精度最高的团队，而挑战赛的获胜者则是在最多对象类别上获胜的团队。<sup>8</sup>

*Difference with PASCAL VOC.* 评估图像中占据极少像素的物体实例的定位是一项挑战。PASCAL VOC 的方法是将此类实例标记为“困难”并在评估时忽略它们。然而，由于 ILSVRC 包含了更多样化的物体类别，例如“指甲”和“乒乓球”等存在许多极小实例的类别，因此在评估中纳入即使是非常小的物体实例也至关重要。

在算法2中，若  $\text{IOU}(b, B) \geq \text{thr}(B)$ ，则预测边界框  $b$  被认为被真实边界框  $B$  正确定位。PASCAL VOC 指标采用阈值  $\text{thr}(B) = 0.5$ 。然而对于小物体，即使几个像素的偏差根据此阈值也显得不可接受。例如，考虑一个尺寸为  $10 \times 10$  像素的物体  $B$ ，若检测窗口为  $20 \times 20$  像素且完全包含该物体。此时每个维度的误差约为5像素，这接近人类标注的平均误差水平。但此情况下的交并比 (IOU) 仅为  $100/400 = 0.25$ ，远低于0.5的阈值。因此

对于较小的物体，我们在ILSVRC中放宽了阈值，允许标注在物体周围每个方向上平均延伸最多5个像素。具体来说，如果真实边界框  $B$  的尺寸为  $w \times h$ ，那么

$$\text{thr}(B) = \min \left( 0.5, \frac{wh}{(w+10)(h+10)} \right) \quad (5)$$

实际上，这仅改变了尺寸小于约  $25 \times$  像素的物体的阈值，并影响了检测验证集中5.5%的物体。

*Practical consideration.* ILSVRC检测评估中一个额外的实际考虑是微妙且直接源于ILSVRC的规模。在PASCAL中，算法通常会在测试集上为每个类别返回许多检测结果，包括那些置信度较低的检测。这使得算法至少在极低精确度的范围内能够达到高召回率。而在ILSVRC检测测试集上，如果算法为每张图像的每个物体返回10个边界框，这将导致  $10 \times 200 \times 40 \times 80 \text{M}$  次检测。每次检测包含图像索引、类别索引、4个边界框坐标和置信度分数，因此每次检测大约需要28字节。完整的检测结果集将需要  $2 \times 24 \text{Gb}$  来存储并提交给评估服务器，这是不切实际的。这意味着算法被隐含要求将其预测限制在仅最有信心的位置。

## 5 Methods

ILSVRC数据集及其竞赛在大规模图像识别与检索领域推动了显著的算法进步。

### 5.1 挑战条目

本节按时间顺序组织，重点介绍每年参与ILSVRC竞赛中尤为创新且成效显著的方法。表5、表6和表7列出了所有参赛团队。我们看到2012年随着大规模卷积神经网络的发展出现了一个转折点。

*ILSVRC2010.* 挑战赛的第一年只包含分类任务。NEC团队的获胜方案 (Lin等人, 2011) 采用了SIFT (Lowe, 2004) 和LBP (Ahonen等人, 2006) 特征，并结合了两种非线性编码表示方法 (Zhou等人, 2010; Wang

<sup>8</sup> In this paper we focus on the mean average precision across all categories as the measure of a team's performance. This is done for simplicity and is justified since the ordering of teams by mean average precision was always the same as the ordering by object categories won.



et al., 2010) and a stochastic SVM. The honorable mention XRCE team (Perronnin et al., 2010) used an improved Fisher vector representation (Perronnin and Dance, 2007) along with PCA dimensionality reduction and data compression followed by a linear SVM. Fisher vector-based methods have evolved over five years of the challenge and continued performing strongly in every ILSVRC from 2010 to 2014.

*ILSVRC2011.* The winning classification entry in 2011 was the 2010 runner-up team XRCE, applying high-dimensional image signatures (Perronnin et al., 2010) with compression using product quantization (Sanchez and Perronnin, 2011) and one-vs-all linear SVMs. The single-object localization competition was held for the first time, with two brave entries. The winner was the UvA team using a selective search approach to generate class-independent object hypothesis regions (van de Sande et al., 2011b), followed by dense sampling and vector quantization of several color SIFT features (van de Sande et al., 2010), pooling with spatial pyramid matching (Lazebnik et al., 2006), and classifying with a histogram intersection kernel SVM (Maji and Malik, 2009) trained on a GPU (van de Sande et al., 2011a).

*ILSVRC2012.* This was a turning point for large-scale object recognition, when large-scale deep neural networks entered the scene. The undisputed winner of both the classification and localization tasks in 2012 was the SuperVision team. They trained a large, deep convolutional neural network on RGB values, with 60 million parameters using an efficient GPU implementation and a novel hidden-unit dropout trick (Krizhevsky et al., 2012; Hinton et al., 2012). The second place in image classification went to the ISI team, which used Fisher vectors (Sanchez and Perronnin, 2011) and a streamlined version of Graphical Gaussian Vectors (Harada and Kuniyoshi, 2012), along with linear classifiers using Passive-Aggressive (PA) algorithm (Crammer et al., 2006). The second place in single-object localization went to the VGG, with an image classification system including dense SIFT features and color statistics (Lowe, 2004), a Fisher vector representation (Sanchez and Perronnin, 2011), and a linear SVM classifier, plus additional insights from (Arandjelovic and Zisserman, 2012; Sanchez et al., 2012). Both ISI and VGG used (Felzenszwalb et al., 2010) for object localization; SuperVision used a regression model trained to predict bounding box locations. Despite the weaker detection model, SuperVision handily won the object localization task. A detailed analysis and comparison of the SuperVision and VGG submissions on the single-object localization task can be found in (Russakovsky et al., 2013).

The influence of the success of the SuperVision model can be clearly seen in ILSVRC2013 and ILSVRC2014.

*ILSVRC2013.* There were 24 teams participating in the ILSVRC2013 competition, compared to 21 in the previous three years *combined*. Following the success of the deep learning-based method in 2012, the vast majority of entries in 2013 used deep convolutional neural networks in their submission. The winner of the classification task was Clarifai, with several large deep convolutional networks averaged together. The network architectures were chosen using the visualization technique of (Zeiler and Fergus, 2013), and they were trained on the GPU following (Zeiler et al., 2011) using the dropout technique (Krizhevsky et al., 2012).

The winning single-object localization OverFeat submission was based on an integrated framework for using convolutional networks for classification, localization and detection with a multiscale sliding window approach (Sermanet et al., 2013). They were the only team tackling all three tasks.

The winner of object detection task was UvA team, which utilized a new way of efficient encoding (van de Sande et al., 2014) densely sampled color descriptors (van de Sande et al., 2010) pooled using a multi-level spatial pyramid in a selective search framework (Uijlings et al., 2013). The detection results were rescored using a full-image convolutional network classifier.

*ILSVRC2014.* 2014 attracted the most submissions, with 36 teams submitting 123 entries compared to just 24 teams in 2013 – a 1.5x increase in participation.<sup>9</sup> As in 2013 almost all teams used convolutional neural networks as the basis for their submission. Significant progress has been made in just one year: image classification error was almost halved since ILSVRC2013 and object detection mean average precision almost doubled compared to ILSVRC2013. Please refer to Section 6.1 for details.

In 2014 teams were allowed to use outside data for training their models in the competition, so there were six tracks: provided and outside data tracks in each of image classification, single-object localization, and object detection tasks.

The winning image classification with provided data team was GoogLeNet, which explored an improved convolutional neural network architecture combining the multi-scale idea with intuitions gained from the Hebbian principle. Additional dimension reduction layers allowed them to increase both the depth and the width

<sup>9</sup> Table 7 omits 4 teams which submitted results but chose not to officially participate in the challenge.



等人, 2010) 以及一个随机支持向量机。荣誉提名XRCE团队 (Perronnin等人, 2010) 采用了改进的Fisher向量表示 (Perronnin和Dance, 2007), 并结合了PCA降维和数据压缩, 随后使用线性支持向量机。基于Fisher向量的方法在该挑战赛的五年间不断发展, 并在2010年至2014年的每一届ILSVRC中持续表现强劲。

*ILSVRC2011.* 2011年的获胜分类参赛作品是2010年的亚军团队XRCE, 他们应用了高维图像签名 (Perronnin等人, 2010年), 并通过产品量化进行压缩 (Sanchez和Perronnin, 2011年), 以及一对多线性支持向量机。单目标定位竞赛首次举办, 有两支勇敢的团队参赛。获胜者是UvA团队, 他们采用选择性搜索方法生成类别无关的目标假设区域 (van de Sande等人, 2011b), 随后对多种颜色SIFT特征进行密集采样和向量量化 (van de Sande等人, 2010年), 通过空间金字塔匹配进行池化 (Lazebnik等人, 2006年), 并使用在GPU上训练的直方图交集核支持向量机进行分类 (Maji和Malik, 2009年; van de Sande等人, 2011a)。

*ILSVRC2012.* 这是大规模物体识别的一个转折点, 当时大规模深度神经网络开始崭露头角。2012年分类与定位任务无可争议的赢家是SuperVision团队。他们基于RGB值训练了一个大型深度卷积神经网络, 该网络包含6000万个参数, 采用了高效的GPU实现和一种新颖的隐藏单元丢弃技巧 (Krizhevsky等人, 2012; Hinton等人, 2012)。图像分类的第二名由ISI团队获得, 他们使用了费舍尔向量 (Sanchez与Perronnin, 2011) 和精简版的高斯图向量 (Harada与Kuniyoshi, 2012), 并搭配采用被动-主动 (PA) 算法的线性分类器 (Crammer等人, 2006)。单物体定位的第二名则归属VGG团队, 其图像分类系统包含密集SIFT特征与颜色统计 (Lowe, 2004)、费舍尔向量表示 (Sanchez与Perronnin, 2011) 以及线性SVM分类器, 并融合了 (Arandjelovic与Zisserman, 2012; Sanchez等人, 2012) 的额外洞见。ISI和VGG均采用 (Felzenszwalb等人, 2010) 的方法进行物体定位; 而SuperVision使用了训练来预测边界框位置的回归模型。尽管检测模型相对较弱, SuperVision仍以显著优势赢得了物体定位任务。关于SuperVision与VGG在单物体定位任务上提交方案的详细分析与比较, 可参阅 (Russakovsky等人, 2013)。

SuperVision模型成功的影响在ILSVRC2013和ILSVRC2014中清晰可见。

*ILSVRC2013.* ILSVRC2013竞赛共有24支队伍参与, 而前三年仅有21支 *combined*。继2012年基于深度学习的方法取得成功后, 2013年绝大多数参赛作品都在提交中使用了深度卷积神经网络。分类任务的获胜者是Clarifai, 其方案采用了多个大型深度卷积神经网络进行集成平均。网络架构的选择借鉴了 (Zeiler和Fergus, 2013) 的可视化技术, 并遵循 (Zeiler等人, 2011) 在GPU上使用dropout技术 (Krizhevsky等人, 2012) 进行训练。

获胜的单目标定位OverFeat方案基于一个集成框架, 该框架采用多尺度滑动窗口方法 (Sermanet等人, 2013年), 利用卷积网络进行分类、定位和检测。他们是唯一一个同时应对所有三项任务的团队。

目标检测任务的获胜者是UvA团队, 他们采用了一种新颖的高效编码方法 (van de Sande等人, 2014年), 在选择性搜索框架 (Uijlings等人, 2013年) 中, 通过多级空间金字塔对密集采样的颜色描述符 (van de Sande等人, 2010年) 进行池化处理。检测结果通过全图像卷积网络分类器进行了重新评分。

*ILSVRC2014.* 2014年吸引了最多的参赛作品, 共有36支团队提交了123项参赛作品, 而2013年仅有24支团队——参与度增长了1.5倍。<sup>9</sup> 与2013年类似, 几乎所有团队都使用卷积神经网络作为其参赛作品的基础。仅一年时间就取得了显著进展: 图像分类错误率较ILSVRC2013几乎减半, 而目标检测的平均精度均值相比ILSVRC2013几乎翻倍。详情请参阅第6.1节。

2014年, 竞赛允许团队使用外部数据训练模型, 因此设立了六个赛道: 图像分类、单目标定位和目标检测任务各设提供数据与外部数据两个赛道。

使用提供数据获胜的图像分类团队是GoogLeNet, 该团队探索了一种改进的卷积神经网络架构, 将多尺度理念与从赫布原理获得的直觉相结合。额外的降维层使他们能够同时增加网络的深度和宽度。

<sup>9</sup> Table 7 omits 4 teams which submitted results but chose not to officially participate in the challenge.

## ILSVRC 2010

| Codename     | CLS         | Institutions                                                                                                   | Contributors and references                                                                                                                                                                                                                                                                                                                    |
|--------------|-------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hminmax      | 54.4        | Massachusetts Institute of Technology                                                                          | Jim Mutch, Sharat Chikkerur, Hristo Paskov, Ruslan Salakhutdinov, Stan Bileschi, Hueihan Jhuang                                                                                                                                                                                                                                                |
| IBM          | 70.1        | IBM research <sup>†</sup> , Georgia Tech <sup>‡</sup>                                                          | Lexing Xie <sup>†</sup> , Hua Ouyang <sup>‡</sup> , Apostol Natsev <sup>†</sup>                                                                                                                                                                                                                                                                |
| ISIL         | 44.6        | Intelligent Systems and Informatics Lab., The University of Tokyo                                              | Tatsuya Harada, Hideki Nakayama, Yoshitaka Ushiku, Yuya Yamashita, Jun Imura, Yasuo Kuniyoshi                                                                                                                                                                                                                                                  |
| ITNLP        | 78.7        | Harbin Institute of Technology                                                                                 | Deyuan Zhang, Wenfeng Xuan, Xiaolong Wang, Bingquan Liu, Chengjie Sun                                                                                                                                                                                                                                                                          |
| LIG          | 60.7        | Laboratoire d'Informatique de Grenoble                                                                         | Georges Quénot                                                                                                                                                                                                                                                                                                                                 |
| NEC          | <b>28.2</b> | NEC Labs America <sup>†</sup> , University of Illinois at Urbana-Champaign <sup>‡</sup> , Rutgers <sup>‡</sup> | Yuanqing Lin <sup>†</sup> , Fengjun Lv <sup>†</sup> , Shenghuo Zhu <sup>†</sup> , Ming Yang <sup>†</sup> , Timothee Cour <sup>†</sup> , Kai Yu <sup>†</sup> , LiangLiang Cao <sup>‡</sup> , Zhen Li <sup>‡</sup> , Min-Hsuan Tsai <sup>‡</sup> , Xi Zhou <sup>‡</sup> , Thomas Huang <sup>‡</sup> , Tong Zhang <sup>‡</sup> (Lin et al., 2011) |
| NII          | 74.2        | National Institute of Informatics, Tokyo, Japan <sup>†</sup> , Hefei Normal Univ. Hefei, China <sup>‡</sup>    | Cai-Zhi Zhu <sup>†</sup> , Xiao Zhou <sup>‡</sup> , Shinichi Satoh <sup>†</sup>                                                                                                                                                                                                                                                                |
| NTU          | 58.3        | CeMNet, SCE, NTU, Singapore                                                                                    | Zhengxiang Wang, Liang-Tien Chia                                                                                                                                                                                                                                                                                                               |
| Regularities | 75.1        | SRI International                                                                                              | Omid Madani, Brian Burns                                                                                                                                                                                                                                                                                                                       |
| UCI          | 46.6        | University of California Irvine                                                                                | Hamed Pirsiavash, Deva Ramanan, Charles Fowlkes                                                                                                                                                                                                                                                                                                |
| XRCE         | 33.6        | Xerox Research Centre Europe                                                                                   | Jorge Sanchez, Florent Perronnin, Thomas Mensink (Perronnin et al., 2010)                                                                                                                                                                                                                                                                      |

## ILSVRC 2011

| Codename | CLS         | LOC         | Institutions                                                             | Contributors and references                                                                                                                                                                                                |
|----------|-------------|-------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISI      | 36.0        | -           | Intelligent Systems and Informatics lab, University of Tokyo             | Tatsuya Harada, Asako Kanezaki, Yoshitaka Ushiku, Yuya Yamashita, Sho Inaba, Hiroshi Muraoka, Yasuo Kuniyoshi                                                                                                              |
| NII      | 50.5        | -           | National Institute of Informatics, Japan                                 | Duy-Dinh Le, Shinichi Satoh                                                                                                                                                                                                |
| UvA      | 31.0        | <b>42.5</b> | University of Amsterdam <sup>†</sup> , University of Trento <sup>‡</sup> | Koen E. A. van de Sande <sup>†</sup> , Jasper R. R. Uijlings <sup>‡</sup> , Arnold W. M. Smeulders <sup>†</sup> , Theo Gevers <sup>†</sup> , Nicu Sebe <sup>‡</sup> , Cees Snoek <sup>†</sup> (van de Sande et al., 2011b) |
| XRCE     | <b>25.8</b> | 56.5        | Xerox Research Centre Europe <sup>†</sup> , CHII <sup>‡</sup>            | Florent Perronnin <sup>†</sup> , Jorge Sanchez <sup>†‡</sup> (Sanchez and Perronnin, 2011)                                                                                                                                 |

## ILSVRC 2012

| Codename    | CLS         | LOC         | Institutions                                                                      | Contributors and references                                                                                                                                                                                                                       |
|-------------|-------------|-------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISI         | 26.2        | 53.6        | University of Tokyo <sup>†</sup> , JST PRESTO <sup>‡</sup>                        | Naoyuki Gunji <sup>†</sup> , Takayuki Higuchi <sup>†</sup> , Koki Yasumoto <sup>†</sup> , Hiroshi Muraoka <sup>†</sup> , Yoshitaka Ushiku <sup>†</sup> , Tatsuya Harada <sup>†‡</sup> , Yasuo Kuniyoshi <sup>†</sup> (Harada and Kuniyoshi, 2012) |
| LEAR        | 34.5        | -           | LEAR INRIA Grenoble <sup>†</sup> , TVPA Xerox Research Centre Europe <sup>‡</sup> | Thomas Mensink <sup>†‡</sup> , Jakob Verbeek <sup>†</sup> , Florent Perronnin <sup>†</sup> , Gabriela Csurka <sup>‡</sup> (Mensink et al., 2012)                                                                                                  |
| VGG         | 27.0        | 50.0        | University of Oxford                                                              | Karen Simonyan, Yusuf Aytar, Andrea Vedaldi, Andrew Zisserman (Arandjelovic and Zisserman, 2012; Sanchez et al., 2012)                                                                                                                            |
| SuperVision | <b>16.4</b> | <b>34.2</b> | University of Toronto                                                             | Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton (Krizhevsky et al., 2012)                                                                                                                                                                        |
| UvA         | 29.6        | -           | University of Amsterdam                                                           | Koen E. A. van de Sande, Amir Habibian, Cees G. M. Snoek (Sanchez and Perronnin, 2011; Scheirer et al., 2012)                                                                                                                                     |
| XRCE        | 27.1        | -           | Xerox Research Centre Europe <sup>†</sup> , LEAR INRIA <sup>‡</sup>               | Florent Perronnin <sup>†</sup> , Zeynep Akata <sup>†‡</sup> , Zaid Harchaoui <sup>‡</sup> , Cordelia Schmid <sup>‡</sup> (Perronnin et al., 2012)                                                                                                 |

**Table 5** Teams participating in ILSVRC2010-2012, ordered alphabetically. Each method is identified with a codename used in the text. We report flat top-5 classification and single-object localization error, in percents (lower is better). For teams which submitted multiple entries we report the best score. In 2012, SuperVision also submitted entries trained with the extra data from the ImageNet Fall 2011 release, and obtained 15.3% classification error and 33.5% localization error. Key references are provided where available. More details about the winning entries can be found in Section 5.1.

| Codename     | CLS         | LOC | Institutions                                                                                                   | Contributors and references                                                                                                                                                                                             |
|--------------|-------------|-----|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hminmax      | 54.4        |     | Massachusetts Institute of Technology                                                                          | Jim Mutch, Sharat Chikkerur, Hristo Paskov, Ruslan Sala                                                                                                                                                                 |
| IBM          | 70.1        |     | IBM research <sup>†</sup> , Georgia Tech <sup>‡</sup>                                                          | Lexing Xie <sup>†</sup> , Hua Ouyang <sup>‡</sup> , Apostol Natsev <sup>†</sup>                                                                                                                                         |
| ISIL         | 44.6        |     | Intelligent Systems and Informatics Lab., The University of Tokyo                                              | Tatsuya Harada, Hideki Nakayama, Yoshitaka Ushiku, Yuya                                                                                                                                                                 |
| ITNLP        | 78.7        |     | Harbin Institute of Technology                                                                                 | Deyuan Zhang, Wenteng Xuan, Xiaolong Wang, Bingquan                                                                                                                                                                     |
| LIG          | 60.7        |     | Laboratoire d'Informatique de Grenoble                                                                         | Georges Quénot                                                                                                                                                                                                          |
| NEC          | <b>28.2</b> |     | NEC Labs America <sup>†</sup> , University of Illinois at Urbana-Champaign <sup>‡</sup> , Rutgers <sup>±</sup> | Yuangqing Lin <sup>†</sup> , Pengjun Lv <sup>†</sup> , Shenghuo Zhu <sup>†</sup> , Ming Yan, Zhen Li <sup>‡</sup> , Min-Hsuan Tsai <sup>‡</sup> , Xi Zhou <sup>‡</sup> , Thomas Huang <sup>‡</sup> , (Lin et al., 2011) |
| NII          | 74.2        |     | National Institute of Informatics, Tokyo, Japan <sup>†</sup> , Hefei Normal Univ. Hefei, China <sup>‡</sup>    | Cai-Zhi Zhu <sup>†</sup> , Xiao Zhou <sup>‡</sup> , Shinichi Satoh <sup>†</sup>                                                                                                                                         |
| NTU          | 58.3        |     | CeMNet, SCE, NTU, Singapore                                                                                    | Zhengxiang Wang, Liang-Tien Chia                                                                                                                                                                                        |
| Regularities | 75.1        |     | SRI International                                                                                              | Omid Madani, Brian Burns                                                                                                                                                                                                |
| UCI          | 46.6        |     | University of California Irvine                                                                                | Hamed Pirsiavash, Deva Ramanan, Charles Fowlkes                                                                                                                                                                         |
| XRCE         | 33.6        |     | Xerox Research Centre Europe                                                                                   | Jorge Sanchez, Florent Perronnin, Thomas Mensink (Perronnin et al., 2010)                                                                                                                                               |

#### ILSVRC 2011

| Codename | CLS         | LOC         | Institutions                                                             | Contributors and references                                                                                                                |
|----------|-------------|-------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| ISI      | 36.0        | -           | Intelligent Systems and Informatics lab, University of Tokyo             | Tatsuya Harada, Asako Kanazaki, Yoshitaka Ushiku, Yuya Kuniyoshi                                                                           |
| NII      | 50.5        | -           | National Institute of Informatics, Japan                                 | Duy-Dinh Le, Shinichi Satoh                                                                                                                |
| UvA      | 31.0        | <b>42.5</b> | University of Amsterdam <sup>†</sup> , University of Trento <sup>‡</sup> | Koen E. A. van de Sande <sup>†</sup> , Jasper R. R. Uijlings <sup>‡</sup> , Arnold Cees Snoek <sup>†</sup><br>(van de Sande et al., 2011b) |
| XRCE     | <b>25.8</b> | 56.5        | Xerox Research Centre Europe <sup>†</sup> , CHU <sup>‡</sup>             | Florent Perronnin <sup>†</sup> , Jorge Sanchez <sup>†</sup> <sup>‡</sup><br>(Sanchez and Perronnin, 2011)                                  |

#### ILSVRC 2012

| Codename    | CLS         | LOC         | Institutions                                                                      | Contributors and references                                                                                                                                                  |
|-------------|-------------|-------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISI         | 26.2        | 53.6        | University of Tokyo <sup>†</sup> , JST PRESTO <sup>‡</sup>                        | Naoyuki Gunji <sup>†</sup> , Takeyuki Higuchi <sup>†</sup> , Koki Yasunoto <sup>†</sup> , Harada <sup>‡</sup> , Yasuo Kuniyoshi <sup>†</sup><br>(Harada and Kuniyoshi, 2012) |
| LEAR        | 34.5        | -           | LEAR INRIA Grenoble <sup>†</sup> , TVPA Xerox Research Centre Europe <sup>‡</sup> | Thomas Mensink <sup>†</sup> , Jakob Verbeek <sup>†</sup> , Florent Perronnin <sup>‡</sup> , (Mensink et al., 2012)                                                           |
| VGG         | 27.0        | 50.0        | University of Oxford                                                              | Karen Simonyan, Yusanf Ayar, Andrea Vedaldi, Andrew Zisserman, 2012; Sanchez et al., 2012)                                                                                   |
| SuperVision | <b>16.4</b> | <b>34.2</b> | University of Toronto                                                             | Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton                                                                                                                             |
| UvA         | 29.6        | -           | University of Amsterdam                                                           | Koen E. A. van de Sande, Amir Habibi, Cees G. M. Snoek (Sanchez and Perronnin, 2011; Scheijer et al., 2012)                                                                  |
| XRCE        | 27.1        | -           | Xerox Research Centre Europe <sup>†</sup> , LEAR INRIA <sup>‡</sup>               | Florent Perronnin <sup>†</sup> , Zeynep Akata <sup>†</sup> , Zaid Harchaout <sup>‡</sup> , Co (Perronnin et al., 2012)                                                       |

**Table 5** 参与ILSVRC2010-2012的团队按字母顺序排列。每种方法均以文本中使用的代号标识。我们以百分比形式报告平面top-5分类和于提交多个参赛结果的团队，我们报告其最佳成绩。2012年，SuperVision团队还提交了使用ImageNet 2011年秋季发布版额外数据训练的参5%的定位错误率。在可能的情况下提供了关键参考文献。获胜参赛方案的更多细节详见第5.1节。

| Codename  | CLS         | LOC         | DET         | Institutions                                                                                                                                                                                 | Contributors and references                                                                                                                                                                                                                          |
|-----------|-------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adobe     | 15.2        | -           | -           | Adobe <sup>†</sup> , University of Illinois at Urbana-Champaign <sup>‡</sup>                                                                                                                 | Hailin Jin <sup>†</sup> , Zhe Lin <sup>†</sup> , Jianchao Yang <sup>†</sup> , Tom Paine <sup>‡</sup><br>(Krizhevsky et al., 2012)                                                                                                                    |
| AHoward   | 13.6        | -           | -           | Andrew Howard Consulting                                                                                                                                                                     | Andrew Howard                                                                                                                                                                                                                                        |
| BUPT      | 25.2        | -           | -           | Beijing University of Posts and Telecommunications <sup>†</sup> , Orange Labs International Center Beijing <sup>‡</sup>                                                                      | Chong Huang <sup>†</sup> , Yunlong Bian <sup>†</sup> , Hongliang Bai <sup>‡</sup> , Bo Liu <sup>†</sup> , Yanchao Feng <sup>†</sup> , Yuan Dong <sup>†</sup>                                                                                         |
| Clarifai  | <b>11.7</b> | -           | -           | Clarifai                                                                                                                                                                                     | Matthew Zeiler<br>(Zeiler and Fergus, 2013; Zeiler et al., 2011)                                                                                                                                                                                     |
| CogVision | 16.1        | -           | -           | Microsoft Research <sup>†</sup> , Harbin Institute of Technology <sup>‡</sup>                                                                                                                | Kuiyuan Yang <sup>†</sup> , Yalong Bai <sup>†</sup> , Yong Rui <sup>‡</sup>                                                                                                                                                                          |
| decaf     | 19.2        | -           | -           | University of California Berkeley                                                                                                                                                            | Yangqing Jia, Jeff Donahue, Trevor Darrell<br>(Donahue et al., 2013)                                                                                                                                                                                 |
| Deep Punx | 20.9        | -           | -           | Saint Petersburg State University                                                                                                                                                            | Evgeny Smirnov, Denis Timoshenko, Alexey Korolev<br>(Krizhevsky et al., 2012; Wan et al., 2013; Tang, 2013)                                                                                                                                          |
| Delta     | -           | -           | <b>6.1</b>  | National Tsing Hua University                                                                                                                                                                | Che-Rung Lee, Hwann-Tzong Chen, Hao-Ping Kang, Tzu-Wei Huang, Ci-Hong Deng, Hao-Che Kao                                                                                                                                                              |
| IBM       | 20.7        | -           | -           | University of Illinois at Urbana-Champaign <sup>†</sup> , IBM Watson Research Center <sup>‡</sup> , IBM Haifa Research Center <sup>‡</sup>                                                   | Zhicheng Yan <sup>†</sup> , Liangliang Cao <sup>‡</sup> , John R Smith <sup>‡</sup> , Noel Codella <sup>‡</sup> , Michele Merler <sup>‡</sup> , Sharath Pankanti <sup>‡</sup> , Sharon Alpert <sup>‡</sup> , Yochay Tzur <sup>‡</sup> ,              |
| MIL       | 24.4        | -           | -           | University of Tokyo                                                                                                                                                                          | Masatoshi Hidaka, Chie Kamada, Yusuke Mukuta, Naoyuki Gunji, Yoshitaka Ushiku, Tatsuya Harada                                                                                                                                                        |
| Minerva   | 21.7        |             |             | Peking University <sup>†</sup> , Microsoft Research <sup>‡</sup> , Shanghai Jiao Tong University <sup>‡</sup> , XiDian University <sup>§</sup> , Harbin Institute of Technology <sup>§</sup> | Tianjun Xiao <sup>†‡</sup> , Minjie Wang <sup>‡‡</sup> , Jianpeng Li <sup>§‡</sup> , Yalong Bai <sup>§‡</sup> , Jiaying Zhang <sup>‡</sup> , Kuiyuan Yang <sup>‡</sup> , Chuntao Hong <sup>‡</sup> , Zheng Zhang <sup>‡</sup><br>(Wang et al., 2014) |
| NEC       | -           | -           | <b>19.6</b> | NEC Labs America <sup>†</sup> , University of Missouri <sup>‡</sup>                                                                                                                          | Xiaoyu Wang <sup>†</sup> , Miao Sun <sup>‡</sup> , Tianbao Yang <sup>†</sup> , Yuanqing Lin <sup>†</sup> , Tony X. Han <sup>‡</sup> , Shenghuo Zhu <sup>†</sup><br>(Wang et al., 2013)                                                               |
| NUS       | 13.0        |             |             | National University of Singapore                                                                                                                                                             | Min Lin*, Qiang Chen*, Jian Dong, Junshi Huang, Wei Xia, Shuicheng Yan (* = equal contribution)<br>(Krizhevsky et al., 2012)                                                                                                                         |
| Orange    | 25.2        |             |             | Orange Labs International Center Beijing <sup>†</sup> , Beijing University of Posts and Telecommunications <sup>‡</sup>                                                                      | Hongliang Bai <sup>†</sup> , Lezi Wang <sup>‡</sup> , Shusheng Cen <sup>‡</sup> , YiNan Liu <sup>‡</sup> , Kun Tao <sup>†</sup> , Wei Liu <sup>†</sup> , Peng Li <sup>†</sup> , Yuan Dong <sup>†</sup>                                               |
| OverFeat  | 14.2        | <b>30.0</b> | (19.4)      | New York University                                                                                                                                                                          | Pierre Sermanet, David Eigen, Michael Mathieu, Xiang Zhang, Rob Fergus, Yann LeCun<br>(Sermanet et al., 2013)                                                                                                                                        |
| Quantum   | 82.0        | -           | -           | Self-employed <sup>†</sup> , Student in Troy High School, Fullerton, CA <sup>‡</sup>                                                                                                         | Henry Shu <sup>†</sup> , Jerry Shu <sup>‡</sup><br>(Batra et al., 2013)                                                                                                                                                                              |
| SYSU      | -           | -           | <b>10.5</b> | Sun Yat-Sen University, China.                                                                                                                                                               | Xiaolong Wang<br>(Felzenszwalb et al., 2010)                                                                                                                                                                                                         |
| Toronto   | -           | -           | <b>11.5</b> | University of Toronto                                                                                                                                                                        | Yichuan Tang*, Nitish Srivastava*, Ruslan Salakhutdinov (* = equal contribution)                                                                                                                                                                     |
| Trimps    | 26.2        | -           | -           | The Third Research Institute of the Ministry of Public Security, P.R. China                                                                                                                  | Jie Shao, Xiaoteng Zhang, Yanfeng Shang, Wenfei Wang, Lin Mei, Chuanping Hu                                                                                                                                                                          |
| UCLA      | -           | -           | <b>9.8</b>  | University of California Los Angeles                                                                                                                                                         | Yukun Zhu, Jun Zhu, Alan Yuille                                                                                                                                                                                                                      |
| UIUC      | -           | -           | <b>1.0</b>  | University of Illinois at Urbana-Champaign                                                                                                                                                   | Thomas Paine, Kevin Shih, Thomas Huang<br>(Krizhevsky et al., 2012)                                                                                                                                                                                  |
| UvA       | 14.3        | -           | <b>22.6</b> | University of Amsterdam, Euvision Technologies                                                                                                                                               | Koen E. A. van de Sande, Daniel H. F. Fontijne, Cees G. M. Snoek, Harro M. G. Stokman, Arnold W. M. Smeulders<br>(van de Sande et al., 2014)                                                                                                         |
| VGG       | 15.2        | <b>46.4</b> | -           | Visual Geometry Group, University of Oxford                                                                                                                                                  | Karen Simonyan, Andrea Vedaldi, Andrew Zisserman<br>(Simonyan et al., 2013)                                                                                                                                                                          |
| ZF        | 13.5        | -           | -           | New York University                                                                                                                                                                          | Matthew D Zeiler, Rob Fergus<br>(Zeiler and Fergus, 2013; Zeiler et al., 2011)                                                                                                                                                                       |

**Table 6** Teams participating in ILSVRC2013, ordered alphabetically. Each method is identified with a codename used in the text. For classification and single-object localization we report flat top-5 error, in percents (lower is better). For detection we report mean average precision, in percents (higher is better). Even though the winner of the challenge was determined by the number of object categories won, this correlated strongly with mAP. Parentheses indicate the team used outside training data and was not part of the official competition. Some competing teams also submitted entries trained with outside data: Clarifai with 11.2% classification error, NEC with 20.9% detection mAP. Key references are provided where available. More details about the winning entries can be found in Section 5.1.

ILSVRC 2013

| Codename         | CLS  | LOC  | DET    | Institutions                                                                                                                                                                                 | Contributors and references                                                                             |
|------------------|------|------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Adobe            | 15.2 | -    | -      | Adobe <sup>†</sup> , University of Illinois at Urbana-Champaign <sup>‡</sup>                                                                                                                 | Hailin Jin <sup>†</sup> , Zhe Lin <sup>†</sup> , Jianchao Yang <sup>†</sup> , (Krizhevsky et al., 2012) |
| AHoward          | 13.6 | -    | -      | Andrew Howard Consulting                                                                                                                                                                     | Andrew Howard                                                                                           |
| BUP <sup>†</sup> | 25.2 | -    | -      | Beijing University of Posts and Telecommunications <sup>†</sup> , Orange Labs International Center Beijing <sup>‡</sup>                                                                      | Chong Huang <sup>†</sup> , Yanlong Bian <sup>†</sup> , Honglata                                         |
| Clarifai         | 11.7 | -    | -      | Clarifai                                                                                                                                                                                     | Matthew Zeiler                                                                                          |
| CogVision        | 16.1 | -    | -      | Microsoft Research <sup>†</sup> , Harbin Institute of Technology <sup>‡</sup>                                                                                                                | (Zeiler and Fergus, 2013; Zeiler et al., 2010)                                                          |
| decaf            | 19.2 | -    | -      | University of California Berkeley                                                                                                                                                            | Kuiyuan Yang <sup>†</sup> , Yalong Bai <sup>†</sup> , Yong Rui;                                         |
| Deep Pmx         | 20.9 | -    | -      | Saint Petersburg State University                                                                                                                                                            | Yangqing Jia, Jeff Donahue, Trevor Dar                                                                  |
| Delta            | -    | -    | 6.1    | National Tsing Hua University                                                                                                                                                                | (Donahue et al., 2013)                                                                                  |
| IBM              | 20.7 | -    | -      | University of Illinois at Urbana-Champaign <sup>†</sup> , IBM Watson Research Center <sup>‡</sup> , IBM Haifa Research Center <sup>†</sup>                                                   | Yangqing Jia, Jeff Donahue, Trevor Dar                                                                  |
| MIL              | 24.4 | -    | -      | University of Tokyo                                                                                                                                                                          | Evgeny Sminov, Denis Timoshenko, Ale                                                                    |
| Minerva          | 21.7 |      |        | Peking University <sup>†</sup> , Microsoft Research <sup>‡</sup> , Shanghai Jiao Tong University <sup>†</sup> , Xidian University <sup>§</sup> , Harbin Institute of Technology <sup>5</sup> | (Krizhevsky et al., 2012; Wan et al., 201                                                               |
| NEC              | -    | -    | 19.6   | NEC Labs America <sup>†</sup> , University of Missouri <sup>‡</sup>                                                                                                                          | Che-Rung Lee, Hwann-Tzong Chen, Hao                                                                     |
| NUS              | 13.0 |      |        | National University of Singapore                                                                                                                                                             | Che Kao                                                                                                 |
| Orange           | 25.2 |      |        | Orange Labs International Center Beijing <sup>†</sup> , Beijing University of Posts and Telecommunications <sup>‡</sup>                                                                      | Zhicheng Yan <sup>†</sup> , Liangliang Cao <sup>†</sup> , John F                                        |
| OverFeat         | 14.2 | 30.0 | (19.4) | New York University                                                                                                                                                                          | Pankanti <sup>‡</sup> , Sharon Alpert <sup>†</sup> , Yochay Tz                                          |
| Quantum          | 82.0 | -    | -      | Self-employed <sup>†</sup> , Student in Troy High School, Pullerton, CA <sup>†</sup>                                                                                                         | Masatoshi Hidaka, Chie Kamada, Yusuke                                                                   |
| SYSU             | -    | -    | 10.5   | Sun Yat-Sen University, China.                                                                                                                                                               | Tianjun Xiao <sup>†</sup> , Minjie Wang <sup>†</sup> , Jianpu                                           |
| Toronto          | -    | -    | 11.5   | University of Toronto                                                                                                                                                                        | Yang <sup>†</sup> , Chuntee Hong <sup>‡</sup> , Zheng Zhang <sup>†</sup>                                |
| Thimps           | 26.2 | -    | -      | The Third Research Institute of the Ministry of Public Security, P.R. China                                                                                                                  | (Wang et al., 2014)                                                                                     |
| UCLA             | -    | -    | 9.8    | University of California Los Angeles                                                                                                                                                         | XiaoYu Wang <sup>†</sup> , Miao Sun <sup>†</sup> , Tianbao Yan                                          |
| UUC              | -    | -    | 1.0    | University of Illinois at Urbana-Champaign                                                                                                                                                   | (Wang et al., 2013)                                                                                     |
| UVA              | 14.3 | -    | 22.6   | University of Amsterdam, Euvision Technologies                                                                                                                                               | Min Lin*, Qiang Chen*, Jian Dong, Ju                                                                    |
| VGG              | 15.2 | 46.4 | -      | Visual Geometry Group, University of Oxford                                                                                                                                                  | (Krizhevsky et al., 2012)                                                                               |
| ZF               | 13.5 | -    | -      | New York University                                                                                                                                                                          | Hongliang Bai <sup>†</sup> , Lezi Wang <sup>†</sup> , Shusheng                                          |

**Table 6** 参与ILSVRC2013的团队按字母顺序排列。每种方法在文中以代号标识。对于分类和单目标定位任务，我们报告的是扁平化前5任务，我们报告的是平均精度均值（百分比，越高越好）。尽管挑战赛的优胜者由获胜的目标类别数量决定，但这与mAP高度相关。括号内为正式竞赛。部分参赛团队也提交了使用外部数据训练的模型：Clarifai的分类错误率为11.2%，NEC的检测mAP为20.9%。关键参考文献已见前节。



ILSVRC 2014

| Codename    | CLS        | CLSo        | LOC         | LOCo        | DET         | DETo        | Institutions                                                                                                     | Contributors and references                                                                                                                                                                                                                              |
|-------------|------------|-------------|-------------|-------------|-------------|-------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adobe       | -          | 11.6        | -           | <b>30.1</b> | -           | -           | Adobe <sup>†</sup> , UIUC <sup>‡</sup>                                                                           | Hailin Jin <sup>†</sup> , Zhaowen Wang <sup>‡</sup> , Jianchao Yang <sup>†</sup> , Zhe Lin <sup>†</sup>                                                                                                                                                  |
| AHoward     | 8.1        | -           | o           | -           | -           | -           | Howard Vision Technologies                                                                                       | Andrew Howard (Howard, 2014)                                                                                                                                                                                                                             |
| BDC         | 11.3       | -           | o           | -           | -           | -           | Institute for Infocomm Research <sup>†</sup> , Universit  Pierre et Marie Curie <sup>‡</sup>                     | Olivier Morre <sup>†‡</sup> , Hanlin Goh <sup>†</sup> , Antoine Veillard <sup>‡</sup> , Vijay Chandrasekhar <sup>†</sup> (Krizhevsky et al., 2012)                                                                                                       |
| Berkeley    | -          | -           | -           | -           | -           | 34.5        | UC Berkeley                                                                                                      | Ross Girshick, Jeff Donahue, Sergio Guadarrama, Trevor Darrell, Jitendra Malik (Girshick et al., 2013, 2014)                                                                                                                                             |
| BREIL       | 16.0       | -           | o           | -           | -           | -           | KAIST department of EE                                                                                           | Jun-Cheol Park, Yunhun Jang, Hyungwon Choi, JaeYoung Jun (Chatfield et al., 2014; Jia, 2013)                                                                                                                                                             |
| Brno        | 17.6       | -           | 52.0        | -           | -           | -           | Brno University of Technology                                                                                    | Martin Kol  , Michal Hradi , Pavel Svoboda (Krizhevsky et al., 2012; Mikolov et al., 2013; Jia, 2013)                                                                                                                                                    |
| CASIA-2     | -          | -           | -           | -           | 28.6        | -           | Chinese Academy of Science <sup>†</sup> , South-east University <sup>‡</sup>                                     | Peihao Huang <sup>†</sup> , Yongzhen Huang <sup>†</sup> , Feng Liu <sup>‡</sup> , Zifeng Wu <sup>†</sup> , Fang Zhao <sup>†</sup> , Liang Wang <sup>†</sup> , Tieniu Tan <sup>†</sup> (Girshick et al., 2014)                                            |
| CASIAWS     | -          | <b>11.4</b> | -           | o           | -           | -           | CRIPAC, CASIA                                                                                                    | Weiqiang Ren, Chong Wang, Yanhua Chen, Kaiqi Huang, Tieniu Tan (Arbel ez et al., 2014)                                                                                                                                                                   |
| Cldi        | 13.9       | -           | 46.9        | -           | -           | -           | KAIST <sup>†</sup> , Cldi Inc. <sup>‡</sup>                                                                      | Kyunghyun Paeng <sup>†</sup> , Donggeun Yoo <sup>†</sup> , Sunggyun Park <sup>†</sup> , Jungin Lee <sup>‡</sup> , Anthony S. Paek <sup>‡</sup> , In So Kweon <sup>†</sup> , Seong Dae Kim <sup>†</sup> (Krizhevsky et al., 2012; Perronnin et al., 2010) |
| CUHK        | -          | -           | -           | -           | -           | 40.7        | The Chinese University of Hong Kong                                                                              | Wanli Ouyang, Ping Luo, Xingyu Zeng, Shi Qiu, Yonglong Tian, Hongsheng Li, Shuo Yang, Zhe Wang, Yuanjun Xiong, Chen Qian, Zhenyao Zhu, Ruohui Wang, Chen-Change Loy, Xiaogang Wang, Xiaoou Tang (Ouyang et al., 2014; Ouyang and Wang, 2013)             |
| DeepCNet    | 17.5       | -           | o           | -           | -           | -           | University of Warwick                                                                                            | Ben Graham (Graham, 2013; Schmidhuber, 2012)                                                                                                                                                                                                             |
| DeepInsight | -          | -           | -           | -           | -           | 40.5        | NLPR <sup>†</sup> , HKUST <sup>‡</sup>                                                                           | Junjie Yan <sup>†</sup> , Naiyan Wang <sup>‡</sup> , Stan Z. Li <sup>†</sup> , Dit-Yan Yeung <sup>‡</sup> (Girshick et al., 2014)                                                                                                                        |
| FengjunLv   | 17.4       | -           | o           | -           | -           | -           | Fengjun Lv Consulting                                                                                            | Fengjun Lv (Krizhevsky et al., 2012; Harel et al., 2007)                                                                                                                                                                                                 |
| GoogLeNet   | <b>6.7</b> | -           | 26.4        | -           | -           | <b>43.9</b> | Google                                                                                                           | Christian Szegedy, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed, Drago Anguelov, Dumitru Erhan, Andrew Rabinovich (Szegedy et al., 2014)                                                                                                           |
| HKUST       | -          | -           | -           | -           | 28.9        | -           | Hong Kong U. of Science and Tech. <sup>†</sup> , Chinese U. of H. K. <sup>‡</sup> , Stanford U. <sup> </sup>     | Cewu Lu <sup>†</sup> , Hei Law <sup>*†</sup> , Hao Chen <sup>*‡</sup> , Qifeng Chen <sup>* </sup> , Yao Xiao <sup>*†</sup> Chi Keung Tang <sup>†</sup> (Uijlings et al., 2013; Girshick et al., 2013; Perronnin et al., 2010; Felzenszwalb et al., 2010) |
| libccv      | 16.0       | -           | o           | -           | -           | -           | libccv.org                                                                                                       | Liu Liu (Zeiler and Fergus, 2013)                                                                                                                                                                                                                        |
| MIL         | 18.3       | -           | 33.7        | -           | -           | 30.4        | The University of Tokyo <sup>†</sup> , IIT Guwahati <sup>‡</sup>                                                 | Senthil Purushwalkam <sup>†‡</sup> , Yuichiro Tsuchiya <sup>†</sup> , Atsushi Kanehira <sup>†</sup> , Asako Kanezaki <sup>†</sup> , Tatsuya Harada <sup>†</sup> (Kanezaki et al., 2014; Girshick et al., 2013)                                           |
| MPG.UT      | -          | -           | -           | -           | -           | 26.4        | The University of Tokyo                                                                                          | Riku Togashi, Keita Iwamoto, Tomoaki Iwase, Hideki Nakayama (Girshick et al., 2014)                                                                                                                                                                      |
| MSRA        | 8.1        | -           | 35.5        | -           | 35.1        | -           | Microsoft Research <sup>†</sup> , Xi'an Jiaotong U. <sup>‡</sup> , U. of Science and Tech. of China <sup> </sup> | Kaiming He <sup>†</sup> , Xiangyu Zhang <sup>‡</sup> , Shaoqing Ren <sup> </sup> , Jian Sun <sup>†</sup> (He et al., 2014)                                                                                                                               |
| NUS         | -          | -           | -           | -           | <b>37.2</b> | -           | National University of Singapore <sup>†</sup> , IBM Research Australia <sup>‡</sup>                              | Jian Dong <sup>†</sup> , Yunchao Wei <sup>†</sup> , Min Lin <sup>†</sup> , Qiang Chen <sup>‡</sup> , Wei Xia <sup>†</sup> , Shuicheng Yan <sup>†</sup> (Lin et al., 2014a; Chen et al., 2014)                                                            |
| NUS-BST     | 9.8        | -           | o           | -           | -           | -           | National Univ. of Singapore <sup>†</sup> , Beijing Samsung Telecom R&D Center <sup>†</sup>                       | Min Lin <sup>†</sup> , Jian Dong <sup>†</sup> , Hanjiang Lai <sup>†</sup> , Junjun Xiong <sup>‡</sup> , Shuicheng Yan <sup>†</sup> (Lin et al., 2014a; Howard, 2014; Krizhevsky et al., 2012)                                                            |
| Orange      | 15.2       | 14.8        | 42.8        | 42.7        | -           | 27.7        | Orange Labs Beijing <sup>†</sup> , BUPT China <sup>‡</sup>                                                       | Hongliang Bai <sup>†</sup> , Yinan Liu <sup>†</sup> , Bo Liu <sup>‡</sup> , Yanchao Feng <sup>‡</sup> , Kun Tao <sup>†</sup> , Yuan Dong <sup>†</sup> (Girshick et al., 2014)                                                                            |
| PassBy      | 16.7       | -           | o           | -           | -           | -           | LENOVO <sup>†</sup> , HKUST <sup>‡</sup> , U. of Macao <sup> </sup>                                              | Lin Sun <sup>†‡</sup> , Zhanghui Kuang <sup>†</sup> , Cong Zhao <sup>†</sup> , Kui Jia <sup> </sup> , Oscar C. Au <sup>‡</sup> (Jia, 2013; Krizhevsky et al., 2012)                                                                                      |
| SCUT        | 18.8       | -           | o           | -           | -           | -           | South China Univ. of Technology                                                                                  | Guo Lihua, Liao Qijun, Ma Qianli, Lin Junbin                                                                                                                                                                                                             |
| Southeast   | -          | -           | -           | -           | 30.5        | -           | Southeast U. <sup>†</sup> , Chinese A. of Sciences <sup>‡</sup>                                                  | Feng Liu <sup>†</sup> , Zifeng Wu <sup>‡</sup> , Yongzhen Huang <sup>‡</sup>                                                                                                                                                                             |
| SYSU        | 14.4       | -           | 31.9        | -           | -           | -           | Sun Yat-Sen University                                                                                           | Liliang Zhang, Tianshui Chen, Shuye Zhang, Wanglan He, Liang Lin, Dengguang Pang, Lingbo Liu                                                                                                                                                             |
| Trimps      | -          | 11.5        | -           | 42.2        | -           | 33.7        | The Third Research Institute of the Ministry of Public Security                                                  | Jie Shao, Xiaoteng Zhang, JianYing Zhou, Jian Wang, Jian Chen, Yanfeng Shang, Wenfei Wang, Lin Mei, Chuanping Hu (Girshick et al., 2014; Manen et al., 2013; Howard, 2014)                                                                               |
| TTIC        | 10.2       | -           | 48.3        | -           | -           | -           | Toyota Technological Institute at Chicago <sup>†</sup> , Ecole Centrale Paris <sup>‡</sup>                       | George Papandreou <sup>†</sup> , Iasonas Kokkinos <sup>‡</sup> (Papandreou, 2014; Papandreou et al., 2014; Jojic et al., 2003; Krizhevsky et al., 2012; Sermanet et al., 2013; Dubout and Fleuret, 2012; Iandola et al., 2014)                           |
| UI          | 99.5       | -           | o           | -           | -           | -           | University of Isfahan                                                                                            | Fatemeh Shafizadegan, Elham Shabaninia (Yang et al., 2009)                                                                                                                                                                                               |
| UvA         | 12.1       | -           | o           | -           | 32.0        | 35.4        | U. of Amsterdam and Euvision Tech.                                                                               | Koen van de Sande, Daniel Fontijn, Cees Snoek, Harro Stokman, Arnold Smeulders (van de Sande et al., 2014)                                                                                                                                               |
| VGG         | 7.3        | -           | <b>25.3</b> | -           | -           | -           | University of Oxford                                                                                             | Karen Simonyan, Andrew Zisserman (Simonyan and Zisserman, 2014)                                                                                                                                                                                          |
| XYZ         | 11.2       | -           | o           | -           | -           | -           | The University of Queensland                                                                                     | Zhongwen Xu and Yi Yang (Krizhevsky et al., 2012; Jia, 2013; Zeiler and Fergus, 2013; Lin et al., 2014a)                                                                                                                                                 |

**Table 7** Teams participating in ILSVRC2014, ordered alphabetically. Each method is identified with a codename used in the text. For classification and single-object localization we report flat top-5 error, in percents (lower is better). For detection we report mean average precision, in percents (higher is better). CLSo, LOCo, DETo corresponds to entries using outside training data (officially allowed in ILSVRC2014). o means localization error greater than 60% (localization submission was required with every classification submission). Key references are provided where available. More details about the winning entries can be found in Section 5.1.

| ILSVRC 2014 |      |      |      |      |      |      |                                                                                                                  |  |  |
|-------------|------|------|------|------|------|------|------------------------------------------------------------------------------------------------------------------|--|--|
| Codename    | CLS  | CLSo | LOC  | LOCo | DET  | DETo | Institutions                                                                                                     |  |  |
| Adobe       | -    | 11.6 | -    | 30.1 | -    | -    | Adobe <sup>†</sup> , UTIC <sup>‡</sup>                                                                           |  |  |
| AHoward     | 8.1  | -    | 0    | -    | -    | -    | Howard Vision Technologies                                                                                       |  |  |
| BDC         | 11.3 | -    | 0    | -    | -    | -    | Institute for Infocomm Research <sup>†</sup> , Universit  Pierre et Marie Curie <sup>‡</sup>                     |  |  |
| Berkeley    | -    | -    | -    | -    | -    | 34.5 | UC Berkeley                                                                                                      |  |  |
| BREIL       | 16.0 | -    | 0    | -    | -    | -    | KAIST department of EE                                                                                           |  |  |
| Brio        | 17.6 | -    | 52.0 | -    | -    | -    | Brno University of Technology                                                                                    |  |  |
| CASIA-2     | -    | -    | -    | -    | 28.6 | -    | Chinese Academy of Science <sup>†</sup> , Southeast University <sup>‡</sup>                                      |  |  |
| CASIAWS     | -    | 11.4 | -    | 0    | -    | -    | CRTPAC, CASIA                                                                                                    |  |  |
| Cidi        | 13.9 | -    | 46.9 | -    | -    | -    | KAIST <sup>†</sup> , Cidi Inc. <sup>‡</sup>                                                                      |  |  |
| CUHK        | -    | -    | -    | -    | -    | 40.7 | The Chinese University of Hong Kong                                                                              |  |  |
| DeepCNet    | 17.5 | -    | 0    | -    | -    | -    | University of Warwick                                                                                            |  |  |
| DeepInsight | -    | -    | -    | -    | -    | 40.5 | NLPRI, HKUST <sup>‡</sup>                                                                                        |  |  |
| FengjunLv   | 17.4 | -    | 0    | -    | -    | -    | Fengjun Lv Consulting                                                                                            |  |  |
| GoogleNet   | 6.7  | -    | 26.4 | -    | -    | 43.9 | Google                                                                                                           |  |  |
| HKUST       | -    | -    | -    | -    | 28.9 | -    | Hong Kong U. of Science and Tech <sup>†</sup> , Chinese U. of H. K. <sup>‡</sup> , Stanford U. <sup>‡</sup>      |  |  |
| libccv      | 16.0 | -    | 0    | -    | -    | -    | libccv.org                                                                                                       |  |  |
| MIL         | 18.3 | -    | 33.7 | -    | -    | 30.4 | The University of Tokyo <sup>†</sup> , HIT Guwahati <sup>‡</sup>                                                 |  |  |
| MPG-UT      | -    | -    | -    | -    | -    | 26.4 | The University of Tokyo                                                                                          |  |  |
| MSRA        | 8.1  | -    | 35.5 | -    | 35.1 | -    | Microsoft Research <sup>†</sup> , Xi'an Jiaotong U. <sup>‡</sup> , U. of Science and Tech. of China <sup>‡</sup> |  |  |
| NUS         | -    | -    | -    | -    | 37.2 | -    | National University of Singapore <sup>†</sup> , IBM Research Australia <sup>‡</sup>                              |  |  |
| NUS-BST     | 9.8  | -    | 0    | -    | -    | -    | National Univ. of Singapore <sup>†</sup> , Beijing Samsung Telecom R&D Center <sup>‡</sup>                       |  |  |
| Orange      | 15.2 | 14.8 | 42.8 | 42.7 | -    | 27.7 | Orange Labs Beijing <sup>†</sup> , BUPT China <sup>‡</sup>                                                       |  |  |
| PassBy      | 16.7 | -    | 0    | -    | -    | -    | LENOVO <sup>†</sup> , HKUST <sup>‡</sup> , U. of Macao <sup>‡</sup>                                              |  |  |
| SCUT        | 18.8 | -    | 0    | -    | -    | -    | South China Univ. of Technology                                                                                  |  |  |
| Southeast   | -    | -    | -    | -    | 30.5 | -    | Southeast U. <sup>†</sup> , Chinese A. of Sciences <sup>‡</sup>                                                  |  |  |
| SYSU        | 14.4 | -    | 31.9 | -    | -    | -    | Sun Yat-Sen University                                                                                           |  |  |
| Trimps      | -    | 11.5 | -    | 42.2 | -    | 33.7 | The Third Research Institute of the Ministry of Public Security                                                  |  |  |
| TTIC        | 10.2 | -    | 48.3 | -    | -    | -    | Toyota Technological Institute at Chicago <sup>†</sup> , Ecole Centrale Paris <sup>‡</sup>                       |  |  |
| UI          | 99.5 | -    | 0    | -    | -    | -    | University of Isfahan                                                                                            |  |  |
| UvA         | 12.1 | -    | 0    | -    | 32.0 | 35.4 | U. of Amsterdam and Eindhoven Tech.                                                                              |  |  |
| VGG         | 7.3  | -    | 25.3 | -    | -    | -    | University of Oxford                                                                                             |  |  |
| XYZ         | 11.2 | -    | 0    | -    | -    | -    | The University of Queensland                                                                                     |  |  |

**Table 7** 参与ILSVRC2014的团队按字母顺序排列。每种方法在文中以代号标识。对于分类和单目标定位任务，我们报告的是扁平化前）。对于检测任务，我们报告的是平均精度均值，以百分比表示（数值越高越好）。CLSo、LOCo、DETo对应使用外部训练数据的参赛方定位错误率超过60%（每项分类提交都必须附带定位结果）。尽可能提供了关键参考文献。获胜方案的更多细节详见第5.1节。

of the network significantly without incurring significant computational overhead. In the image classification with external data track, CASIAWS won by using weakly supervised object localization from only classification labels to improve image classification. MCG region proposals (Arbeláez et al., 2014) pretrained on PASCAL VOC 2012 data are used to extract region proposals, regions are represented using convolutional networks, and a multiple instance learning strategy is used to learn weakly supervised object detectors to represent the image.

In the single-object localization with provided data track, the winning team was VGG, which explored the effect of convolutional neural network depth on its accuracy by using three different architectures with up to 19 weight layers with rectified linear unit non-linearity, building off of the implementation of Caffe (Jia, 2013). For localization they used per-class bounding box regression similar to OverFeat (Sermanet et al., 2013). In the single-object localization with external data track, Adobe used 2000 additional ImageNet classes to train the classifiers in an integrated convolutional neural network framework for both classification and localization, with bounding box regression. At test time they used k-means to find bounding box clusters and rank the clusters according to the classification scores.

In the object detection with provided data track, the winning team NUS used the RCNN framework (Girshick et al., 2013) with the network-in-network method (Lin et al., 2014a) and improvements of (Howard, 2014). Global context information was incorporated following (Chen et al., 2014). In the object detection with external data track, the winning team was GoogLeNet (which also won image classification with provided data). It is truly remarkable that the same team was able to win at both image classification and object detection, indicating that their methods are able to not only classify the image based on scene information but also accurately localize multiple object instances. Just like most teams participating in this track, GoogLeNet used the image classification dataset as extra training data.

## 5.2 Large scale algorithmic innovations

ILSVRC over the past five years has paved the way for several breakthroughs in computer vision.

The field of categorical object recognition has dramatically evolved in the large-scale setting. Section 5.1 documents the progress, starting from coded SIFT features and evolving to large-scale convolutional neural networks dominating at all three tasks of image classification, single-object localization, and object detection. With the availability of so much training data (along

with an efficient algorithmic implementation and GPU computing resources) it became possible to learn neural networks directly from the image data, without needing to create multi-stage hand-tuned pipelines of extracted features and discriminative classifiers. The major breakthrough came in 2012 with the win of the SuperVision team on image classification and single-object localization tasks (Krizhevsky et al., 2012), and by 2014 all of the top contestants were relying heavily on convolutional neural networks.

Further, over the past few years there has been a lot of focus on large-scale recognition in the computer vision community. Best paper awards at top vision conferences in 2013 were awarded to large-scale recognition methods: at CVPR 2013 to "Fast, Accurate Detection of 100,000 Object Classes on a Single Machine" (Dean et al., 2013) and at ICCV 2013 to "From Large Scale Image Categorization to Entry-Level Categories" (Ordonez et al., 2013). Additionally, several influential lines of research have emerged, such as large-scale weakly supervised localization work of (Kuettel et al., 2012) which was awarded the best paper award in ECCV 2012 and large-scale zero-shot learning, e.g., (Frome et al., 2013).

## 6 Results and analysis

### 6.1 Improvements over the years

State-of-the-art accuracy has improved significantly from ILSVRC2010 to ILSVRC2014, showcasing the massive progress that has been made in large-scale object recognition over the past five years. The performance of the winning ILSVRC entries for each task and each year are shown in Figure 9. The improvement over the years is clearly visible. In this section we quantify and analyze this improvement.

#### 6.1.1 Image classification and single-object localization improvement over the years

There has been a 4.2x reduction in image classification error (from 28.2% to 6.7%) and a 1.7x reduction in single-object localization error (from 42.5% to 25.3%) since the beginning of the challenge. For consistency, here we consider only teams that use the provided training data. Even though the exact object categories have changed (Section 3.1.1), the large scale of the dataset has remained the same (Table 2), making the results comparable across the years. The dataset has not changed since 2012, and there has been a 2.4x reduction in image classification error (from 16.4% to 6.7%) and a 1.3x in

在不显著增加计算开销的情况下，显著提升了网络的性能。在借助外部数据的图像分类任务中，CASIAWS团队通过仅利用分类标签进行弱监督目标定位，从而提升了图像分类效果。该方法采用基于PASCAL VOC 2012数据预训练的MCG区域建议算法（Arbeláez等人，2014）提取候选区域，使用卷积网络表征区域特征，并采用多示例学习策略训练弱监督目标检测器以表征图像。

在提供数据轨迹的单目标定位任务中，获胜团队VG G探索了卷积神经网络深度对其准确性的影响。他们基于Caffe框架（Jia, 2013）实现了三种不同架构，最多采用19个带有线性整流单元非线性的权重层。对于定位任务，他们采用了类似OverFeat（Sermanet等人，2013）的逐类边界框回归方法。在允许使用外部数据的单目标定位任务中，Adobe团队额外使用了2000个ImageNet类别，在统一的卷积神经网络框架中同时训练分类和定位模型，并采用边界框回归技术。测试阶段，他们使用k均值算法寻找边界框聚类，并根据分类得分对聚类进行排序。

在提供数据的目标检测赛道中，获胜团队NUS采用了RCNN框架（Girshick等人，2013年），结合了网络中的网络方法（Lin等人，2014a）以及（Howard，2014年）的改进。全局上下文信息则借鉴了（Chen等人，2014年）的方法进行整合。在使用外部数据的目标检测赛道中，获胜团队是GoogLeNet（该团队同时在提供数据的图像分类赛道中获胜）。值得注意的是，同一团队能同时在图像分类和目标检测中获胜，这表明他们的方法不仅能基于场景信息对图像进行分类，还能准确定位多个目标实例。与此赛道的大多数参赛团队类似，GoogLeNet将图像分类数据集作为额外训练数据使用。

## 5.2 大规模算法创新

过去五年中，ILSVRC为计算机视觉领域的多项突破铺平了道路。

类别物体识别领域在大规模场景下经历了显著的发展。第5.1节记录了从编码SIFT特征开始，到大规模卷积神经网络在图像分类、单物体定位和物体检测这三项任务中占据主导地位的演进历程。随着海量训练数据的可获得性（以及

随着高效的算法实现和GPU计算资源的应用，现在可以直接从图像数据中学习神经网络，而无需构建多阶段、手动调整的特征提取和判别分类器流程。这一领域的重大突破发生在2012年，SuperVision团队在图像分类和单目标定位任务中获胜（Krizhevsky等人，2012年），到2014年，所有顶尖参赛者都已高度依赖卷积神经网络。

此外，过去几年计算机视觉领域对大规模识别给予了大量关注。2013年顶级视觉会议的最佳论文奖均授予了大规模识别方法：CVPR 2013的获奖论文是《单机快速准确检测10万个物体类别》（Dean等人，2013），ICCV 2013的获奖论文是《从大规模图像分类到入门级类别》（Ordonez等人，2013）。同时，还涌现出多个具有影响力的研究方向，例如（Kuettel等人，2012）提出的大规模弱监督定位工作曾获ECCV 2012最佳论文奖，以及（Frome等人，2013）等研究者开展的大规模零样本学习。

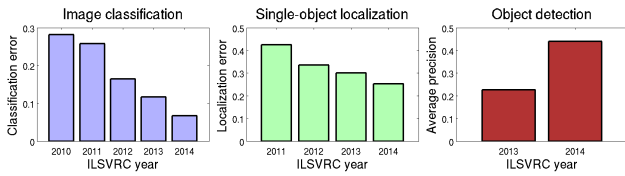
## 6 Results and analysis

### 6.1 多年来的改进

从ILSVRC2010到ILSVRC2014，最先进的准确率显著提升，展示了过去五年在大规模物体识别领域取得的巨大进展。每年各项任务的获胜ILSVRC参赛作品性能如图9所示，逐年进步清晰可见。本节我们将对这一进步进行量化分析。

#### 6.1.1 Image classification and single-object localization improvement over the years

自挑战开始以来，图像分类错误率降低了4.2倍（从28.2%降至6.7%），单目标定位错误率降低了1.7倍（从42.5%降至25.3%）。为保持一致性，此处仅考虑使用官方提供训练数据的团队。尽管具体目标类别有所调整（第3.1.1节），但数据集的大规模特性始终保持不变（表2），这使得历年结果具有可比性。自2012年以来数据集未作变动，其间图像分类错误率下降2.4倍（从16.4%降至6.7%），单目标定位错误率下降1.3倍



**Fig. 9** Performance of winning entries in the ILSVRC2010-2014 competitions in each of the three tasks (details about the entries and numerical results are in Section 5.1). There is a steady reduction of error every year in object classification and single-object localization tasks, and a 1.9x improvement in mean average precision in object detection. There are two considerations in making these comparisons. (1) The object categories used in ILSVRC changed between years 2010 and 2011, and between 2011 and 2012. However, the large scale of the data (1000 object categories, 1.2 million training images) has remained the same, making it possible to compare results. Image classification and single-object localization entries shown here use only provided training data. (2) The size of the object detection training data has increased significantly between years 2013 and 2014 (Section 3.3). Section 6.1 discusses the relative effects of training data increase versus algorithmic improvements.

single-object localization error (from 33.5% to 25.3%) in the past three years.

#### 6.1.2 Object detection improvement over the years

Object detection accuracy as measured by the mean average precision (mAP) has increased 1.9x since the introduction of this task, from 22.6% mAP in ILSVRC2013 to 43.9% mAP in ILSVRC2014. However, these results are not directly comparable for two reasons. First, the size of the object detection training data has increased significantly from 2013 to 2014 (Section 3.3). Second, the 43.9% mAP result was obtained with the addition of the image classification and single-object localization training data. Here we attempt to understand the relative effects of the training set size increase versus algorithmic improvements. All models are evaluated on the same ILSVRC2013-2014 object detection test set.

First, we quantify the effects of increasing detection training data between the two challenges by comparing the same model trained on ILSVRC2013 detection data versus ILSVRC2014 detection data. The UvA team’s framework from 2013 achieved 22.6% with ILSVRC2013 data (Table 6) and 26.3% with ILSVRC2014 data and no other modifications.<sup>10</sup> The absolute increase in mAP was 3.7%. The RCNN model achieved 31.4% mAP with ILSVRC2013 detection plus image classification data (Girshick et al., 2013) and 34.5% mAP with ILSVRC2014 detection plus image classification data (Berkeley team in Table 7). The absolute

increase in mAP by expanding ILSVRC2013 detection data to ILSVRC2014 was 3.1%.

Second, we quantify the effects of adding in the external data for training object detection models. The NEC model in 2013 achieved 19.6% mAP trained on ILSVRC2013 detection data alone and 20.9% mAP trained on ILSVRC2013 detection plus classification data (Table 6). The absolute increase in mAP was 1.3%. The UvA team’s best entry in 2014 achieved 32.0% mAP trained on ILSVRC2014 detection data and 35.4% mAP trained on ILSVRC2014 detection plus classification data. The absolute increase in mAP was 3.4%.

Thus, we conclude based on the evidence so far that expanding the ILSVRC2013 detection set to the ILSVRC2014 set, as well as adding in additional training data from the classification task, all account for approximately 1 – 4% in absolute mAP improvement for the models. For comparison, we can also attempt to quantify the effect of algorithmic innovation. The UvA team’s 2013 framework achieved 26.3% mAP on ILSVRC2014 data as mentioned above, and their improved method in 2014 obtained 32.0% mAP (Table 7). This is 5.8% absolute increase in mAP over just one year from algorithmic innovation alone.

In summary, we conclude that the absolute 21.3% increase in mAP between winning entries of ILSVRC2013 (22.6% mAP) and of ILSVRC2014 (43.9% mAP) is the result of impressive algorithmic innovation and not just a consequence of increased training data. However, increasing the ILSVRC2014 object detection training dataset further *is* likely to produce additional improvements in detection accuracy for current algorithms.

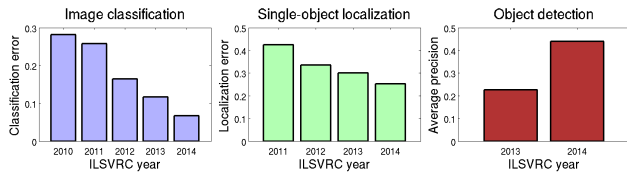
## 6.2 Statistical significance

One important question to ask is whether results of different submissions to ILSVRC are statistically significantly different from each other. Given the large scale, it is no surprise that even minor differences in accuracy are statistically significant; we seek to quantify exactly how much of a difference is enough.

Following the strategy employed by PASCAL VOC (Everingham et al., 2014), for each method we obtain a confidence interval of its score using bootstrap sampling. During each bootstrap round, we sample  $N$  images with replacement from all the available  $N$  test images and evaluate the performance of the algorithm on those sampled images. This can be done very efficiently by precomputing the accuracy on each image. Given the results of all the bootstrapping rounds we discard the lower and the upper  $\alpha$  fraction. The range of the remaining results represents the  $1 - 2\alpha$  confi-

<sup>10</sup> Personal communication with members of the UvA team.





**Fig. 9** ILSVRC2010-2014竞赛中各项任务获胜参赛作品的性能表现（参赛作品详情与数值结果见第5.1节）。在物体分类与单物体定位任务中，误差率逐年稳步下降；物体检测任务的均值平均精度则提升了1.9倍。进行此类比较时需注意两点：（1）ISLVR C使用的物体类别在2010-2011年及2011-2012年间有所变动，但由于数据规模始终保持一致（1000个物体类别、120万训练图像），结果仍具可比性。图中展示的分类与单物体定位参赛作品仅使用官方提供的训练数据。（2）物体检测训练数据量在2013-2014年间显著增加（第3.3节）。第6.1节将讨论训练数据增长与算法改进的相对影响。

单目标定位误差（从33.5%降至25.3%）在过去三年中。

#### 6.1.2 Object detection improvement over the years

以平均精度均值（mAP）衡量的目标检测准确率自该任务提出以来已提升1.9倍，从ILSVRC2013的22.6% mAP上升至ILSVRC2014的43.9% mAP。然而，这些结果因以下两个原因无法直接比较：首先，目标检测训练数据的规模在2013至2014年间显著增加（第3.3节）；其次，43.9% mAP的结果是在加入图像分类与单目标定位训练数据后取得的。本文旨在解析训练集规模扩大与算法改进的相对影响。所有模型均在相同的ILSVRC2013-2014目标检测测试集上进行评估。

首先，我们通过比较同一模型在ILSVRC2013检测数据与ILSVRC2014检测数据上的训练效果，量化了两次挑战间增加检测训练数据的影响。UvA团队2013年的框架使用ILSVRC2013数据时达到22.6%（表6），而使用ILSVRC2014数据且未作其他修改时达到26.3%<sup>10</sup>，mAP绝对值提升3.7%。RCNN模型使用ILSVRC2013检测数据加图像分类数据时获得31.4% mAP（Girshick等，2013），使用ILSVRC2014检测数据加图像分类数据时获得34.5% mAP（表7中伯克利团队结果）。

将ILSVRC2013检测数据扩展至ILSVRC2014带来的mAP提升为3.1%。

其次，我们量化了添加外部数据对训练目标检测模型的影响。2013年的NEC模型仅使用ILSVRC2013检测数据训练时达到19.6% mAP，而使用ILSVRC2013检测加分类数据训练时达到20.9% mAP（表6）。mAP的绝对提升为1.3%。2014年UvA团队的最佳参赛模型使用ILSVRC2014检测数据训练时达到32.0% mAP，使用ILSVRC2014检测加分类数据训练时达到35.4% mAP。mAP的绝对提升为3.4%。

因此，根据目前的证据，我们得出结论：将ILSVRC2013检测集扩展至ILSVRC2014集，以及从分类任务中增加额外的训练数据，共同为模型带来了约1–4%的绝对mAP提升。作为对比，我们也可以尝试量化算法创新的影响。如上所述，UvA团队2013年的框架在ILSVRC2014数据上取得了26.3%的mAP，而他们2014年的改进方法则达到了32.0%的mAP（表7）。这意味着仅通过算法创新，一年内就实现了5.8%的绝对mAP增长。

总而言之，我们得出结论，ILSVRC2013获胜作品（22.6% mAP）与ILSVRC2014获胜作品（43.9% mAP）之间mAP绝对值21.3%的提升，是卓越算法创新的成果，而不仅仅是训练数据增加的结果。然而，进一步扩大ILSVRC2014目标检测训练数据集可能会为现有算法带来检测准确率的额外提升。

#### 6.2 统计显著性

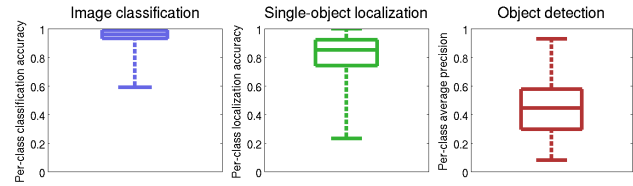
一个重要的问题是，ILSVRC不同提交结果之间是否存在统计学上的显著差异。鉴于其规模之大，即使准确率上的微小差异也具有统计显著性并不令人意外；我们需要精确量化多大的差异才算足够。

遵循PASCAL VOC（Everingham等人，2014年）采用的策略，对于每种方法，我们通过自助采样获得其得分的置信区间。在每一轮自助采样中，我们从所有可用的 $N$ 张测试图像中有放回地抽取 $N$ 张图像，并在这些抽样图像上评估算法的性能。通过预先计算每张图像的准确率，这一过程可以高效完成。根据所有自助采样轮次的结果，我们剔除最低和最高的 $\alpha$ 比例部分。剩余结果的范围代表了 $1-2\alpha$ 置信

<sup>10</sup> Personal communication with members of the UvA team.

| Image classification       |                          |                 |                      |
|----------------------------|--------------------------|-----------------|----------------------|
| Year                       | Codename                 | Error (percent) | 99.9% Conf Int       |
| <b>2014</b>                | <b>GoogLeNet</b>         | <b>6.66</b>     | <b>6.40 - 6.92</b>   |
| 2014                       | VGG                      | 7.32            | 7.05 - 7.60          |
| 2014                       | MSRA                     | 8.06            | 7.78 - 8.34          |
| 2014                       | AHoward                  | 8.11            | 7.83 - 8.39          |
| 2014                       | DeeperVision             | 9.51            | 9.21 - 9.82          |
| 2013                       | Clarifai <sup>†</sup>    | 11.20           | 10.87 - 11.53        |
| 2014                       | CASIAWS <sup>†</sup>     | 11.36           | 11.03 - 11.69        |
| 2014                       | Trimps <sup>†</sup>      | 11.46           | 11.13 - 11.80        |
| 2014                       | Adobe <sup>†</sup>       | 11.58           | 11.25 - 11.91        |
| <b>2013</b>                | <b>Clarifai</b>          | <b>11.74</b>    | <b>11.41 - 12.08</b> |
| 2013                       | NUS                      | 12.95           | 12.60 - 13.30        |
| 2013                       | ZF                       | 13.51           | 13.14 - 13.87        |
| 2013                       | AHoward                  | 13.55           | 13.20 - 13.91        |
| 2013                       | OverFeat                 | 14.18           | 13.83 - 14.54        |
| 2014                       | Orange <sup>†</sup>      | 14.80           | 14.43 - 15.17        |
| 2012                       | SuperVision <sup>†</sup> | 15.32           | 14.94 - 15.69        |
| <b>2012</b>                | <b>SuperVision</b>       | <b>16.42</b>    | <b>16.04 - 16.80</b> |
| 2012                       | ISI                      | 26.17           | 25.71 - 26.65        |
| 2012                       | VGG                      | 26.98           | 26.53 - 27.43        |
| 2012                       | XRCE                     | 27.06           | 26.60 - 27.52        |
| 2012                       | UvA                      | 29.58           | 29.09 - 30.04        |
| Single-object localization |                          |                 |                      |
| Year                       | Codename                 | Error (percent) | 99.9% Conf Int       |
| <b>2014</b>                | <b>VGG</b>               | <b>25.32</b>    | <b>24.87 - 25.78</b> |
| 2014                       | GoogLeNet                | 26.44           | 25.98 - 26.92        |
| <b>2013</b>                | <b>OverFeat</b>          | <b>29.88</b>    | <b>29.38 - 30.35</b> |
| 2014                       | Adobe <sup>†</sup>       | 30.10           | 29.61 - 30.58        |
| 2014                       | SYSU                     | 31.90           | 31.40 - 32.40        |
| 2012                       | SuperVision <sup>†</sup> | 33.55           | 33.05 - 34.04        |
| 2014                       | MIL                      | 33.74           | 33.24 - 34.25        |
| <b>2012</b>                | <b>SuperVision</b>       | <b>34.19</b>    | <b>33.67 - 34.69</b> |
| 2014                       | MSRA                     | 35.48           | 34.97 - 35.99        |
| 2014                       | Trimps <sup>†</sup>      | 42.22           | 41.69 - 42.75        |
| 2014                       | Orange <sup>†</sup>      | 42.70           | 42.18 - 43.24        |
| 2013                       | VGG                      | 46.42           | 45.90 - 46.95        |
| 2012                       | VGG                      | 50.03           | 49.50 - 50.57        |
| 2012                       | ISI                      | 53.65           | 53.10 - 54.17        |
| 2014                       | CASIAWS <sup>†</sup>     | 61.96           | 61.44 - 62.48        |
| Object detection           |                          |                 |                      |
| Year                       | Codename                 | AP (percent)    | 99.9% Conf Int       |
| 2014                       | GoogLeNet <sup>†</sup>   | 43.93           | 42.92 - 45.65        |
| 2014                       | CUHK <sup>†</sup>        | 40.67           | 39.68 - 42.30        |
| 2014                       | DeepInsight <sup>†</sup> | 40.45           | 39.49 - 42.06        |
| <b>2014</b>                | <b>NUS</b>               | <b>37.21</b>    | <b>36.29 - 38.80</b> |
| 2014                       | UvA <sup>†</sup>         | 35.42           | 34.63 - 36.92        |
| 2014                       | MSRA                     | 35.11           | 34.36 - 36.70        |
| 2014                       | Berkeley <sup>†</sup>    | 34.52           | 33.67 - 36.12        |
| 2014                       | UvA                      | 32.03           | 31.28 - 33.49        |
| 2014                       | Southeast                | 30.48           | 29.70 - 31.93        |
| 2014                       | HKUST                    | 28.87           | 28.03 - 30.20        |
| <b>2013</b>                | <b>UvA</b>               | <b>22.58</b>    | <b>22.00 - 23.82</b> |
| 2013                       | NEC <sup>†</sup>         | 20.90           | 20.40 - 22.15        |
| 2013                       | NEC                      | 19.62           | 19.14 - 20.85        |
| 2013                       | OverFeat <sup>†</sup>    | 19.40           | 18.82 - 20.61        |
| 2013                       | Toronto                  | 11.46           | 10.98 - 12.34        |
| 2013                       | SYSU                     | 10.45           | 10.04 - 11.32        |
| 2013                       | UCLA                     | 9.83            | 9.48 - 10.77         |

**Table 8** We use bootstrapping to construct 99.9% confidence intervals around the result of up to top 5 submissions to each ILSVRC task in 2012-2014. <sup>†</sup> means the entry used external training data. The winners using the provided data for each track and each year are bolded. The difference between the winning method and the runner-up each year is significant even at the 99.9% level.



**Fig. 10** For each object class, we consider the best performance of any entry submitted to ILSVRC2012-2014, including entries using additional training data. The plots show the distribution of these “optimistic” per-class results. Performance is measured as accuracy for image classification (left) and for single-object localization (middle), and as average precision for object detection (right). While the results are very promising in image classification, the ILSVRC datasets are far from saturated: many object classes continue to be challenging for current algorithms.

dence interval. We run a large number of bootstrapping rounds (from 20,000 until convergence). Table 8 shows the results of the top entries to each task of ILSVRC2012-2014. The winning methods are statistically significantly different from the other methods, even at the 99.9% level.

### 6.3 Current state of categorical object recognition

Besides looking at just the average accuracy across hundreds of object categories and tens of thousands of images, we can also delve deeper to understand where mistakes are being made and where researchers’ efforts should be focused to expedite progress.

To do so, in this section we will be analyzing an “optimistic” measurement of state-of-the-art recognition performance instead of focusing on the differences in individual algorithms. For each task and each object class, we compute the best performance of *any* entry submitted to *any* ILSVRC2012-2014, including methods using additional training data. Since the test sets have remained the same, we can directly compare all the entries in the past three years to obtain the most “optimistic” measurement of state-of-the-art accuracy on each category.

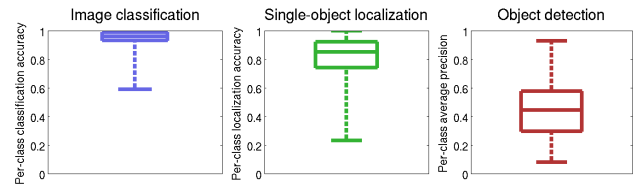
For consistency with the object detection metric (higher is better), in this section we will be using image classification and single-object localization *accuracy* instead of error, where  $accuracy = 1 - error$ .

#### 6.3.1 Range of accuracy across object classes

Figure 10 shows the distribution of accuracy achieved by the “optimistic” models across the object categories. The image classification model achieves 94.6% accuracy on average (or 5.4% error), but there remains a 41.0% absolute difference inaccuracy between the most

| Image classification       |                          |                 |                      |
|----------------------------|--------------------------|-----------------|----------------------|
| Year                       | Codename                 | Error (percent) | 99.9% Conf Int       |
| <b>2014</b>                | <b>GoogLeNet</b>         | <b>6.66</b>     | <b>6.40 - 6.92</b>   |
| 2014                       | VGG                      | 7.32            | 7.05 - 7.60          |
| 2014                       | MSRA                     | 8.06            | 7.78 - 8.34          |
| 2014                       | AHoward                  | 8.11            | 7.83 - 8.39          |
| 2014                       | DeeperVision             | 9.51            | 9.21 - 9.82          |
| 2013                       | Clarifai <sup>†</sup>    | 11.20           | 10.87 - 11.53        |
| 2014                       | CASIAWS <sup>†</sup>     | 11.36           | 11.03 - 11.69        |
| 2014                       | Trimps <sup>†</sup>      | 11.46           | 11.13 - 11.80        |
| 2014                       | Adobe <sup>†</sup>       | 11.58           | 11.25 - 11.91        |
| <b>2013</b>                | <b>Clarifai</b>          | <b>11.74</b>    | <b>11.41 - 12.08</b> |
| 2013                       | NUS                      | 12.95           | 12.60 - 13.30        |
| 2013                       | ZF                       | 13.51           | 13.14 - 13.87        |
| 2013                       | AHoward                  | 13.55           | 13.20 - 13.91        |
| 2013                       | OverFeat                 | 14.18           | 13.83 - 14.54        |
| 2014                       | Orange <sup>†</sup>      | 14.80           | 14.43 - 15.17        |
| 2012                       | SuperVision <sup>†</sup> | 15.32           | 14.94 - 15.69        |
| <b>2012</b>                | <b>SuperVision</b>       | <b>16.42</b>    | <b>16.04 - 16.80</b> |
| 2012                       | ISI                      | 26.17           | 25.71 - 26.65        |
| 2012                       | VGG                      | 26.98           | 26.53 - 27.43        |
| 2012                       | XRCE                     | 27.06           | 26.60 - 27.52        |
| 2012                       | UvA                      | 29.58           | 29.09 - 30.04        |
| Single-object localization |                          |                 |                      |
| Year                       | Codename                 | Error (percent) | 99.9% Conf Int       |
| <b>2014</b>                | <b>VGG</b>               | <b>25.32</b>    | <b>24.87 - 25.78</b> |
| 2014                       | GoogLeNet                | 26.44           | 25.98 - 26.92        |
| <b>2013</b>                | <b>OverFeat</b>          | <b>29.88</b>    | <b>29.38 - 30.35</b> |
| 2014                       | Adobe <sup>†</sup>       | 30.10           | 29.61 - 30.58        |
| 2014                       | SYSU                     | 31.90           | 31.40 - 32.40        |
| 2012                       | SuperVision <sup>†</sup> | 33.55           | 33.05 - 34.04        |
| 2014                       | MIL                      | 33.74           | 33.24 - 34.25        |
| <b>2012</b>                | <b>SuperVision</b>       | <b>34.19</b>    | <b>33.67 - 34.69</b> |
| 2014                       | MSRA                     | 35.48           | 34.97 - 35.99        |
| 2014                       | Trimps <sup>†</sup>      | 42.22           | 41.69 - 42.75        |
| 2014                       | Orange <sup>†</sup>      | 42.70           | 42.18 - 43.24        |
| 2013                       | VGG                      | 46.42           | 45.90 - 46.95        |
| 2012                       | VGG                      | 50.03           | 49.50 - 50.57        |
| 2012                       | ISI                      | 53.65           | 53.10 - 54.17        |
| 2014                       | CASIAWS <sup>†</sup>     | 61.96           | 61.44 - 62.48        |
| Object detection           |                          |                 |                      |
| Year                       | Codename                 | AP (percent)    | 99.9% Conf Int       |
| 2014                       | GoogLeNet <sup>†</sup>   | 43.93           | 42.92 - 45.65        |
| 2014                       | CUHK <sup>†</sup>        | 40.67           | 39.68 - 42.30        |
| 2014                       | DeepInsight <sup>†</sup> | 40.45           | 39.49 - 42.06        |
| <b>2014</b>                | <b>NUS</b>               | <b>37.21</b>    | <b>36.29 - 38.80</b> |
| 2014                       | UvA <sup>†</sup>         | 35.42           | 34.63 - 36.92        |
| 2014                       | MSRA                     | 35.11           | 34.36 - 36.70        |
| 2014                       | Berkeley <sup>†</sup>    | 34.52           | 33.67 - 36.12        |
| 2014                       | UvA                      | 32.03           | 31.28 - 33.49        |
| 2014                       | Southeast                | 30.48           | 29.70 - 31.93        |
| 2014                       | HKUST                    | 28.87           | 28.03 - 30.20        |
| <b>2013</b>                | <b>UvA</b>               | <b>22.58</b>    | <b>22.00 - 23.82</b> |
| 2013                       | NEC <sup>†</sup>         | 20.90           | 20.40 - 22.15        |
| 2013                       | NEC                      | 19.62           | 19.14 - 20.85        |
| 2013                       | OverFeat <sup>†</sup>    | 19.40           | 18.82 - 20.61        |
| 2013                       | Toronto                  | 11.46           | 10.98 - 12.34        |
| 2013                       | SYSU                     | 10.45           | 10.04 - 11.32        |
| 2013                       | UCLA                     | 9.83            | 9.48 - 10.77         |

**Table 8** 我们采用自助法 (bootstrapping) 为2012-2014年间每项ILSVRC任务中最多前5名的提交结果构建了99.9%的置信区间。<sup>†</sup>表示该参赛作品使用了外部训练数据。每年每赛道中使用规定数据的获胜者已用粗体标出。每年获胜方法与亚军之间的差异即使在99.9%的置信水平上仍具有显著性。



**Fig. 10** 对于每个物体类别，我们考虑了提交至ILSVRC2012-2014竞赛的所有参赛方案（包括使用额外训练数据的方案）中的最佳性能。这些图表展示了这些“乐观”的每类结果分布。性能通过图像分类准确率（左图）、单物体定位准确率（中图）以及物体检测平均精度（右图）进行衡量。尽管图像分类的结果非常令人鼓舞，但ILSVRC数据集远未达到饱和状态：许多物体类别对当前算法而言仍具挑战性。

置信区间。我们进行了大量自助抽样轮次（从20,000次直至收敛）。表8展示了ILSVRC2012-2014各任务中顶尖参赛方法的结果。即使在99.9%的置信水平下，优胜方法与其他方法仍存在统计学上的显著差异。

### 6.3 类别物体识别的当前状态

除了观察数百个物体类别和数万张图像的平均准确率外，我们还可以深入探究错误发生的位置，以及研究人员应集中精力以加速进展的方向。

为此，在本节中，我们将分析一种对当前最先进识别性能的“乐观”评估，而非聚焦于单个算法的差异。针对每项任务和每个对象类别，我们计算了提交至ILSVRC2012-2014的*any*参赛方案中的最佳性能，包括使用额外训练数据的方法。由于测试集保持不变，我们可以直接比较过去三年的所有参赛方案，从而获得对每个类别上最先进准确率的“乐观”评估。

为了与目标检测指标（越高越好）保持一致，在本节中我们将使用图像分类和单目标定位 $accuracy$ 而非误差，其中 $accuracy = 1 - error$ 。

#### 6.3.1 Range of accuracy across object classes

图10展示了“乐观”模型在各个物体类别上达到的准确率分布。图像分类模型平均达到94.6%的准确率（或5.4%的误差），但最

and least accurate object class. The single-object localization model achieves 81.5% accuracy on average (or 18.5% error), with a 77.0% range in accuracy across the object classes. The object detection model achieves 44.7% average precision, with an 84.7% range across the object classes. It is clear that the ILSVRC dataset is far from saturated: performance on many categories has remained poor despite the strong overall performance of the models.

### 6.3.2 Qualitative examples of easy and hard classes

Figures 11 and 12 show the easiest and hardest classes for each task, i.e., classes with the best and worst results obtained with the “optimistic” models.

For image classification, 121 out of 1000 object classes have 100% image classification accuracy according to the optimistic estimate. Figure 11 (top) shows a random set of 10 of them. They contain a variety of classes, such as mammals like “red fox” and animals with distinctive structures like “stingray”. The hardest classes in the image classification task, with accuracy as low as 59.0%, include metallic and see-through man-made objects, such as “hook” and “water bottle,” the material “velvet” and the highly varied scene class “restaurant.”

For single-object localization, the 10 easiest classes with 99.0 – 100% accuracy are all mammals and birds. The hardest classes include metallic man-made objects such as “letter opener” and “ladle”, plus thin structures such as “pole” and “spacebar” and highly varied classes such as “wing”. The most challenging class “spacebar” has a only 23.0% localization accuracy.

Object detection results are shown in Figure 12. The easiest classes are living organisms such as “dog” and “tiger”, plus “basketball” and “volleyball” with distinctive shape and color, and a somewhat surprising “snowplow.” The easiest class “butterfly” is not yet perfectly detected but is very close with 92.7% AP. The hardest classes are as expected small thin objects such as “flute” and “nail”, and the highly varied “lamp” and “backpack” classes, with as low as 8.0% AP.

### 6.3.3 Per-class accuracy as a function of image properties

We now take a closer look at the image properties to try to understand why current algorithms perform well on some object classes but not others. One hypothesis is that variation in accuracy comes from the fact that instances of some classes tend to be much smaller in images than instances of other classes, and smaller objects may be harder for computers to recognize. In this section we argue that while accuracy is correlated with

object scale in the image, not all variation in accuracy can be accounted for by scale alone.

For every object class, we compute its *average scale*, or the average fraction of image area occupied by an instance of the object class on the ILSVRC2012-2014 validation set. Since the images and object classes in the image classification and single-object localization tasks are the same, we use the bounding box annotations of the single-object localization dataset for both tasks. In that dataset the object classes range from “swimming trunks” with scale of 1.5% to “spider web” with scale of 85.6%. In the object detection validation dataset the object classes range from “sunglasses” with scale of 1.3% to “sofa” with scale of 44.4%.

Figure 13 shows the performance of the “optimistic” method as a function of the average scale of the object in the image. Each dot corresponds to one object class. We observe a very weak positive correlation between object scale and image classification accuracy:  $\rho = 0.14$ . For single-object localization and object detection the correlation is stronger, at  $\rho = 0.40$  and  $\rho = 0.41$  respectively. It is clear that not all variation in accuracy can be accounted for by scale alone. Nevertheless, in the next section we will normalize for object scale to ensure that this factor is not affecting our conclusions.

### 6.3.4 Per-class accuracy as a function of object properties.

Besides considering image-level properties we can also observe how accuracy changes as a function of intrinsic object properties. We define three properties inspired by human vision: the real-world size of the object, whether it’s deformable within instance, and how textured it is. For each property, the object classes are assigned to one of a few bins (listed below). These properties are illustrated in Figure 1.

Human subjects annotated each of the 1000 image classification and single-object localization object classes from ILSVRC2012-2014 with these properties. (Rusakovsky et al., 2013). By construction (see Section 3.3.1), each of the 200 object detection classes is either also one of 1000 object classes or is an ancestor of one or more of the 1000 classes in the ImageNet hierarchy. To compute the values of the properties for each object detection class, we simply average the annotated values of the descendant classes.

In this section we draw the following conclusions about state-of-the-art recognition accuracy as a function of these object properties:

- **Real-world size:** *XS* for extra small (e.g. nail), small (e.g. fox), medium (e.g. bookcase), large (e.g. car) or *XL* for extra large (e.g. church)

以及准确率最低的物体类别。单物体定位模型的平均准确率达到81.5%（即18.5%的误差），不同物体类别间的准确率差异范围为77.0%。物体检测模型的平均精度为44.7%，各类别间的精度差异范围达84.7%。显然ILSVRC数据集远未达到饱和：尽管模型整体表现强劲，但许多类别的性能仍然较差。

### 6.3.2 Qualitative examples of easy and hard classes

图11和图12展示了每个任务中最简单和最困难的类别，即使用“乐观”模型获得最佳和最差结果的类别。

在图像分类任务中，根据乐观估计，1000个物体类别中有121个达到了100%的图像分类准确率。图11（顶部）展示了随机选取的其中10个类别。这些类别多种多样，包括哺乳动物如“红狐”，以及具有独特结构的动物如“黄貂鱼”。图像分类任务中最难的类别准确率低至59.0%，包括金属和透明的人造物体，如“挂钩”和“水瓶”，材质类别“天鹅绒”，以及高度多样化的场景类别“餐厅”。

在单目标定位任务中，10个最容易的类别（准确率达99.0–100%）均为哺乳动物和鸟类。最具挑战性的类别包括金属人造物（如“开信刀”和“长柄勺”）、细长结构物（如“杆子”和“空格键”）以及形态多变的类别（如“翅膀”）。其中最困难的“空格键”类别定位准确率仅为23.0%。

目标检测结果如图12所示。最容易识别的类别包括生物体如“狗”和“老虎”，以及具有独特形状和颜色的“篮球”和“排球”，还有略显意外的“扫雪机”。最简单的类别“蝴蝶”虽未达到完美检测，但已非常接近，AP值为92.7%。最困难的类别如预期所示，是细小的物体如“长笛”和“钉子”，以及形态高度多变的“灯具”和“背包”类别，其AP值低至8.0%。

### 6.3.3 Per-class accuracy as a function of image properties

我们现在更仔细地观察图像属性，试图理解为什么当前算法在某些物体类别上表现良好，而在其他类别上则不然。一种假设是，准确率的差异源于某些类别的实例在图像中往往比其他类别的实例小得多，而较小的物体可能对计算机来说更难识别。在本节中，我们认为虽然准确率与

图像中的物体尺度，并非所有准确度的变化都能仅由尺度来解释。

对于每个物体类别，我们计算其*average scale*，即在ILSVRC2012-2014验证集上该物体类别单个实例所占图像面积的平均比例。由于图像分类任务和单物体定位任务中的图像及物体类别相同，我们对两项任务均使用单物体定位数据集的边界框标注。在该数据集中，物体类别的尺度范围从“游泳裤”的1.5%到“蜘蛛网”的85.6%。在物体检测验证数据集中，物体类别的尺度范围则从“太阳镜”的1.3%到“沙发”的44.4%。

图13展示了“乐观”方法的性能随图像中物体平均尺度的变化情况。每个点对应一个物体类别。我们观察到物体尺度与图像分类准确率之间存在极弱的正相关性： $\rho = 0.14$ 。对于单物体定位和物体检测任务，相关性更强，分别为 $\rho = 0.40$ 和 $\rho = 0.41$ 。显然，准确率的变化不能仅由尺度因素完全解释。尽管如此，在下一节我们将对物体尺度进行归一化处理，以确保该因素不会影响我们的结论。

### 6.3.4 Per-class accuracy as a function of object properties.

除了考虑图像层面的属性，我们还可以观察准确率如何随内在物体属性的变化而变化。我们定义了三种受人类视觉启发的属性：物体的真实世界尺寸、它是否在实例内可变形，以及其纹理丰富程度。针对每种属性，物体类别被分配到几个分组之一（如下所列）。这些属性在图1中进行了说明。

人类受试者使用这些属性对ILSVRC2012-2014中的1000个图像分类和单目标定位对象类别逐一进行了标注（Rusakovsky等人，2013年）。根据设计（见第3.3.1节），200个目标检测类别中的每一个要么是1000个对象类别之一，要么是ImageNet层次结构中一个或多个类别的祖先。为计算每个目标检测类别的属性值，我们直接对后代类别的标注值进行平均处理。

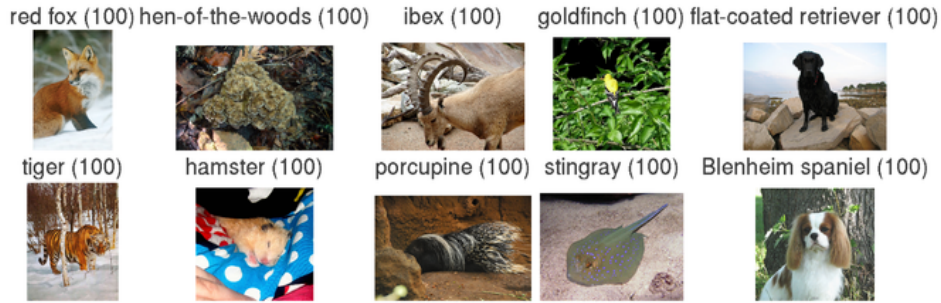
在本节中，我们针对这些物体属性，就最先进的识别精度得出以下结论：

- **Real-world size:** *XS* for extra small (e.g. nail), small (e.g. fox), medium (e.g. bookcase), large (e.g. car) or *XL* for extra large (e.g. church)



## Image classification

### Easiest classes



### Hardest classes



## Single-object localization

### Easiest classes



### Hardest classes



**Fig. 11** For each object category, we take the best performance of any entry submitted to ILSVRC2012-2014 (including entries using additional training data). Given these “optimistic” results we show the easiest and harder classes for each task. The numbers in parentheses indicate classification and localization accuracy. For image classification the 10 easiest classes are randomly selected from among 121 object classes with 100% accuracy. Object detection results are shown in Figure 12.

## Image classification

### Easiest classes



### Hardest classes



## Single-object localization

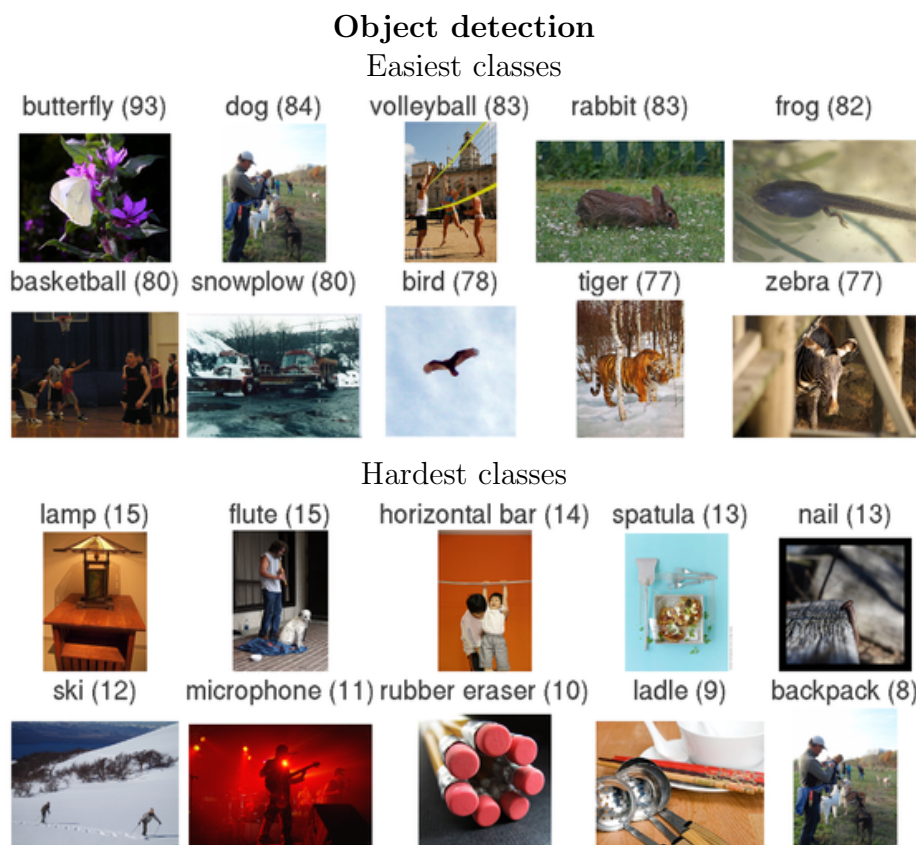
### Easiest classes



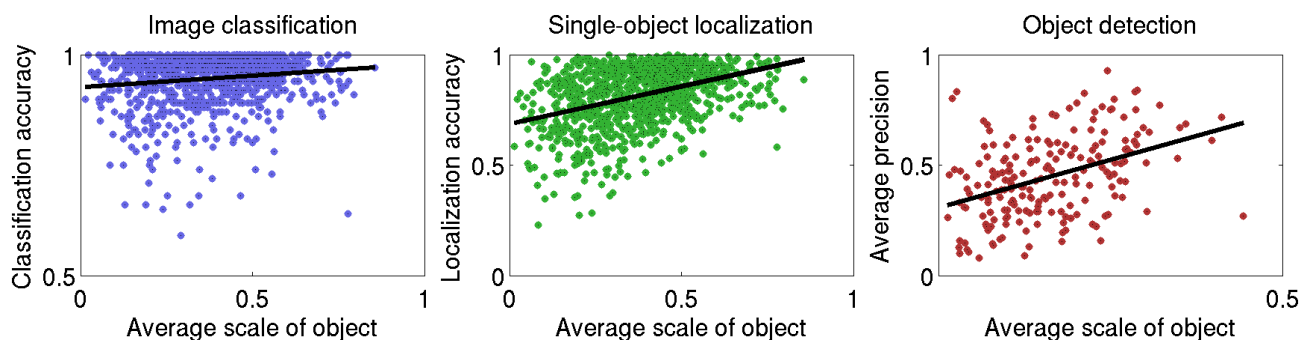
### Hardest classes



**Fig. 11** 对于每个物体类别，我们选取ILSVRC2012-2014竞赛中提交的任何参赛方案（包括使用额外训练数据的方案）的最佳性能。基于这些“乐观”结果，我们展示了每项任务中最简单和较难的类别。括号中的数字表示分类和定位准确率。在图像分类任务中，10个最简单类别是从准确率达到100%的121个物体类别中随机选取的。物体检测结果如图12所示。

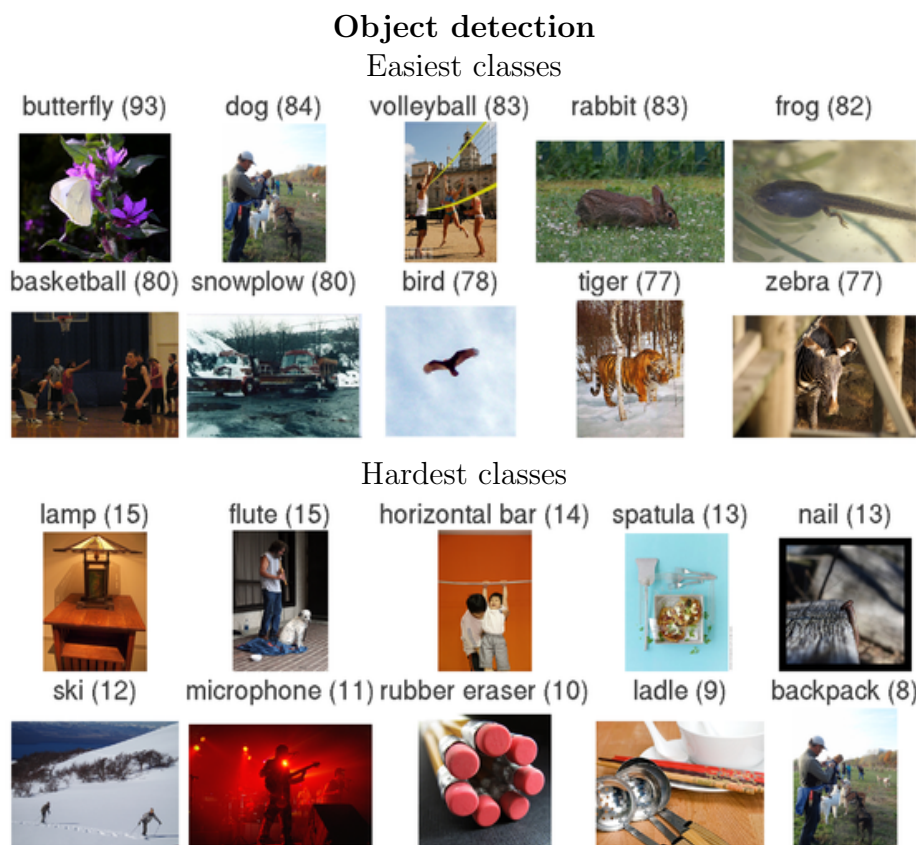


**Fig. 12** For each object category, we take the best performance of any entry submitted to ILSVRC2012-2014 (including entries using additional training data). Given these “optimistic” results we show the easiest and harder classes for the object detection task, i.e., classes with best and worst results. The numbers in parentheses indicate average precision. Image classification and single-object localization results are shown in Figure 11.

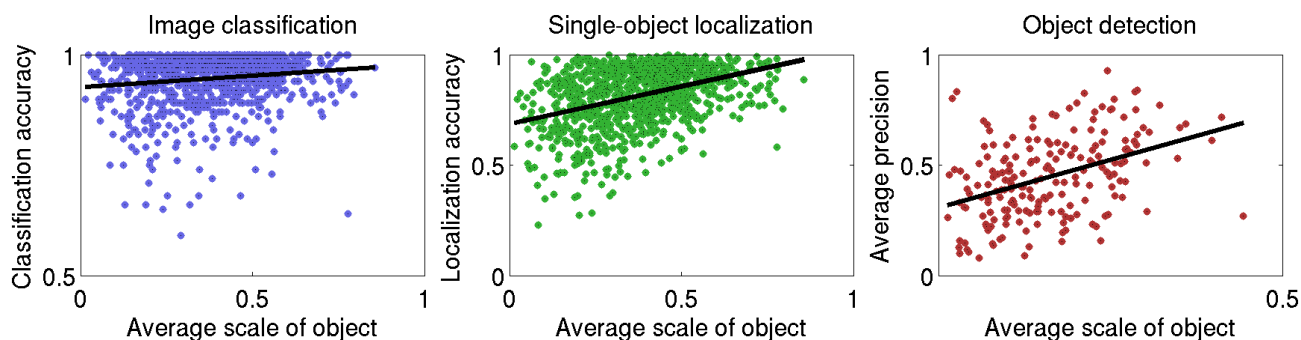


**Fig. 13** Performance of the “optimistic” method as a function of object scale in the image, on each task. Each dot corresponds to one object class. Average scale (x-axis) is computed as the average fraction of the image area occupied by an instance of that object class on the ILSVRC2014 validation set. “Optimistic” performance (y-axis) corresponds to the best performance on the test set of any entry submitted to ILSVRC2012-2014 (including entries with additional training data). The test set has remained the same over these three years. We see that accuracy tends to increase as the objects get bigger in the image. However, it is clear that far from all the variation in accuracy on these classes can be accounted for by scale alone.





**Fig. 12** 对于每个物体类别，我们选取ILSVRC2012-2014期间提交的任何参赛方案（包括使用额外训练数据的方案）中的最佳性能。基于这些“乐观”结果，我们展示了物体检测任务中最简单和最困难的类别，即结果最佳和最差的类别。括号中的数字表示平均精度。图像分类和单物体定位结果如图11所示。



**Fig. 13** “乐观”方法在各项任务中作为图像内物体尺度函数的性能表现。每个点对应一个物体类别。平均尺度（x轴）的计算方式为：该物体类别在ILSVRC2014验证集上单个实例所占图像面积的平均比例。“乐观”性能（y轴）对应ILSVRC2012-2014期间提交的任何参赛方案（含使用额外训练数据的方案）在测试集上的最佳性能。该测试集在这三年间保持不变。我们可以观察到，当图像中物体尺寸增大时，准确率往往随之提升。然而显然可见，仅凭尺度因素远不能解释这些类别在准确率上的全部差异。

The image classification and single-object localization “optimistic” models performs better on large and extra large real-world objects than on smaller ones. The “optimistic” object detection model surprisingly performs better on extra small objects than on small or medium ones.

- **Deformability within instance:** *Rigid (e.g., mug) or deformable (e.g., water snake)*

The “optimistic” model on each of the three tasks performs statistically significantly better on deformable objects compared to rigid ones. However, this effect disappears when analyzing natural objects separately from man-made objects.

- **Amount of texture:** *none (e.g. punching bag), low (e.g. horse), medium (e.g. sheep) or high (e.g. honeycomb)*

The “optimistic” model on each of the three tasks is significantly better on objects with at least low level of texture compared to untextured objects.

These and other findings are justified and discussed in detail below.

*Experimental setup.* We observed in Section 6.3.3 that objects that occupy a larger area in the image tend to be somewhat easier to recognize. To make sure that differences in object scale are not influencing results in this section, we normalize each bin by object scale. We discard object classes with the largest scales from each bin as needed until the average object scale of object classes in each bin across one property is the same (or as close as possible). For real-world size property for example, the resulting average object scale in each of the five bins is 31.6% – 31.7% in the image classification and single-object localization tasks, and 12.9% – 13.4% in the object detection task.<sup>11</sup>

Figure 14 shows the average performance of the “optimistic” model on the object classes that fall into each bin for each property. We analyze the results in detail below. Unless otherwise specified, the reported accuracies below are after the scale normalization step.

To evaluate statistical significance, we compute the 95% confidence interval for accuracy using bootstrapping: we repeatedly sample the object classes within the bin with replacement, discard some as needed to normalize by scale, and compute the average accuracy of the “optimistic” model on the remaining classes. We report the 95% confidence intervals (CI) in parentheses.

<sup>11</sup> For rigid versus deformable objects, the average scale in each bin is 34.1% – 34.2% for classification and localization, and 13.5% – 13.7% for detection. For texture, the average scale in each of the four bins is 31.1% – 31.3% for classification and localization, and 12.7% – 12.8% for detection.

*Real-world size.* In Figure 14(top, left) we observe that in the image classification task the “optimistic” model tends to perform significantly better on objects which are larger in the real-world. The classification accuracy is 93.6% – 93.9% on XS, S and M objects compared to 97.0% on L and 96.4% on XL objects. Since this is after normalizing for scale and thus can’t be explained by the objects’ size in the image, we conclude that either (1) larger real-world objects are easier for the model to recognize, or (2) larger real-world objects usually occur in images with very distinctive backgrounds.

To distinguish between the two cases we look Figure 14(top, middle). We see that in the single-object localization task, the L objects are easy to localize at 82.4% localization accuracy. XL objects, however, tend to be the hardest to localize with only 73.4% localization accuracy. We conclude that the appearance of L objects must be easier for the model to learn, while XL objects tend to appear in distinctive backgrounds. The image background make these XL classes easier for the image-level classifier, but the individual instances are difficult to accurately localize. Some examples of L objects are “killer whale,” “schooner,” and “lion,” and some examples of XL objects are “boathouse,” “mosque,” “toyshop” and “steel arch bridge.”

In Figure 14(top, right) corresponding to the object detection task, the influence of real-world object size is not as apparent. One of the key reasons is that many of the XL and L object classes of the image classification and single-object localization datasets were removed in constructing the detection dataset (Section 3.3.1) since they were not basic categories well-suited for detection. There were only 3 XL object classes remaining in the dataset (“train,” “airplane” and “bus”), and none after scale normalization. We omit them from the analysis. The average precision of XS, S, M objects (44.5%, 39.0%, and 38.5% mAP respectively) is statistically insignificant from average precision on L objects: 95% confidence interval of L objects is 37.5% – 59.5%. This may be due to the fact that there are only 6 L object classes remaining after scale normalization; all other real-world size bins have at least 18 object classes.

Finally, it is interesting that performance on XS objects of 44.5% mAP (CI 40.5% – 47.6%) is statistically significantly better than performance on S or M objects with 39.0% mAP and 38.5% mAP respectively. Some examples of XS objects are “strawberry,” “bow tie” and “rugby ball.”

*Deformability within instance.* In Figure 14(second row) it is clear that the “optimistic” model performs statistically significantly worse on rigid objects than on deformable objects. Image classification accuracy is 93.2%



图像分类和单目标定位的“乐观”模型在大型和超大型现实世界物体上的表现优于较小物体。令人惊讶的是，“乐观”目标检测模型在超小物体上的表现优于中小型物体。

– **Deformability within instance:** *Rigid (e.g., mug) or deformable (e.g., water snake)* 在这三项任务中，“乐观”模型在可变形物体上的表现均具有统计学意义上的显著优势。

与刚性物体相比的物体。然而，当将自然物体与人造物体分开分析时，这种效应就消失了。

– **Amount of texture:** *none (e.g. punching bag), low (e.g. horse), medium (e.g. sheep) or high (e.g. honeycomb)* 在三个任务中，针对至少具有低纹理水平的物体，“乐观”模型的表现显著优于无纹理物体。

这些及其他发现将在下文中得到详细论证和讨论。

*Experimental setup.* 我们在6.3.3节中观察到，图像中占据面积较大的物体往往更容易被识别。为确保物体尺度的差异不影响本节结果，我们按物体尺度对每个分组进行归一化处理。我们会根据需要舍弃每个分组中尺度最大的物体类别，直到各组中物体类别在某一属性上的平均物体尺度相同（或尽可能接近）。以真实世界尺寸属性为例，经过处理后，图像分类和单物体定位任务中五个分组的平均物体尺度分别为31.6%–31.7%，而物体检测任务中则为12.9%–13.4%。<sup>11</sup>

图14展示了“乐观”模型在每个属性对应的各个区间内物体类别的平均性能。我们将在下文中详细分析这些结果。除非另有说明，以下报告的准确率均为经过尺度归一化处理后的结果。

为了评估统计显著性，我们使用自助法计算准确率的95%置信区间：我们在箱内对物体类别进行重复抽样（有放回），根据需要舍弃部分样本以按尺度归一化，并计算“乐观”模型在剩余类别上的平均准确率。我们在括号中报告95%置信区间（CI）。

<sup>11</sup> For rigid versus deformable objects, the average scale in each bin is 34.1% – 34.2% for classification and localization, and 13.5%–13.7% for detection. For texture, the average scale in each of the four bins is 31.1% – 31.3% for classification and localization, and 12.7% – 12.8% for detection.

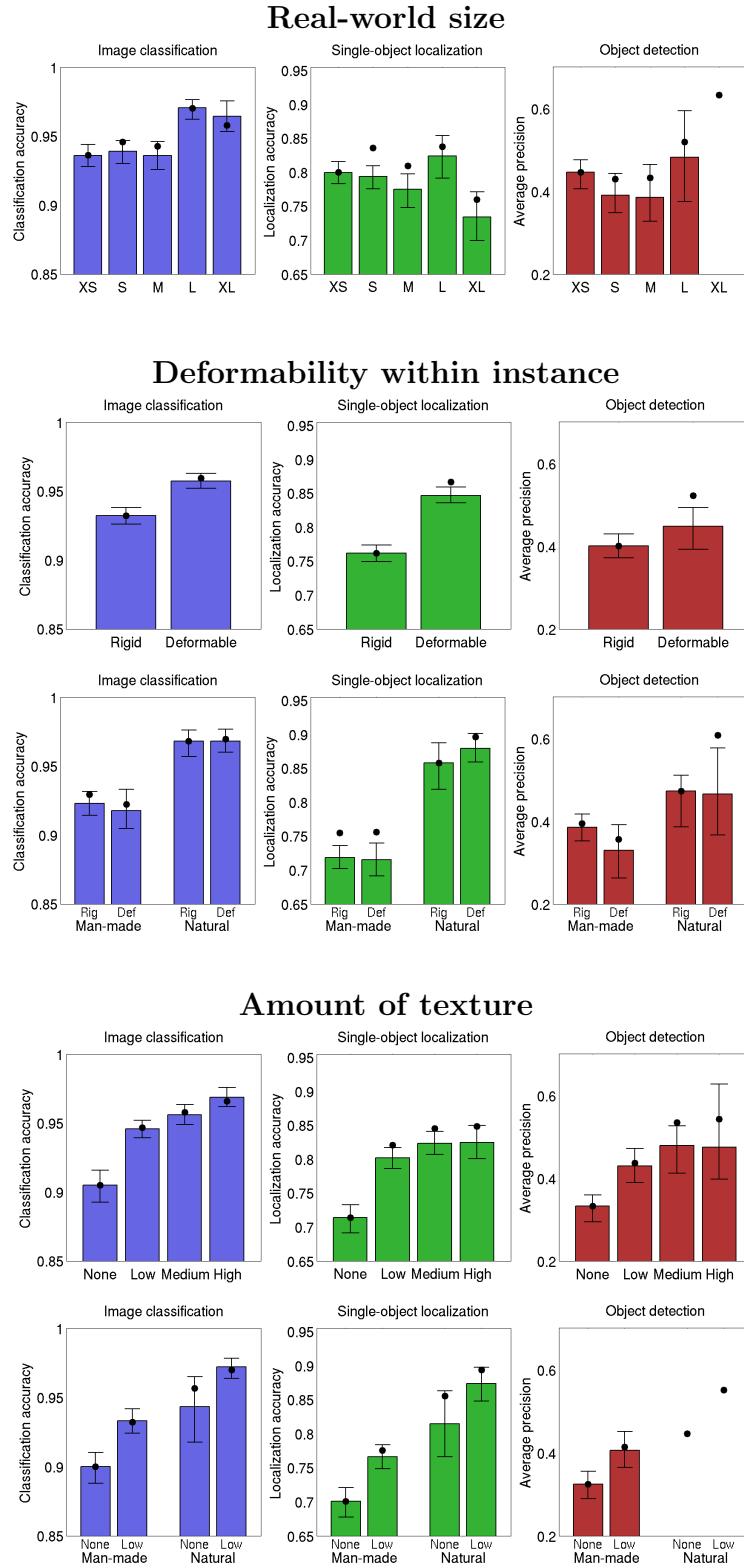
*Real-world size.* 在图14（上，左）中我们观察到，在图像分类任务中，“乐观”模型在现实世界中体积较大的物体上往往表现显著更优。对于XS、S和M尺寸的物体，分类准确率为93.6% – 93.9%，而L尺寸物体达到97.0%，XL尺寸物体则为96.4%。由于数据已进行尺度归一化处理，无法通过物体在图像中的尺寸解释这一现象，因此我们得出结论：要么（1）现实世界中较大的物体更容易被模型识别，要么（2）较大的现实世界物体通常出现在具有非常独特背景的图像中。

为了区分这两种情况，我们查看图14（上，中）。我们发现，在单目标定位任务中，L类物体易于定位，定位准确率达到82.4%。然而，XL类物体往往最难定位，定位准确率仅为73.4%。我们得出结论：模型必然更容易学习L类物体的外观特征，而XL类物体往往出现在独特的背景中。图像背景使得这些XL类别对图像级分类器而言更易识别，但单个实例却难以精确定位。L类物体的例子包括“虎鲸”、“纵帆船”和“狮子”，而XL类物体的例子则有“船库”、“清真寺”、“玩具店”和“钢拱桥”。

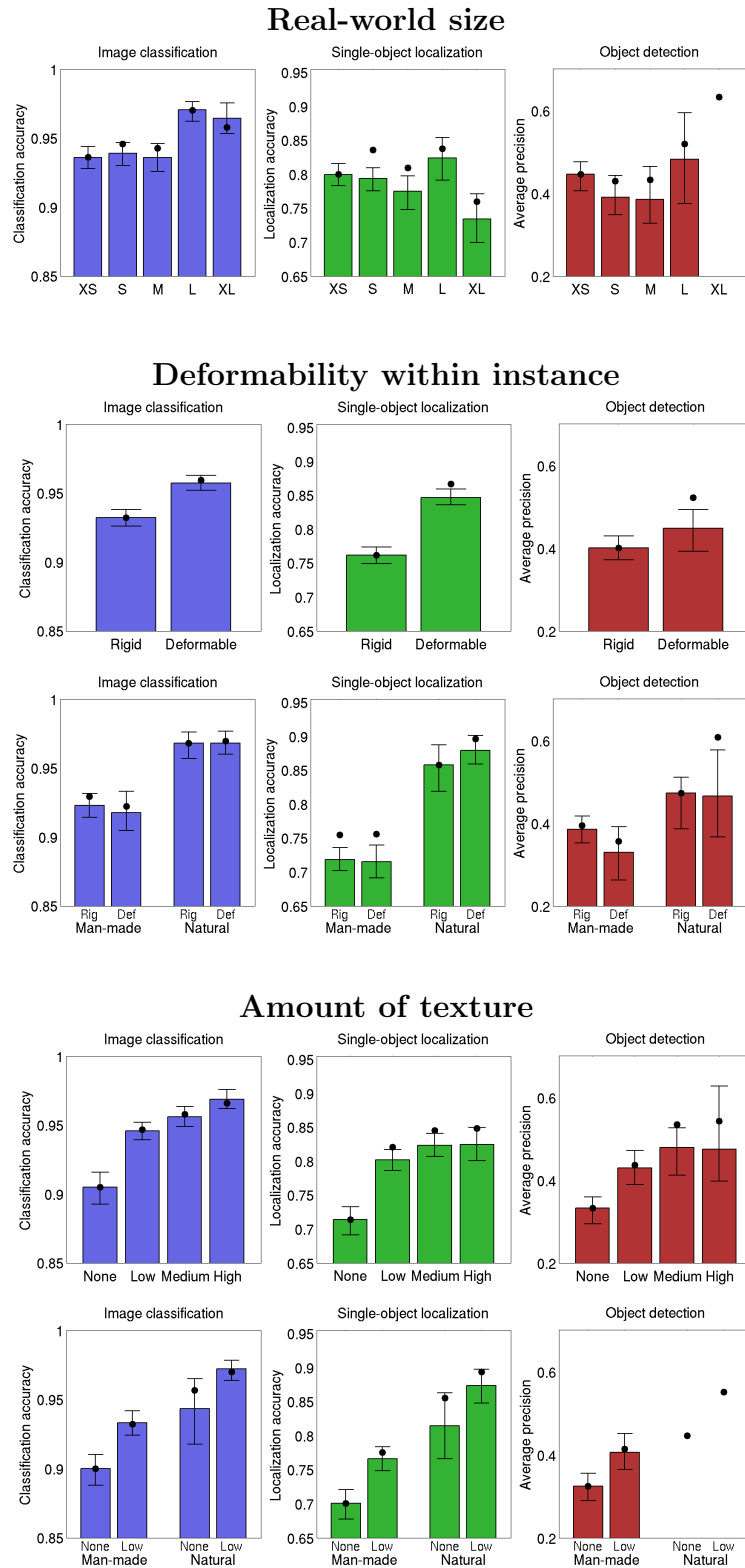
在图14（顶部，右侧）对应的物体检测任务中，真实世界物体尺寸的影响并不明显。一个关键原因在于，在构建检测数据集时（第3.3.1节），图像分类和单物体定位数据集中许多XL和L类别的物体类别被移除了，因为它们并非适合检测的基本类别。数据集中仅保留了3个XL物体类别（“火车”、“飞机”和“巴士”），且在尺度归一化后已不存在XL类别。我们将其排除在分析之外。XS、S、M物体的平均精度（分别为44.5%、39.0%和38.5% mAP）与L物体的平均精度在统计上无显著差异：L物体的95%置信区间为37.5% – 59.5%。这可能是因为尺度归一化后仅剩6个L物体类别；而其他真实世界尺寸分箱中至少包含18个物体类别。

最后，有趣的是，在XS物体上44.5%的mAP（置信区间40.5% – 47.6%）表现，在统计上显著优于S或M物体上分别为39.0% mAP和38.5% mAP的表现。XS物体的一些例子包括“草莓”、“领结”和“橄榄球”。

*Deformability within instance.* 在图14（第二行）中，可以明显看出“乐观”模型在刚性物体上的表现统计显著差于可变形物体。图像分类准确率为93.2%



**Fig. 14** Performance of the “optimistic” computer vision model as a function of object properties. The x-axis corresponds to object properties annotated by human labelers for each object class (Russakovsky et al., 2013) and illustrated in Figure 1. The y-axis is the average accuracy of the “optimistic” model. Note that the range of the y-axis is different for each task to make the trends more visible. The black circle is the average accuracy of the model on all object classes that fall into each bin. The color bars show the model accuracy averaged across the remaining classes. Error bars show the 95% confidence interval obtained with bootstrapping. Some bins are missing color bars because less than 5 object classes remained in the bin after scale normalization. For example, the bar for XL real-world object detection classes is missing because that bin has only 3 object classes (airplane, bus, train) and after normalizing by scale no classes remain.



**Fig. 14** “乐观”计算机视觉模型性能随物体属性变化的函数关系。x轴对应人工标注者为每个物体类别标注的属性（Russakovsky等人，2013年），如图1所示。y轴表示“乐观”模型的平均准确率。请注意，为使趋势更明显，每个任务的y轴范围各不相同。黑色圆点表示模型在落入每个分箱的所有物体类别上的平均准确率。我们通过对每个分箱内的物体尺度进行归一化来控制物体尺度的影响（详见第6.3.4节）。彩色条形表示模型在剩余类别上的平均准确率。误差条显示通过自助法获得的95%置信区间。部分分箱缺少彩色条形，是因为尺度归一化后该分箱中剩余的物体类别少于5个。例如，XL现实世界物体检测类别的条形缺失，是因为该分箱仅包含3个物体类别（飞机、公交车、火车），经过尺度归一化后已无剩余类别。

on rigid objects (CI 92.6% – 93.8%), much smaller than 95.7% on deformable ones. Single-object localization accuracy is 76.2% on rigid objects (CI 74.9% – 77.4%), much smaller than 84.7% on deformable ones. Object detection mAP is 40.1% on rigid objects (CI 37.2% – 42.9%), much smaller than 44.8% on deformable ones.

We can further analyze the effects of deformability after separating object classes into “natural” and “man-made” bins based on the ImageNet hierarchy. Deformability is highly correlated with whether the object is natural or man-made: 0.72 correlation for image classification and single-object localization classes, and 0.61 for object detection classes. Figure 14(third row) shows the effect of deformability on performance of the model for man-made and natural objects separately.

Man-made classes are significantly harder than natural classes: classification accuracy 92.8% (CI 92.3% – 93.3%) for man-made versus 97.0% for natural, localization accuracy 75.5% (CI 74.3% – 76.5%) for man-made versus 88.5% for natural, and detection mAP 38.7% (CI 35.6 – 41.3%) for man-made versus 50.9% for natural. However, whether the classes are rigid or deformable within this subdivision is no longer significant in most cases. For example, the image classification accuracy is 92.3% (CI 91.4% – 93.1%) on man-made rigid objects and 91.8% on man-made deformable objects – not statistically significantly different.

There are two cases where the differences in performance *are* statistically significant. First, for single-object localization, natural deformable objects are easier than natural rigid objects: localization accuracy of 87.9% (CI 85.9% – 90.1%) on natural deformable objects is higher than 85.8% on natural rigid objects – falling slightly outside the 95% confidence interval. This difference in performance is likely because deformable natural animals tend to be easier to localize than rigid natural fruit.

Second, for object detection, man-made rigid objects are easier than man-made deformable objects: 38.5% mAP (CI 35.2% – 41.7%) on man-made rigid objects is higher than 33.0% mAP on man-made deformable objects. This is because man-made rigid objects include classes like “traffic light” or “car” whereas the man-made deformable objects contain challenging classes like “plastic bag,” “swimming trunks” or “stethoscope.”

*Amount of texture.* Finally, we analyze the effect that object texture has on the accuracy of the “optimistic” model. Figure 14(fourth row) demonstrates that the model performs better as the amount of texture on the object increases. The most significant difference is between the performance on untextured objects and the performance on objects with low texture. Image clas-

sification accuracy is 90.5% on untextured objects (CI 89.3% – 91.6%), lower than 94.6% on low-textured objects. Single-object localization accuracy is 71.4% on untextured objects (CI 69.1% – 73.3%), lower than 80.2% on low-textured objects. Object detection mAP is 33.2% on untextured objects (CI 29.5% – 35.9%), lower than 42.9% on low-textured objects.

Texture is correlated with whether the object is natural or man-made, at 0.35 correlation for image classification and single-object localization, and 0.46 correlation for object detection. To determine if this is a contributing factor, in Figure 14(bottom row) we break up the object classes into natural and man-made and show the accuracy on objects with no texture versus objects with low texture. We observe that the model is still statistically significantly better on low-textured object classes than on untextured ones, both on man-made and natural object classes independently.<sup>12</sup>

## 6.4 Human accuracy on large-scale image classification

Recent improvements in state-of-the-art accuracy on the ILSVRC dataset are easier to put in perspective when compared to human-level accuracy. In this section we compare the performance of the leading large-scale image classification method with the performance of humans on this task.

To support this comparison, we developed an interface that allowed a human labeler to annotate images with up to five ILSVRC target classes. We compare human errors to those of the winning ILSRC2014 image classification model, GoogLeNet (Section 5.1). For this analysis we use a random sample of 1500 ILSVRC2012-2014 image classification test set images.

*Annotation interface.* Our web-based annotation interface consists of one test set image and a list of 1000 ILSVRC categories on the side. Each category is described by its title, such as “cowboy boot.” The categories are sorted in the topological order of the ImageNet hierarchy, which places semantically similar concepts nearby in the list. For example, all motor vehicle-related classes are arranged contiguously in the list. Every class category is additionally accompanied by a row of 13 examples images from the training set to allow for faster visual scanning. The user of the interface selects 5 categories from the list by clicking on the desired items.

<sup>12</sup> Natural object detection classes are removed from this analysis because there are only 3 and 13 natural untextured and low-textured classes respectively, and none remain after scale normalization. All other bins contain at least 9 object classes after scale normalization.

在刚性物体上（置信区间 92.6% – 93.8%），远低于可变形物体上的 95.7%。单物体定位准确率在刚性物体上为 76.2%（置信区间 74.9% – 77.4%），远低于可变形物体上的 84.7%。物体检测 mAP 在刚性物体上为 40.1%（置信区间 37.2% – 42.9%），远低于可变形物体上的 44.8%。

我们可以根据ImageNet层次结构将物体类别分为“自然”和“人造”两类，进一步分析形变能力的影响。形变能力与物体属于自然类还是人造类高度相关：图像分类和单物体定位类别的相关系数为0.72，物体检测类别的相关系数为0.61。图14（第三行）分别展示了形变能力对模型处理人造物体和自然物体时性能的影响。

人造类别的识别难度显著高于自然类别：分类准确率人造类为92.8%（置信区间92.3% – 93.3%），自然类为97.0%；定位准确率人造类为75.5%（置信区间74.3% – 76.5%），自然类为88.5%；检测mAP人造类为38.7%（置信区间35.6 – 41.3%），自然类为50.9%。然而在此细分维度下，类别属于刚性或可变形物体在多数情况下不再具有显著影响。例如，人造刚性物体的图像分类准确率为92.3%（置信区间91.4% – 93.1%），人造可变形物体为91.8%——两者在统计学上无显著差异。

在两种情况下，性能差异{v\*}具有统计学显著性。首先，在单目标定位任务中，自然可变形物体比自然刚性物体更容易定位：自然可变形物体的定位准确率为87{v\*}9%（置信区间85{v\*}9% {v\*} 90{v\*}1%），高于自然刚性物体的85{v\*}8%——该数值略超出95%置信区间。这种性能差异很可能是因为可变的自然动物通常比刚性的自然水果更容易定位。

其次，在物体检测方面，人造刚性物体比人造可变形物体更容易检测：人造刚性物体的38.5% mAP（置信区间35.2% – 41.7%）高于人造可变形物体的33.0% mAP。这是因为人造刚性物体包含“交通信号灯”或“汽车”等类别，而人造可变形物体则包含“塑料袋”、“泳裤”或“听诊器”等具有挑战性的类别。

*Amount of texture.* 最后，我们分析了物体纹理对“乐观”模型精度的影响。图14（第四行）表明，随着物体表面纹理的增加，模型表现更好。最显著的差异出现在无纹理物体与低纹理物体的性能之间。图像分类—

在无纹理物体上的分类准确率为90.5%（置信区间89.3%至91.6%），低于低纹理物体的94.6%。单物体定位准确率在无纹理物体上为71.4%（置信区间69.1%至73.3%），低于低纹理物体的80.2%。无纹理物体的目标检测mAP为33.2%（置信区间29.5%至35.9%），低于低纹理物体的42.9%。

纹理与物体是自然还是人造相关，在图像分类和单物体定位中相关性为0.35，在物体检测中相关性为0.46。为判断这是否是一个影响因素，我们在图14（底行）中将物体类别分为自然和人造，并展示了无纹理物体与低纹理物体的准确率。我们观察到，无论是在人造还是自然物体类别中，模型在低纹理物体类别上的表现仍然在统计上显著优于无纹理物体类别。<sup>12</sup>

#### 6.4 人类在大规模图像分类上的准确率

ILSVRC数据集上最先进精度的最新进展，与人类水平精度相比较时，更容易被客观看待。本节将领先的大规模图像分类方法在此任务上的表现与人类的表现进行对比。

为了支持这一比较，我们开发了一个界面，允许人工标注者为图像标注最多五个ILSVRC目标类别。我们将人类标注错误与ILSRC2014图像分类比赛的获胜模型GoogLeNet（第5.1节）的错误进行了对比。在此分析中，我们随机抽取了1500张ILSVRC2012-2014图像分类测试集的图片作为样本。

*Annotation interface.* 我们的基于网络的标注界面由一张测试集图像和侧边栏上的1000个ILSVRC类别列表组成。每个类别通过其标题进行描述，例如“牛仔靴”。这些类别按照ImageNet层次结构的拓扑顺序排序，使得语义上相似的概念在列表中相邻排列。例如，所有与机动车辆相关的类别在列表中连续排列。每个类别还附带一行来自训练集的13张示例图像，以便进行更快速的视觉扫描。界面用户通过点击所需项目从列表中选择5个类别。

<sup>12</sup> Natural object detection classes are removed from this analysis because there are only 3 and 13 natural untextured and low-textured classes respectively, and none remain after scale normalization. All other bins contain at least 9 object classes after scale normalization.



Since our interface is web-based, it allows for natural scrolling through the list, and also search by text.

*Annotation protocol.* We found the task of annotating images with one of 1000 categories to be an extremely challenging task for an untrained annotator. The most common error that an untrained annotator is susceptible to is a failure to consider a relevant class as a possible label because they are unaware of its existence.

Therefore, in evaluating the human accuracy we relied primarily on expert annotators who learned to recognize a large portion of the 1000 ILSVRC classes. During training, the annotators labeled a few hundred validation images for practice and later switched to the test set images.

#### 6.4.1 Quantitative comparison of human and computer accuracy on large-scale image classification

We report results based on experiments with two expert annotators. The first annotator (A1) trained on 500 images and annotated 1500 test images. The second annotator (A2) trained on 100 images and then annotated 258 test images. The average pace of labeling was approximately 1 image per minute, but the distribution is strongly bimodal: some images are quickly recognized, while some images (such as those of fine-grained breeds of dogs, birds, or monkeys) may require multiple minutes of concentrated effort.

The results are reported in Table 9.

*Annotator 1.* Annotator A1 evaluated a total of 1500 test set images. The GoogLeNet classification error on this sample was estimated to be 6.8% (recall that the error on full test set of 100,000 images is 6.7%, as shown in Table 7). The human error was estimated to be **5.1%**. Thus, annotator A1 achieves a performance superior to GoogLeNet, by approximately 1.7%. We can analyze the statistical significance of this result under the null hypothesis that they are from the same distribution. In particular, comparing the two proportions with a z-test yields a one-sided  $p$ -value of  $p = 0.022$ . Thus, we can conclude that this result is statistically significant at the 95% confidence level.

*Annotator 2.* Our second annotator (A2) trained on a smaller sample of only 100 images and then labeled 258 test set images. As seen in Table 9, the final classification error is significantly worse, at approximately 12.0% Top-5 error. The majority of these errors (48.8%) can be attributed to the annotator failing to spot and consider the ground truth label as an option.

| Relative Confusion                       | A1   | A2    |
|------------------------------------------|------|-------|
| Human succeeds, GoogLeNet succeeds       | 1352 | 219   |
| Human succeeds, GoogLeNet fails          | 72   | 8     |
| Human fails, GoogLeNet succeeds          | 46   | 24    |
| Human fails, GoogLeNet fails             | 30   | 7     |
| Total number of images                   | 1500 | 258   |
| Estimated GoogLeNet classification error | 6.8% | 5.8%  |
| Estimated human classification error     | 5.1% | 12.0% |

**Table 9** Human classification results on the ILSVRC2012-2014 classification test set, for two expert annotators A1 and A2. We report top-5 classification error.

Thus, we conclude that a significant amount of training time is necessary for a human to achieve competitive performance on ILSVRC. However, with a sufficient amount of training, a human annotator is still able to outperform the GoogLeNet result ( $p = 0.022$ ) by approximately 1.7%.

*Annotator comparison.* We also compare the prediction accuracy of the two annotators. Of a total of 204 images that both A1 and A2 labeled, 174 (85%) were correctly labeled by both A1 and A2, 19 (9%) were correctly labeled by A1 but not A2, 6 (3%) were correctly labeled by A2 but not A1, and 5 (2%) were incorrectly labeled by both. These include 2 images that we consider to be incorrectly labeled in the ground truth.

In particular, our results suggest that the human annotators do not exhibit strong overlap in their predictions. We can approximate the performance of an “optimistic” human classifier by assuming an image to be correct if at least one of A1 or A2 correctly labeled the image. On this sample of 204 images, we approximate the error rate of an “optimistic” human annotator at 2.4%, compared to the GoogLeNet error rate of 4.9%.

#### 6.4.2 Analysis of human and computer errors on large-scale image classification

We manually inspected both human and GoogLeNet errors to gain an understanding of common error types and how they compare. For purposes of this section, we only discuss results based on the larger sample of 1500 images that were labeled by annotator A1. Examples of representative mistakes are in Figure 15. The analysis and insights below were derived specifically from GoogLeNet predictions, but we suspect that many of the same errors may be present in other methods.

*Types of errors in both computer and human annotations:*

1. **Multiple objects.** Both GoogLeNet and humans struggle with images that contain multiple ILSVRC

由于我们的界面是基于网页的，它允许自然滚动浏览列表，并且也支持通过文本进行搜索。

*Annotation protocol.* 我们发现，对于未经训练的标注者来说，为图像标注1000个类别中的一项是极具挑战性的任务。未经训练的标注者最容易犯的错误是未能将相关类别视为可能的标签，因为他们不知道其存在。

因此，在评估人类准确度时，我们主要依赖学习了如何识别1000个ILSVRC类别中大部分类别的专家标注员。在训练期间，标注员先标注了几百张验证图像进行练习，随后转向测试集图像。

#### 6.4.1 Quantitative comparison of human and computer accuracy on large-scale image classification

我们报告了基于两位专家标注员实验的结果。第一位标注员（A1）在500张图像上进行了训练，并标注了1500张测试图像。第二位标注员（A2）在100张图像上训练后，标注了258张测试图像。平均标注速度约为每分钟1张图像，但分布呈现明显的双峰特征：部分图像可快速识别，而另一些图像（例如细分类别的狗、鸟类或猴子图像）可能需要数分钟的集中精力才能完成标注。

结果报告于表9中。

*Annotator 1.* 标注员A1共评估了1500张测试集图像。在此样本上，GoogLeNet的分类错误率估计为6.8%（回顾表7所示，在包含100,000张图像的完整测试集上错误率为6.7%）。人工错误率估计为5.1%。因此，标注员A1的表现优于GoogLeNet，幅度约为1.7%。我们可以在“两者来自同一分布”的零假设下分析这一结果的统计显著性。具体而言，通过z检验比较两个比例，得到单侧p值为 $p = 0.022$ 。因此，我们可以在95%的置信水平上判定该结果具有统计显著性。

*Annotator 2.* 我们的第二位标注员（A2）仅用100张图像的小样本进行训练，随后标注了258张测试集图像。如表9所示，其最终分类误差显著更高，Top-5错误率约为12.0%。这些错误中大部分（48.8%）可归因于标注员未能发现并考虑将真实标签作为备选选项。

| Relative Confusion                       | A1   | A2    |
|------------------------------------------|------|-------|
| Human succeeds, GoogLeNet succeeds       | 1352 | 219   |
| Human succeeds, GoogLeNet fails          | 72   | 8     |
| Human fails, GoogLeNet succeeds          | 46   | 24    |
| Human fails, GoogLeNet fails             | 30   | 7     |
| Total number of images                   | 1500 | 258   |
| Estimated GoogLeNet classification error | 6.8% | 5.8%  |
| Estimated human classification error     | 5.1% | 12.0% |

**Table 9** ILSVRC2012-2014分类测试集上的人类分类结果，针对两位专家标注者A1和A2。我们报告了前五名的分类错误率。

因此，我们得出结论，人类需要在ILSVRC上投入大量的训练时间才能达到有竞争力的表现。然而，通过充分的训练，人类标注者仍能超越GoogLeNet的结果（ $p = 0.022$ ）约1.7%。

*Annotator comparison.* 我们还比较了两位标注者的预测准确性。在A1和A2共同标注的204张图像中，174张（85%）被A1和A2同时正确标注，19张（9%）被A1正确标注但未被A2正确标注，6张（3%）被A2正确标注但未被A1正确标注，5张（2%）被两者均错误标注。其中包括2张我们认为在真实标注中也被错误标记的图像。

具体而言，我们的结果表明人类标注者的预测结果重叠度不高。我们可以通过假设只要A1或A2中至少有一方正确标注图像，即判定该图像正确，来估算“乐观”人类分类器的性能。在此204张图像的样本中，我们估算“乐观”人类标注者的错误率为2.4%，而GoogLeNet的错误率为4.9%。

#### 6.4.2 Analysis of human and computer errors on large-scale image classification

我们手动检查了人类和GoogLeNet的错误，以了解常见错误类型及其对比情况。在本节中，我们仅讨论基于标注者A1标记的1500张较大样本图像的结果。代表性错误示例如图15所示。以下分析和见解专门源自GoogLeNet的预测，但我们怀疑许多相同的错误可能也存在于其他方法中。

*Types of errors in both computer and human annotations:*

**1. Multiple objects.** GoogLeNet和人类都在处理包含多个ILSVRC的图像时遇到困难



**Fig. 15** Representative validation images that highlight common sources of error. For each image, we display the ground truth in blue, and top 5 predictions from GoogLeNet follow (red = wrong, green = right). GoogLeNet predictions on the validation set images were graciously provided by members of the GoogLeNet team. From left to right: Images that contain multiple objects, images of extreme closeups and uncharacteristic views, images with filters, images that significantly benefit from the ability to read text, images that contain very small and thin objects, images with abstract representations, and example of a fine-grained image that GoogLeNet correctly identifies but a human would have significant difficulty with.

classes (usually many more than five), with little indication of which object is the focus of the image. This error is only present in the Classification setting, since every image is constrained to have exactly one correct label. In total, we attribute 24 (24%) of GoogLeNet errors and 12 (16%) of human errors to this category. It is worth noting that humans can have a slight advantage in this error type, since it can sometimes be easy to identify the most salient object in the image.

2. **Incorrect annotations.** We found that approximately 5 out of 1500 images (0.3%) were incorrectly annotated in the ground truth. This introduces an approximately equal number of errors for both humans and GoogLeNet.

*Types of errors that the computer is more susceptible to than the human:*

1. **Object small or thin.** GoogLeNet struggles with recognizing objects that are very small or thin in the image, even if that object is the only object present. Examples of this include an image of a standing person wearing sunglasses, a person holding a quill in their hand, or a small ant on a stem of a flower. We estimate that approximately 22 (21%) of GoogLeNet errors fall into this category, while none of the human errors do. In other words, in our sample of images, no image was mislabeled by a human because they were unable to identify a very small or thin object. This discrepancy can be attributed to the fact that a human can very effectively leverage context and affordances to accurately infer the identity of small objects (for example, a few barely visible feathers near person's hand as very likely belonging to a mostly occluded quill).

2. **Image filters.** Many people enhance their photos with filters that distort the contrast and color distributions of the image. We found that 13 (13%) of the images that GoogLeNet incorrectly classified contained a filter. Thus, we posit that GoogLeNet is not very robust to these distortions. In comparison, only one image among the human errors contained a filter, but we do not attribute the source of the error to the filter.
3. **Abstract representations.** GoogLeNet struggles with images that depict objects of interest in an abstract form, such as 3D-rendered images, paintings, sketches, plush toys, or statues. An example is the abstract shape of a bow drawn with a light source in night photography, a 3D-rendered robotic scorpion, or a shadow on the ground, of a child on a swing. We attribute approximately 6 (6%) of GoogLeNet errors to this type of error and believe that humans are significantly more robust, with no such errors seen in our sample.
4. **Miscellaneous sources.** Additional sources of error that occur relatively infrequently include extreme closeups of parts of an object, unconventional viewpoints such as a rotated image, images that can significantly benefit from the ability to read text (e.g. a featureless container identifying itself as "face powder"), objects with heavy occlusions, and images that depict a collage of multiple images. In general, we found that humans are more robust to all of these types of error.

*Types of errors that the human is more susceptible to than the computer:*

1. **Fine-grained recognition.** We found that humans are noticeably worse at fine-grained recognition (e.g. dogs, monkeys, snakes, birds), even when they are



**Fig. 15** 突出常见错误来源的代表性验证图像。对于每张图像，我们以蓝色显示真实标注，随后展示GoogLeNet的前5个预测结果（红色=为错误预测，绿色=为正确预测）。验证集图像的GoogLeNet预测结果由GoogLeNet团队成员慷慨提供。从左至右分别为：包含多个物体的图像、极端特写与非典型视角图像、使用滤镜处理的图像、需借助文本识别能力才能准确判断的图像、包含微小纤细物体的图像、具有抽象表现的图像，以及GoogLeNet能正确识别但人类难以判别的细粒度图像示例。

类别（通常远多于五个），且几乎没有指示图像中哪个物体是焦点。这种错误仅出现在分类设置中，因为每张图像都被限制为恰好有一个正确标签。总体而言，我们将GoogLeNet错误中的24个（24%）以及人类错误中的12个（16%）归因于此类。值得注意的是，人类在这种错误类型上可能略有优势，因为有时识别图像中最显著的物体可能较为容易。

**2. Incorrect annotations.** 我们发现，在约1500张图像中，大约有5张（0.3%）在基准真值中被错误标注。这为人类和GoogLeNet引入了大致相等数量的错误。

*Types of errors that the computer is more susceptible to than the human:*

**1. Object small or thin.** GoogLeNet在识别图像中非常小或细长的物体时存在困难，即使该物体是图像中唯一的对象。例如，一张戴太阳镜站立的人像、手持羽毛笔的人，或是花朵茎干上的一只小蚂蚁。我们估计，大约有22个（21%）的GoogLeNet错误属于此类，而人类错误中则没有一例。换言之，在我们的图像样本中，没有一张图像因人类无法识别极小或细长物体而被错误标注。这种差异可归因于人类能够非常有效地利用上下文信息和可供性，来准确推断小物体的身份（例如，人手附近几乎不可见的几根羽毛，很可能属于一支大部分被遮挡的羽毛笔）。

**2. Image filters.** 许多人使用滤镜来增强照片效果，这些滤镜会扭曲图像的对比度和颜色分布。我们发现，在GoogLeNet错误分类的图像中，有13张（13%）使用了滤镜。因此，我们认为GoogLeNet对这些扭曲的鲁棒性不强。相比之下，在人类分类错误的图像中只有一张使用了滤镜，但我们并不认为该错误源于滤镜。

**3. Abstract representations.** GoogLeNet在处理以抽象形式描绘感兴趣对象的图像时表现不佳，例如3D渲染图像、绘画、素描、毛绒玩具或雕像。一个例子是夜间摄影中用光源绘制的蝴蝶结抽象形状、3D渲染的机器蝎子，或是秋千上孩子在地面的影子。我们将GoogLeNet约6%（6%）的错误归因于此类错误，并认为人类在此方面明显更为稳健，在我们的样本中未观察到此类错误。

**4. Miscellaneous sources.** 相对较少出现的其他误差来源包括：物体的局部极端特写、非常规视角（如图像旋转）、需依赖文字识别才能准确解读的图像（例如仅标注“散粉”的无特征容器）、严重遮挡的物体，以及呈现多图拼贴效果的图像。总体而言，我们发现人类对这些类型的误差具有更强的适应能力。

*Types of errors that the human is more susceptible to than the computer:*

**1. Fine-grained recognition.** 我们发现人类在细粒度识别（例如狗、猴子、蛇、鸟类）方面的表现明显较差，即使他们

in clear view. To understand the difficulty, consider that there are more than 120 species of dogs in the dataset. We estimate that 28 (37%) of the human errors fall into this category, while only 7 (7%) of GoogLeNet errors do.

2. **Class unawareness.** The annotator may sometimes be unaware of the ground truth class present as a label option. When pointed out as an ILSVRC class, it is usually clear that the label applies to the image. These errors get progressively less frequent as the annotator becomes more familiar with ILSVRC classes. Approximately 18 (24%) of the human errors fall into this category.
3. **Insufficient training data.** Recall that the annotator is only presented with 13 examples of a class under every category name. However, 13 images are not always enough to adequately convey the allowed class variations. For example, a brown dog can be incorrectly dismissed as a “Kelpie” if all examples of a “Kelpie” feature a dog with black coat. However, if more than 13 images were listed it would have become clear that a “Kelpie” may have brown coat. Approximately 4 (5%) of human errors fall into this category.

#### 6.4.3 Conclusions from human image classification experiments

We investigated the performance of trained human annotators on a sample of 1500 ILSVRC test set images. Our results indicate that a trained human annotator is capable of outperforming the best model (GoogLeNet) by approximately 1.7% ( $p = 0.022$ ).

We expect that some sources of error may be relatively easily eliminated (e.g. robustness to filters, rotations, collages, effectively reasoning over multiple scales), while others may prove more elusive (e.g. identifying abstract representations of objects). On the other hand, a large majority of human errors come from fine-grained categories and class unawareness. We expect that the former can be significantly reduced with fine-grained expert annotators, while the latter could be reduced with more practice and greater familiarity with ILSVRC classes. Our results also hint that human errors are not strongly correlated and that human ensembles may further reduce human error rate.

It is clear that humans will soon outperform state-of-the-art ILSVRC image classification models only by use of significant effort, expertise, and time. One interesting follow-up question for future investigation is how computer-level accuracy compares with human-level accuracy on more complex image understanding tasks.

## 7 Conclusions

In this paper we described the large-scale data collection process of ILSVRC, provided a summary of the most successful algorithms on this data, and analyzed the success and failure modes of these algorithms. In this section we discuss some of the key lessons we learned over the years of ILSVRC, strive to address the key criticisms of the datasets and the challenges we encountered over the years, and conclude by looking forward into the future.

### 7.1 Lessons learned

The key lesson of collecting the datasets and running the challenges for five years is this: **All human intelligence tasks need to be exceptionally well-designed.** We learned this lesson both when annotating the dataset using Amazon Mechanical Turk workers (Section 3) and even when trying to evaluate human-level image classification accuracy using expert labelers (Section 6.4). The first iteration of the labeling interface was always bad – generally meaning *completely unusable*. If there was any inherent ambiguity in the questions posed (and there almost always was), workers found it and accuracy suffered. If there is one piece of advice we can offer to future research, it is to very carefully design, continuously monitor, and extensively sanity-check all crowdsourcing tasks.

The other lesson, already well-known to large-scale researchers, is this: **Scaling up the dataset always reveals unexpected challenges.** From designing complicated multi-step annotation strategies (Section 3.2.1) to having to modify the evaluation procedure (Section 4), we had to continuously adjust to the large-scale setting. On the plus side, of course, the major breakthroughs in object recognition accuracy (Section 5) and the analysis of the strength and weaknesses of current algorithms as a function of object class properties (Section 6.3) would never have been possible on a smaller scale.

### 7.2 Criticism

In the past five years, we encountered three major criticisms of the ILSVRC dataset and the corresponding challenge: (1) the ILSVRC dataset is insufficiently challenging, (2) the ILSVRC dataset contains annotation errors, and (3) the rules of ILSVRC competition are too restrictive. We discuss these in order.

The first criticism is that the objects in the dataset tend to be large and centered in the images, making the dataset insufficiently challenging. In Sections 3.2.2



清晰可见。要理解其中的困难，需考虑到数据集中有超过120种犬类。我们估计，人类错误中有28个（37%）属于此类，而GoogLeNet的错误中仅有7个（7%）属于此类。

**2. Class unawareness.** 标注者有时可能未意识到某个标签选项中存在真实类别。当被指出这是ILSVRC类别时，通常能明显看出该标签适用于图像。随着标注者对ILSVRC类别越来越熟悉，这类错误会逐渐减少。大约有18例（24%）的人工标注错误属于此类。

**3. Insufficient training data.** 回想一下，标注员在每个类别名称下仅看到该类的13个示例。然而，13张图像并不总能充分传达允许的类别变体。例如，如果“凯尔皮犬”的所有示例都是黑色毛发的狗，那么一只棕色的狗可能会被错误地排除为“凯尔皮犬”。然而，如果列出超过13张图像，就会清楚表明“凯尔皮犬”可能有棕色毛发。大约有4例（5%）的人工错误属于此类。

#### 6.4.3 Conclusions from human image classification experiments

我们对1500张ILSVRC测试集图像样本进行了训练有素的人工标注者性能研究。结果表明，经过训练的人工标注者能够超越最佳模型（GoogLeNet）约1.7%（ $p = 0.022$ ）。

我们预计，部分误差来源可能相对容易消除（例如对滤镜、旋转、拼贴的鲁棒性，以及有效跨多尺度推理），而另一些可能更难解决（例如识别物体的抽象表征）。另一方面，绝大多数人类错误源于细粒度类别和类别认知不足。我们认为前者可通过细粒度专家标注显著减少，而后者可通过更多练习和更熟悉ILSVRC类别来降低。我们的结果还暗示，人类错误之间关联性不强，且人类集成方法可能进一步降低人类错误率。

显然，人类很快将仅凭大量努力、专业知识和时间就能超越最先进的ILSVRC图像分类模型。一个值得未来研究的有趣后续问题是，在更复杂的图像理解任务上，计算机级别的准确度与人类级别的准确度相比如何。

## 7 Conclusions

本文阐述了ILSVRC的大规模数据收集过程，总结了该数据集上最成功的算法，并分析了这些算法的成功与失败模式。本节将讨论我们在多年ILSVRC实践中获得的关键经验，着力回应针对该数据集的主要批评意见及我们多年来遇到的挑战，最后展望未来发展方向。

### 7.1 经验教训

收集数据集并运行挑战赛五年的关键教训是：

**All human intelligence tasks need to be exceptionally well-designed.** 这一教训既体现在使用亚马逊 Mechanical Turk 工作者标注数据集时（第3节），也体现在尝试通过专业标注者评估人类水平的图像分类准确度时（第6.4节）。标注界面的首次迭代总是存在缺陷——通常意味着 *completely unusable*。如果所提问题存在任何固有的模糊性（而这几乎总是存在），工作者便会发现它，导致准确度下降。若能为未来研究提供一条建议，那就是：必须极其谨慎地设计、持续监控并广泛进行合理性检验所有众包任务。

另一个教训，对于大规模研究者来说已经众所周知的是：**Scaling up the dataset always reveals unexpected challenges.** 从设计复杂的多步骤标注策略（第3.2.1节）到不得不修改评估流程（第4节），我们必须不断适应大规模研究的设定。当然，从积极的一面看，物体识别准确率的重大突破（第5节）以及根据物体类别属性分析当前算法的优势与不足（第6.3节），在小规模研究中是绝不可能实现的。

### 7.2 批评

在过去的五年中，我们遇到了对ILSVRC数据集及其相应挑战的三大主要批评：(1) ILSVRC数据集挑战性不足，(2) ILSVRC数据集包含标注错误，以及(3) ILSVRC竞赛规则限制过多。我们将依次讨论这些问题。

第一个批评是数据集中物体往往较大且位于图像中心，这使得数据集挑战性不足。在第3.2.2节

and 3.3.4 we tried to put those concerns to rest by analyzing the statistics of the ILSVRC dataset and concluding that it is comparable with, and in many cases much more challenging than, the long-standing PASCAL VOC benchmark (Everingham et al., 2010).

The second is regarding the errors in ground truth labeling. We went through several rounds of in-house post-processing of the annotations obtained using crowd-sourcing, and corrected many common sources of errors (e.g., Appendix E). The major remaining source of annotation errors stem from fine-grained object classes, e.g., labelers failing to distinguish different species of birds. This is a tradeoff that had to be made: in order to annotate data at this scale on a reasonable budget, we had to rely on non-expert crowd labelers. However, overall the dataset is encouragingly clean. By our estimates, 99.7% precision is achieved in the image classification dataset (Sections 3.1.3 and 6.4) and 97.9% of images that went through the bounding box annotation system have all instances of the target object class labeled with bounding boxes (Section 3.2.1).

The third criticism we encountered is over the rules of the competition regarding using external training data. In ILSVRC2010-2013, algorithms had to only use the provided training and validation set images and annotations for training their models. With the growth of the field of large-scale unsupervised feature learning, however, questions began to arise about what exactly constitutes “outside” data: for example, are image features trained on a large pool of “outside” images in an unsupervised fashion allowed in the competition? After much discussion, in ILSVRC2014 we took the first step towards addressing this problem. We followed the PASCAL VOC strategy and created two tracks in the competition: entries using only “provided” data and entries using “outside” data, meaning *any* images or annotations not provided as part of ILSVRC training or validation sets. However, in the future this strategy will likely need to be further revised as the computer vision field evolves. For example, competitions can consider allowing the use of any image features which are publicly available, even if these features were learned on an external source of data.

### 7.3 The future

Given the massive algorithmic breakthroughs over the past five years, we are very eager to see what will happen in the next five years. There are many potential directions of improvement and growth for ILSVRC and other large-scale image datasets.

First, continuing the trend of moving towards richer image understanding (from image classification to single-

object localization to object detection), the next challenge would be to tackle pixel-level object segmentation. The recently released large-scale COCO dataset (Lin et al., 2014b) is already taking a step in that direction.

Second, as datasets grow even larger in scale, it may become impossible to fully annotate them manually. The scale of ILSVRC is already imposing limits on the manual annotations that are feasible to obtain: for example, we had to restrict the number of objects labeled per image in the image classification and single-object localization datasets. In the future, with billions of images, it will become impossible to obtain even one clean label for every image. Datasets such as Yahoo’s Flickr Creative Commons 100M,<sup>13</sup> released with weak human tags but no centralized annotation, will become more common.

The growth of unlabeled or only partially labeled large-scale datasets implies two things. First, algorithms will have to rely more on weakly supervised training data. Second, even evaluation might have to be done *after* the algorithms make predictions, not before. This means that rather than evaluating *accuracy* (how many of the test images or objects did the algorithm get right) or *recall* (how many of the desired images or objects did the algorithm manage to find), both of which require a fully annotated test set, we will be focusing more on *precision*: of the predictions that the algorithm made, how many were deemed correct by humans.

We are eagerly awaiting the future development of object recognition datasets and algorithms, and are grateful that ILSVRC served as a stepping stone along this path.

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## Appendix A ILSVRC2012-2014 image classification and single-object localization object categories

abacus, abaya, academic gown, accordion, acorn, acorn squash, acoustic guitar, admiral, affenpinscher, Afghan hound, African chameleon, African crocodile,

<sup>13</sup> <http://webscope.sandbox.yahoo.com/catalog.php?datatype=i&did=67>

在3.3.4节中，我们试图通过分析ILSVRC数据集的统计数据来消除这些顾虑，并得出结论：该数据集与长期使用的PASCAL VOC基准（Everingham等人，2010年）具有可比性，且在多数情况下更具挑战性。

第二点涉及地面实况标注中的错误。我们对通过众包获得的标注进行了多轮内部后处理，并纠正了许多常见错误来源（例如附录E）。剩余的主要标注错误源于细粒度物体类别，例如标注者未能区分不同种类的鸟类。这是必须做出的权衡：为了在合理预算内实现如此大规模的数据标注，我们不得不依赖非专业的众包标注员。但总体而言，该数据集令人鼓舞地保持了高洁净度。据我们估计，图像分类数据集（章节3.1.3和6.4）达到了99.7%的精确度，而经过边界框标注系统的图像中有97.9%为目标物体类别的所有实例都标注了边界框（章节3.2.1）。

我们遇到的第三个批评是关于竞赛使用外部训练数据的规则。在ILSVRC2010-2013期间，算法只能使用提供的训练集和验证集图像及其标注来训练模型。然而，随着大规模无监督特征学习领域的发展，关于何为“外部”数据的问题开始浮现：例如，是否允许使用通过无监督方式在大量“外部”图像上训练得到的图像特征参赛？经过多次讨论，我们在ILSVRC2014中迈出了解决这一问题的第一步。我们遵循PASCAL VOC的策略，在竞赛中设立了两个赛道：仅使用“提供”数据的参赛作品，以及使用“外部”数据的参赛作品，即使用未作为ILSVRC训练集或验证集部分提供的任何图像或标注。然而，随着计算机视觉领域的发展，这一策略未来很可能需要进一步修订。例如，竞赛可以考虑允许使用任何公开可用的图像特征，即使这些特征是从外部数据源学习得到的。

### 7.3 未来

鉴于过去五年算法上的巨大突破，我们非常期待看到未来五年会发生什么。对于ILSVRC和其他大规模图像数据集来说，存在着许多潜在的改进和发展方向。

首先，延续向更丰富图像理解发展的趋势（从图像分类到单一

从物体定位到物体检测，下一个挑战将是解决像素级的物体分割。最近发布的大规模COCO数据集（Lin等人，2014b）已朝着这个方向迈出了一步。

其次，随着数据集的规模进一步扩大，可能无法完全通过人工方式进行标注。ILSVRC的规模已经对可行的人工标注设定了限制：例如，我们不得不限制图像分类和单目标定位数据集中每张图像标注的对象数量。未来，面对数十亿张图像，甚至为每张图像获取一个准确的标签都将变得不可能。像雅虎的Flickr Creative Commons 100M<sup>13</sup>这样仅带有弱人工标签、缺乏集中标注的数据集，将会变得更加普遍。

未标记或仅部分标记的大规模数据集的增长意味着两件事。首先，算法将不得不更多地依赖弱监督训练数据。其次，甚至评估可能必须在算法做出预测之后进行，而不是之前。这意味着，我们将更多地关注*precision*：在算法做出的预测中，有多少被人类认为是正确的，而不是评估*accuracy*（算法正确识别了多少测试图像或物体）或*recall*（算法成功找到了多少所需图像或物体），这两者都需要一个完全标注的测试集。

我们热切期待物体识别数据集和算法的未来发展，并感谢ILSVRC作为这条道路上的垫脚石。

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### Appendix A ILSVRC2012-2014 image classification and single-object localization object categories

算盘，阿拉伯长袍，学位袍，手风琴，椰子，橡实南瓜，原声吉他，海军上将，猴面犬，阿富汗猎犬，非洲变色龙，非洲鳄鱼。

<sup>13</sup> <http://webscope.sandbox.yahoo.com/catalog.php?datatype=i&did=67>

African elephant, African grey, African hunting dog, agama, agarie, aircraft carrier, Airedale, airliner, airship, albatross, alligator lizard, alp, altar, ambulance, American alligator, American black bear, American chameleon, American coot, American egret, American lobster, American Staffordshire terrier, amphibian, analog clock, anemone fish, Angora, ant, apiary, Appenzeller, apron, Arabian camel, Arctic fox, armadillo, artichoke, ashecan, assault rifle, Australian terrier, axolotl, baboon, backpack, badger, bagel, bakery, balance beam, bald eagle, balloon, ballerina, ballpoint, banana, Band Aid, banded gecko, banjo, bannister, barbell, barber chair, barbershop, barn, barn spider, barometer, barracouta, barrel, barrow, baseball, basenji, basketball, basset, bassinet, bassoon, bath towel, bathing cap, bathtub, beach wagon, beacon, beagle, beaker, bearskin, beaver, Bedlington terrier, bee, bee eater, beer bottle, beer glass, bell cote, bell pepper, Bernese mountain dog, bib, bicycle-built-for-two, bighorn, bikini, binder, binoculars, birdhouse, bison, bittern, black and gold garden spider, black grouse, black stork, black swan, black widow, black-and-tan coonhound, black-footed ferret, Blenheim spaniel, bloodhound, bluetick, boa constrictor, boathouse, bobsled, bolete, bolo tie, bonnet, book jacket, bookcase, bookshop, Border collie, Border terrier, borzoi, Boston bull, bottlecap, Bouvier des Flandres, bow, bow tie, box turtle, boxer, Brabancon griffon, brain coral, brambling, brass, brassiere, breakwater, breastplate, briard, Brittany spaniel, broccoli, broom, brown bear, bubble, bucket, buckeye, buckle, bulbul, bull mastiff, bullet train, bulletproof vest, bullfrog, burrito, bustard, butcher shop, butternut squash, cab, cabbage butterfly, cairn, caldron, can opener, candle, cannon, canoe, capuchin, car mirror, car wheel, carbonara, Cardigan, cardigan, carcoon, carousel, carpenter's kit, carton, cash machine, cassette, cassette player, castle, catamaran, cauliflower, CD player, cello, cellular telephone, centipede, chain, chain mail, chain saw, chain-link fence, chambered nautilus, cheeseburger, cheetah, Chesapeake Bay retriever, chest, chickadee, chiffonier, Chihuahua, chime, chimpanzee, china cabinet, chiton, chocolate sauce, chow, Christmas stocking, church, cicada, cinema, cleaver, cliff, cliff dwelling, cloak, clog, clumber, cock, cocker spaniel, cockroach, cocktail shaker, coffee mug, coffeepot, coho, coil, collie, colobus, combination lock, comic book, common iguana, common newt, computer keyboard, conch, confectionery, consommé, container ship, convertible, coral fungus, coral reef, corkscrew, corn, cornet, coucal, cougar, cowboy boot, cowboy hat, coyote, cradle, crane, crane, crash helmet, crate, crayfish, crib, cricket, Crock Pot, croquet ball, crossword puzzle, crutch, cucumber, cuirass, cup, curly-coated retriever, custard apple, daisy, dalmatian, dam, damselfly, Dandie Dinmont, desk, desktop computer, dhote, dial telephone, diamondback, diaper, digital clock, digital watch, dingo, dining table, dishrag, dishwasher, disk brake, Doberman, dock, dogsled, dome, doormat, dough, dowitcher, dragonfly, drake, drilling platform, drum, drumstick, dugong, dumbbell, dung beetle, Dungeness crab, Dutch oven, ear, earthstar, echidna, eel, eft, eggnog, Egyptian cat, electric fan, electric guitar, electric locomotive, electric ray, English foxhound, English setter, English springer, entertainment center, EntleBucher, envelope, Eskimo dog, espresso, espresso maker, European fire salamander, European gallinule, face powder, feather boa, fiddler crab, fig, file, fire engine, fire screen, fireboat, flagpole, flamingo, flat-coated retriever, flatworm, flute, fly, folding chair, football helmet, forklift, fountain, fountain pen, four-poster, fox squirrel, freight car, French bulldog, French horn, French loaf, frilled lizard, frying pan, fur coat, gar, garbage truck, garden spider, garter snake, gas pump, gasmask, gazelle, German shepherd, German short-haired pointer, geyser, giant panda, giant schnauzer, gibbon, Gila monster, go-kart, goblet, golden retriever, goldfinch, goldfish, golf ball, golfcart, gondola, gong, goose, Gordon setter, gorilla, gown, grand piano, Granny Smith, grasshopper, Great Dane, great grey owl, Great Pyrenees, great white shark, Greater Swiss Mountain dog, green lizard, green mamba, green snake, greenhouse, grey fox, grey whale, greyl, grocery store, groenendael, groom, ground beetle, guacamole, guenon, guillotine, guinea pig, gyromitra, hair slide, hair spray, hair track, hammer, hammerhead, hamper, hamster, hand bowler, handheld computer, handkerchief, hard disc, hare, harmonica, harp, hartebeest, harvester, harvestman, hatchet, hay, head cabbage, hen, hen-of-the-woods, hermit crab, hip, hippopotamus, hog, hognose snake, holster, home theater, honeycomb, hook, hoopskirt, horizontal bar, hornbill, horned viper, horse cart, hot pot, hot dog, hourglass, house finch, howler monkey, hummingbird, hyena, ibex, ibizan hound, ice bear, ice cream, ice lolly, impala, Indian cobra, Indian elephant, indigo bunting, indri, iPod, Irish setter, Irish terrier, Irish water spaniel, Irish wolfhound, iron, isopod, Italian greyhound, jacamar, jack-o'-lantern, jackfruit, jaguar, Japanese spaniel, jay, jean, jeep, jellyfish, jersey, jigsaw puzzle, jinrikisha, joystick, junco, keeshond, kelpie, Kerry blue terrier, killer whale, kimono, king crab, king penguin, king snake, kit fox, kite, knee pad, knot, koala, Komodo dragon, komondor, kuvasz, lab coat, Labrador retriever, lacewing, ladle, ladybug, Lakeland terrier, lakeside, lampshade, langur, laptop, lawn mower, leaf beetle, leafhopper, leatherback turtle, lemon, lens cap, Leonberg, leopard, lesser panda, letter opener, Lhasa, library, lifeboat, lighter, limousine, limpkin, liner, lion, lionfish, lipstick, little blue heron, llama, Loafer, loggerhead, long-horned beetle, lorikeet, lotion, loudspeaker, loupe, lumbermill, lycaenid, lynx, macaque, macaw, Madagascar cat, magnetic compass, magpie, mailbox, mailbag, mail-lot, mailot, malamute, malinois, Maltese dog, manhole cover, mantis, maraca, marimba, marmoset, marmot, mashed potato, mask, matchstick, maypole, maze, measuring cup, meat loaf, medicine chest, meerkat, megalith, menu, Mexican hairless, microphone, microwave, military uniform, milk can, miniature pinscher, miniature poodle, miniature schnauzer, minibus, miniskirt, minivan, mink, missile, mitten, mixing bowl, mobile home, Model T, modem, monarch, monastery, mongoose, monitor, moped, mortar, mortarboard, mosque, mosquito net, motor scooter, mountain bike, mountain tent, mouse, mousetrap, moving van, mud turtle, mushroom, muzzle, nail, neck brace, necklace, nematode, Newfoundland, night snake, nipple, Norfolk terrier, Norwegian elkhound, Norwich terrier, notebook, obelisk, oboe, ocarina, odometer, oil filter, Old English sheepdog, orange, orangutan, organ, oscilloscope, ostrich, otter, otterhound, overskirt, ox, oxcart, oxygen mask, oystercatcher, packet, paddle, paddlewheel, padlock, paintbrush, pajama, palace, panpipe, paper towel, papillon, parachute, parallel bars, park bench, parking meter, partridge, passenger car, patas, patio, pay-phone, peacock, pedestal, Pekinese, pelican, Pembroke, pencil box, pencil sharpener, perfume, Persian cat, Petri dish, photocopy, pick, picklehaube, picket fence, pickup, pier, piggy bank, pill bottle, pillow, pineapple, ping-pong ball, pinwheel, pirate, pitcher, pizza, plane, planetarium, plastic bag, plate, plate rack, platypus, plow, plunger, Polaroid camera, pole, polecat, police van, pomegranate, Pomeranian, poncho, pool table, pot bottle, porcupine, pot, potpie, potter's wheel, power drill, prairie chicken, prayer rug, pretzel, printer, prison, proboscis monkey, projectile, projector, promontory, ptarmigan, puck, puffer, pug, punching bag, purse, quail, quill, quilt, racer, racket, radiator, radio, radio telescope, rain barrel, ram, rapeseed, recreational vehicle, red fox, red wine, red wolf, red-backed sandpiper, red-breasted merganser, redbone, redshank, reel, reflex camera, refrigerator, remote control, restaurant, revolver, rhinoceros beetle, Rhodesian ridgeback, rifle, ringlet, ringneck snake, robin, rock beauty, rock crab, rock python, rocking chair, rottiesser, Rottweiler, rubber eraser, ruddy turnstone, ruffed grouse, rugby ball, rule, running shoe, safe, safety pin, Saint Bernard, saltshaker, Saluki, Samoyed, sandal, sandbar, sarong, sax, scabbard, scale, shipperke, school bus, schooner, scoreboard, scorpion, Scotch terrier, Scottish deerhound, screen, screw, screwdriver, scuba diver, sea anemone, sea cucumber, sea lion, sea slug, sea snake, sea urchin, Sealyham terrier, seashore, seat belt, sewing

machine, Shetland sheepdog, shield, Shih-Tzu, shoe shop, shoji, shopping basket, shopping cart, shovel, shower cap, shower curtain, siamang, Siamese cat, Siberian husky, sidewinder, silky terrier, ski, ski mask, skunk, sleeping bag, slide rule, sliding door, slot, sloth bear, slug, snail, snorkel, snow leopard, snowmobile, snowplow, soap dispenser, soccer ball, sock, soft-coated wheaten terrier, solar dish, sombrero, sorrel, soup bowl, space bar, space heater, space shuttle, spaghetti squash, spatula, speedboat, spider monkey, spider web, spindle, spiny lobster, spoonbill, sports car, spotlight, spotted salamander, squirrel monkey, Staffordshire bullterrier, stage, standard poodle, standard schnauzer, starfish, steam locomotive, steel arch bridge, steel drum, stethoscope, stingray, stinkhorn, stole, stone wall, stopwatch, stove, strainer, strawberry, street sign, streetcar, stretcher, studio couch, stupa, sturgeon, submarine, suit, sulphur butterfly, sulphur-crested cockatoo, sundial, sunglass, sunglasses, sunscreen, suspension bridge, Sussex spaniel, swab, sweatshirt, swimming trunks, swing, switch, syringe, tabby, table lamp, tailed frog, tank, tape player, tarantula, teapot, teddy, television, tench, tennis ball, terrapin, thatch, theater curtain, thimble, three-toed sloth, thresher, throne, thunder snake, Tibetan mastiff, Tibetan terrier, tick, tiger, tiger beetle, tiger cat, tiger shark, tile roof, timber wolf, titi, toaster, tobacco shop, toilet seat, toilet tissue, torch, totem pole, toucan, tow truck, toy poodle, toy terrier, toyshop, tractor, traffic light, trailer truck, tray, tree frog, trench coat, triceratops, tricycle, trifle, trilobite, trimaran, tripod, triumphal arch, trolleybus, trombone, tub, turnstile, tusker, typewriter keyboard, umbrella, unicycle, upright, vacuum, valley, vase, vault, velvet, vending machine, vestment, viaduct, vine snake, violin, vizsla, volcano, volleyball, vulture, waffle iron, Walker hound, walking stick, wall clock, wallaby, wallet, wardrobe, warplane, warthog, washbasin, washer, water bottle, water buffalo, water jug, water ouzel, water snake, water tower, weasel, web site, weevil, Weimaraner, Welsh springer spaniel, West Highland white terrier, whippet, whiptail, whiskey jug, whistle, white stork, white wolf, wig, wild boar, window screen, window shade, Windsor tie, wine bottle, wing, wire-haired fox terrier, wok, wolf spider, wombat, wood rabbit, wooden spoon, wool, worm fence, wreck, yawl, yellow lady's slipper, Yorkshire terrier, yurt, zebra, zucchini

## Appendix B Additional single-object localization dataset statistics

We consider two additional metrics of object localization difficulty: chance performance of localization and the level of clutter. We use these metrics to compare ILSVRC2012-2014 single-object localization dataset to the PASCAL VOC 2012 object detection benchmark. The measures of localization difficulty are computed on the validation set of both datasets. According to both of these measures of difficulty there is a subset of ILSVRC which is as challenging as PASCAL but more than an order of magnitude greater in size. Figure 16 shows the distributions of different properties (object scale, chance performance of localization and level of clutter) across the different classes in the two datasets.

*Chance performance of localization (CPL).* Chance performance on a dataset is a common metric to consider. We define the CPL measure as the expected accuracy of a detector which first randomly samples an object instance of that class and then uses its bounding box directly as the proposed localization window on all other images (after rescaling the images to the same size). Concretely, let  $B_1, B_2, \dots, B_N$  be all the bounding boxes of the object instances within a class, then

$$CPL = \frac{\sum_i \sum_{j \neq i} IOU(B_i, B_j) \geq 0.5}{N(N-1)} \quad (6)$$

Some of the most difficult ILSVRC categories to localize according to this metric are basketball, swimming trunks, ping pong ball and rubber eraser, all with less than 0.2% CPL. This measure correlates strongly ( $\rho = 0.9$ ) with the average scale of the object (fraction of image occupied by object). The average CPL across

非洲象, 非洲灰鹦鹉, 非洲野犬, 鬣蜥, 伞菌, 航空母舰, 艾尔谷犬, 客机, 飞艇, 信天翁, 鳄鱼, 高山, 祭坛, 救护车, 美洲短吻鳄, 美洲黑熊, 美洲变色龙, 美洲白背顶, 美洲白鹭, 美洲龙虾, 美国斯塔福德梗, 两栖动物, 模拟时钟, 海藻鱼, 安哥拉兔, 蚂蚁, 养蜂场, 阿彭策尔犬, 围裙, 阿拉伯骆驼, 北极狐, 狄翁, 洋刺, 垃圾桶, 突击步枪, 澳洲梗, 美国纯口獭, 佛佛, 青包, 獾, 百吉饼, 面包包, 平衡木, 白头海雕, 气球, 棒球运动员, 圆珠笔, 香蕉, 创可贴, 带纹壁虎, 班卓琴, 楼梯扶手, 杠铃, 理发椅, 理发店, 谷仓, 谷仓蜘蛛, 气压计, 南乳鱼, 桶, 手推车, 棒球, 巴仙吉犬, 篮球, 巴吉度犬, 摇篮, 巴松管, 浴巾, 泳帽, 浴缸, 旅行车, 灯塔, 比格犬, 烧杯, 熊皮, 河狸, 贝灵顿梗, 蜜蜂, 蜂虎, 啤酒瓶, 啤酒杯, 钟楼, 甜椒, 伯恩山犬, 围嘴, 双人自行车, 大角羊, 比基尼, 活页夹, 双筒望远镜, 鸟屋, 野牛, 麻鸭, 黑黄圆蛛, 黑琴鸟, 黑熊, 黑天鹅, 黑寡妇蜘蛛, 黑褐猎浣熊犬, 黑足猫, 布伦海姆西班牙猎犬, 寻血猎犬, 蓝斑猎犬, 蟒蛇, 船库, 雪橇, 牛肝菌, 波洛领带, 女帽, 书皮, 书架, 书店, 边境牧羊犬, 边境梗, 俄罗斯猎狼犬, 波士顿梗犬, 瓶塞, 法兰德斯牧羊犬, 弓, 领结, 箱龟, 拳师犬, 布鲁塞尔格里芬犬, 脑珊瑚, 燕雀, 黄铜器, 胸罩, 防波堤, 胸甲, 布里牧羊犬, 布列塔尼猎犬, 西兰花, 扫帚, 棕熊, 气泡, 水桶, 七叶树果实, 扣环, 鸭, 斗牛犬, 子弹头列车, 防弹背心, 牛蛙, 墨西哥卷饼, 鸭, 肉铺, 冬瓜瓜, 出租车, 菜粉蝶, 石象, 大锅, 开罐器, 蜡烛, 大炮, 独木舟, 卷尾猴, 汽车后视镜, 汽车车轮, 培根蛋卷意大利面, 卡迪根犬, 开衫, 刺包菜蓟, 旋转木马, 木工工具箱, 纸箱, 取款机, 磁带, 磁带播放器, 城堡, 双体船, 花椰菜, CD播放器, 大提琴, 手机, 蜈蚣, 链条, 锁子甲, 链锯, 链环栅栏, 鹦鹉螺, 芝士汉堡, 猎豹, 切萨皮克湾寻回犬, 胸部, 山雀, 五斗柜, 吉娃娃, 编织, 黑猩猩, 瓷器柜, 石凳, 巧克力酱, 松狮犬, 圣诞袜, 花椰菜, 蚌, 电影院, 切肉刀, 悬崖, 厨房, 斗篷, 木底鞋, 克伦伯猎犬, 公鸡, 可卡犬, 蜂蝎, 鸡尾酒调制器, 咖啡杯, 咖啡壶, 银链, 线圈, 柯利犬, 疣猴, 密码锁, 漫画书, 普通蜥蜴, 普通螺蛳, 电脑键盘, 海螺, 糖果店, 清汤, 集装箱船, 敞篷车, 珊瑚礁, 珊瑚礁, 开瓶器, 玉米, 短号, 鸚鵡, 美洲狮, 牛仔靴, 牛仔帽, 郊狼, 摇篮, 鹤, 起重机, 防撞头盖, 板条箱, 小龙虾, 婴儿床, 蟋蟀, 慢炖锅, 髓球, 填字游戏, 拐杖, 黄瓜, 胸甲, 杯子, 卷毛寻回犬, 番荔枝, 雏菊, 斑点狗, 水坝, 豆娘, 丹迪丁蒙梗, 书桌, 台式电脑, 针, 拨盘电话, 菱背响尾蛇, 尿布, 数字钟, 电子表, 澳洲野犬, 餐桌, 抹布, 洗碗机, 盘式制动器, 杜宾犬, 码头, 狗拉雪橇, 圆顶, 门垫, 面团, 半蹼鹬, 蜻蜓, 公鸭, 钻井平台, 鼓, 鼓槌, 儒艮, 哑铃, 蜥蜴, 珍宝蟹, 荷兰烤猪, 耳朵, 地星, 针鼹, 鳗鱼, 水蛭, 蛋酒, 埃及猫, 电风扇, 电吉他, 电力机车, 电脑, 英国猎狐犬, 英国塞特犬, 英国史宾格犬, 娱乐中心, 恩斯特赫山地犬, 信封, 爱斯基摩犬, 意式浓缩咖啡, 意式咖啡机, 欧洲火蜥蜴, 欧洲水鸡, 散粉, 羽毛围巾, 招潮蟹, 无花果, 文件, 消防车, 防火屏, 消防船, 旗杆, 火烈鸟, 平毛寻回犬, 扇形伞, 长笛, 苍鹭, 折叠椅, 橄榄球头盔, 叉车, 喷泵, 铜像, 四柱床, 狐松鼠, 货车车厢, 法国斗牛犬, 法国号, 法式长棍面包, 皱褶蜥, 煎锅, 皮大衣, 雀巢, 垃圾车, 园蛛, 束带蛇, 加油站, 防毒面具, 瞪羚, 德国牧羊犬, 德国短毛指示犬, 国歌泉, 大熊猫, 巨型雪纳瑞, 长臂猿, 毒蛇, 卡丁车, 高脚杯, 金毛寻回犬, 金翅雀, 金鱼, 高尔夫球, 高尔夫球车, 多拉拉, 锣, 鹅, 文登塞特犬, 大猩猩, 礼服, 三角钢琴, 青苹果, 蜻蜓, 大丹犬, 马林鸭, 大白熊犬, 大白鲨, 大瑞士山地犬, 绿蜥蜴, 绿曼巴蛇, 绿蛇, 温室, 灰狐, 灰熊, 格棚, 杂货店, 比利时特伏丹犬, 新郎, 步甲, 鸚鵡属, 长尾猴, 断头台, 豚鼠, 鹿花菌, 发夹, 发胶, 半履带车, 锤子, 双髻鲨, 洗衣篮, 仓鼠, 手摇吹风机, 掌上电脑, 手帕, 硬盘, 野兔, 口琴, 竖琴, 魔铃, 收割机, 盲蛛, 短柄小斧, 干草, 卷心菜, 母鸡, 灰树花, 寄居蟹, 臀部, 河马, 猪, 猪鼻蛇, 手枪皮套, 家庭影院, 蜂巢, 钩子, 裙撑, 单杠, 犀鸟, 角蛙, 马车, 火锅, 热狗, 沙漏, 家朱雀, 吼猴, 蜂鸟, 鬣狗, 北山羊, 伊比沙猎犬, 北极熊, 冰淇淋, 冰棒, 黑斑羚, 印度眼镜蛇, 印度象, 靛蓝彩鸦, 大狐猴, iPod, 爱尔兰塞特犬, 爱尔兰水猎犬, 爱尔兰猎狼犬, 鬃犬, 等足目动物, 意大利灵猫, 鸛属, 杰克灯, 菠萝蜜, 美洲豹, 日本种, 松鸦, 牛仔帽, 吉普车, 水母, 运动衫, 拼图, 人力车, 操纵杆, 灯罩, 荷兰毛狮犬, 凯尔皮犬, 凯利蓝梗, 虎鲸, 和服, 帝王蟹, 帝企鹅, 王蛇, 小狐, 风琴, 护膝, 绳结, 考拉, 科莫多巨蜥, 可蒙犬, 库瓦兹犬, 实验服, 拉布拉多寻回犬, 草蛉, 长柄勺, 瓢虫, 湖畔梗, 湖边, 灯罩, 叶猴, 笔记本电脑, 割草机, 叶甲, 叶蝉, 棱皮龟, 柠檬, 镜头盖, 莱昂伯格犬, 豹, 小熊猫, 拆信刀, 拉萨犬, 图书馆, 救生艇, 打火机, 豪华轿车, 秧鸭, 邮轮, 狮子, 狮子鱼, 口红, 小蓝鹭, 羊驼, 乐福鞋, 红海龟, 天平, 吸蜜鹦鹉, 乳液, 扬声器, 放大镜, 锯木厂, 灰蝶, 猓猴, 猕猴桃, 金剛鸚鵡, 马达加斯加猫, 磁罗盘, 喜鹊, 邮袋, 邮箱, 连体泳衣, 连体衣, 阿拉斯加雪橇犬, 马利诺斯犬, 马尔济斯犬, 并盖, 螳螂, 沙球, 马林巴琴, 猴猴, 土拨鼠, 土豆泥, 面具, 火柴, 五月柱, 迷宮, 量杯, 肉糕, 医药箱, 猫鼬, 巨石, 菜单, 墨西哥无毛犬, 麦克风, 微波炉, 军装, 牛奶罐, 迷你杜宾犬, 迷你贵宾犬, 迷你雪纳瑞, 小型巴士, 迷你裙, 小型货车, 水貂, 导弹, 手指手套, 搅拌碗, 活动房屋, T型车, 调制解调器, 黑脉金斑蝶, 修道院, 猫鼬, 显示器, 轻便摩托车, 迫击炮, mortarboard, 清真寺, 蚊帐, 踏板车, 山地自行车, 山地帐篷, 老鼠, 捕鼠器, 搬家货车, 泥龟, 蘑菇, 口套, 钉子, 颈托, 项链, 线虫, 纽芬兰犬, 夜蛇, 乳头, 诺福克梗, 挪威猎鹿犬, 诺里奇梗, 笔记本, 方尖碑, 双簧管, 陶笛, 里程表, 机油滤清器, 古代英国牧羊犬, 橙子, 猩猩, 管风琴, 示波器, 鸵鸟, 水獭, 水獭猎犬, 罩裙, 牛, 牛车, 氧气面罩, 蜥蜴, 小包, 单轮, 挂锁, 画笔, 睡衣, 宫殿, 排箫, 纸巾, 蝴蝶伞, 降落伞, 双杠, 公园长椅, 停车计时器, 鸚鵡, 客车, 赤猴, 露台, 付费电话, 孔雀, 基座, 北京犬, 鸛属, 彭布罗克威尔士柯基犬, 铅笔盒, 卷笔刀, 香水, 波斯猫, 培养皿, 复印机, 锅, 尖顶盔, 尖桩篱栅, 皮卡车, 码头, 存钱罐, 药瓶, 枕头, 菠萝, 乒乓球, 纸风车, 海盜, 水罐, 披萨, 飞机, 天文望远镜, 塑料袋, 盘子, 盘架, 鸭嘴兽, 犁, 皮鞭子, 宝丽来相机, 杆, 臭鼬, 警车, 石榴, 博美犬, 披风, 台球桌, 汽水瓶, 豪猪, 锅, potpie, 陶轮, 电钻, 草原松鸦, 祈祷毯, 椒盐卷饼, 打印机, 监狱, 长鼻猴, 抛射体, 投影仪, 海角, 雷鸟, 冰球, 河豚, 巴哥犬, 沙袋, 钱包, 鸛鸟, 羽茎, 被子, 赛车, 球拍, 散热器, 收音机, 射电望远镜, 雨水桶, 公羊, 油菜籽, 房车, 赤狐, 红葡萄酒, 红狼, 红背滨鹬, 红胸秋沙鸭, 红骨猎浣熊犬, 红脚鹬, 卷轴, 反光照相机, 冰箱, 遥控器, 餐厅, 左轮手枪, 犀龟, 罗德里亚脊背犬, 步枪, 小环, 环颈蛇, 知更鸟, 岩 beauty, 岩蜥, 岩蟒, 摇椅, 旋转烤肉架, 罗威纳犬, 橡皮擦, 翻石鹬, 披肩榛鸡, 橄榄球, 尺子, 跑鞋, 保险箱, 安全别针, 圣伯纳犬, 盐瓶, 萨路基猎犬, 萨摩耶犬, 凉鞋, 沙洲, 沙龙, 萨克斯管, 剑鞘, 秤, 比利时史基伯犬, 校车, 纵帆船, 记分牌, 蝎子, 苏格兰梗, 苏格兰猎鹿犬, 屏幕, 螺丝, 螺丝刀, 潜水员, 海藻, 海参, 海狮, 海蛞蝓, 海蛇, 海胆, 西里汉梗, 海滨, 安全带, 缝纫

机器, 设得兰牧羊犬, 盾牌, 西施犬, 鞋店, 障子, 购物篮, 购物车, 铲子, 浴帽, 浴巾, 合趾猴, 暹罗猫, 西伯利亚哈士奇, 响尾蛇, 丝毛梗, 滑雪板, 滑雪面罩, 臭鼬, 睡袋, 计算尺, 推拉门, 插槽, 懒熊, 蛞蝓, 蜗牛, 呼吸器, 雪豹, 雪地摩托, 扫雪机, 皂液器, 足球, 袜子, 软毛麦色梗, 太阳能碟, 墨西哥宽边帽, 栗色马, 汤碗, 空格键, 取暖器, 航天飞机, 意大利面南瓜, 抹布, 快艇, 蜘蛛猴, 蜘蛛网, 纺锤, 刺龙虾, 琵琶, 跑车, 聚光灯, 斑点蛛蝎, 松鼠猴, 斯塔福德斗牛梗, 舞台, 标准贵宾犬, 标准雪纳瑞, 海星, 蒸汽机车, 钢拱桥, 钢鼓, 听诊器, 黄貂鱼, 鬼笔菌, 披肩, 石墙, 砂表, 炉子, 遮网, 草莓, 路牌, 有轨电车, 担架, 沙发床, 佛塔, 鲟鱼, 潜艇, 西装, 硫磺蝶, 葵花凤头鹦鹉, 日晷, 太阳镜, 太阳镜, 防晒霜, 悬索桥, 苏塞克斯猎犬, 棉签, 运动衫, 游泳裤, 秋千, 开关, 注射器, 虎斑猫, 台灯, 尾蛙, 坦克, 磁带播放器, 狼蛛, 茶壶, 秦帝熊, 电视, 丁鲷, 网球, 水龟, 茅草, 剧院幕布, 顶针, 三趾树懒, 长尾鲨, 王座, 雷蛇, 藏獒, 西藏梗, 蟬虫, 老虎, 虎甲虫, 虎猫, 虎鲨, 瓦屋顶, 森林狼, 伶猴, 烤面包机, 烟草店, 马桶座圈, 卫生纸, 火炬, 图腾柱, 巨嘴鸟, 拖车, 玩具贵宾犬, 玩具梗, 玩具店, 拖拉机, 交通灯, 拖挂卡车, 托盘, 树蛙, 风衣, 三角龙, 三轮车, 松糕, 三叶虫, 三体船, 三脚架, 凯旋门, 无轨电车, 长号, 浴缸, 旋转栅门, 长牙象, 打字机键盘, 雨伞, 独轮车, 立式钢琴, 吸尘器, 山谷, 花瓶, 拱顶, 天鹅绒, 自动售货机, 法衣, 高架桥, 藤蛇, 小提琴, 维兹拉犬, 火山, 排球, 秃鹰, 华夫饼熨斗, 步行猎犬, 手杖, 挂件, 小袋鼠, 钱包, 衣柜, 军用飞机, 疣猪, 洗脸盆, 洗衣机, 水瓶, 水牛, 水罐, 河马, 水蛇, 水塔, 黄鼠狼, 网站, 魏玛犬, 威尔士史宾格犬, 西部高地白梗, 惠比特犬, 鞭尾蜥, 威士忌酒壶, 哨子, 白鹳, 白狼, 假发, 野猪, 纱窗, 百叶窗, 温莎领带, 酒瓶, 翅膀, 刚毛猎狐梗, 炒锅, 狼蛛, 袋熊, 林兔, 木勺, 羊毛, 虫茧, 残骸, 小帆船, 黄灼犬, 约克夏梗, 蒙古包, 斑马, 西葫芦

## Appendix B Additional single-object localization dataset statistics

我们考虑了物体定位难度的两个额外指标：定位的随机表现和杂波水平。我们使用这些指标来比较ILSVRC 2012-2014单物体定位数据集与PASCAL VOC 2012物体检测基准。定位难度的度量是在两个数据集的验证集上计算的。根据这两种难度度量，ILSVRC中存在一个子集，其挑战性与PASCAL相当，但规模要大一个数量级以上。图16展示了两数据集中不同类别间各种属性（物体尺度、定位的随机表现和杂波水平）的分布情况。

*Chance performance of localization (CPL)*. 在数据集上的随机表现是一个常见的考量指标。我们将CPL度量定义为检测器的期望准确率，该检测器首先随机采样该类的一个物体实例，然后直接将其边界框用作所有其他图像上的定位窗口（在将图像调整到相同大小之后）。具体来说，令 $\{v^*\}$ 为某个类别内所有物体实例的边界框，则

$$\text{CPL} = \frac{\sum_i \sum_{j \neq i} \text{IOU}(B_i, B_j) \geq 0.5}{N(N-1)} \quad (6)$$

根据该指标，ILSVRC中最难定位的类别包括篮球、泳裤、乒乓球和橡皮擦，它们的CPL均低于0.2%。这一指标与物体的平均尺度（物体在图像中所占比例）高度相关（ $\rho = 0.9$ ）。整体平均CPL为



the 1000 ILSVRC categories is 20.8%. The 20 PASCAL categories have an average CPL of 8.7%, which is the same as the CPL of the 562 most difficult categories of ILSVRC.

*Clutter.* Intuitively, even small objects are easy to localize on a plain background. To quantify clutter we employ the objectness measure of (Alexe et al., 2012), which is a class-generic object detector evaluating how likely a window in the image contains a coherent object (of any class) as opposed to background (sky, water, grass). For every image  $m$  containing target object instances at positions  $B_1^m, B_2^m, \dots$ , we use the publicly available objectness software to sample 1000 windows  $W_1^m, W_2^m, \dots, W_{1000}^m$ , in order of decreasing probability of the window containing any generic object. Let  $\text{OBJ}(m)$  be the number of generic object-looking windows sampled before localizing an instance of the target category, i.e.,  $\text{OBJ}(m) = \min\{k : \max_i \text{IOU}(W_k^m, B_i^m) \geq 0.5\}$ . For a category containing  $M$  images, we compute the average number of such windows per image and define

$$\text{CLUTTER} = \log_2\left(\frac{1}{M} \sum_m \text{OBJ}(m)\right) \quad (7)$$

The higher the clutter of a category, the harder the objects are to localize according to generic cues. If an object can't be localized with the first 1000 windows (as is the case for 1% of images on average per category in ILSVRC and 5% in PASCAL), we set  $\text{OBJ}(m) = 1001$ . The fact that more than 95% of objects can be localized with these windows imply that the objectness cue is already quite strong, so objects that require many windows on average will be extremely difficult to detect: e.g., ping pong ball (clutter of 9.57, or 758 windows on average), basketball (clutter of 9.21), puck (clutter of 9.17) in ILSVRC. The most difficult object in PASCAL is bottle with clutter score of 8.47. On average, ILSVRC has clutter score of 3.59. The most difficult subset of ILSVRC with 250 object categories has an order of magnitude more categories and the same average amount of clutter (of 5.90) as the PASCAL dataset.

## Appendix C Manually curated queries for obtaining object detection scene images

In Section 3.3.2 we discussed three types of queries we used for collecting the object detection images: (1) single object category name or a synonym; (2) a pair of object category names; (3) a manual query, typically targetting one or more object categories with insufficient data. Here we provide a list of the 129 manually curated queries:

afternoon tea, ant bridge building, armadillo race, armadillo yard, artist studio, auscultation, baby room, banjo orchestra, banjo rehearsal, banjo show, califone headphones & media player sets, camel dessert, camel tourist, carpenter drilling, carpentry, centipede wild, coffee shop, continental breakfast toaster, continental breakfast waffles, crutch walking, desert scorpion, diner, dining room, dining table, dinner, dragonfly friendly, dragonfly kid, dragonfly pond, dragonfly wild, drying hair, dumbbell curl, fan blow wind, fast food, fast food restaurant, firewood chopping, flu shot, goldfish aquarium, goldfish tank, golf cart on golf course, gym dumbbell, hamster drinking water, harmonica orchestra, harmonica rehearsal, harmonica show, harp ensemble, harp orchestra, harp rehearsal, harp show, hedgehog cute, hedgehog floor, hedgehog hidden, hippo bird, hippo friendly, home improvement diy drill, horseback riding, hotel coffee machine, hotel coffee maker, hotel waffle maker, jellyfish scuba, jellyfish snorkeling, kitchen, kitchen counter coffee maker, kitchen counter toaster, kitchenette, koala feed, koala tree, ladybug flower, ladybug yard, laundromat, lion zebra friendly, lunch, mailman, making breakfast, making waffles, mexican food, motorcycle racing, office, office fan, opossum on tree branch, orchestra, panda play, panda tree, pizzeria, pomegranate tree, porcupine climbing trees, power drill carpenter, purse shop, red panda tree, riding competition, riding motor scooters, school supplies, scuba starfish, sea lion beach, sea otter, sea urchin habitat, shopping for school supplies, sitting in front of a fan, skunk and cat, skunk park, skunk wild, skunk yard, snail flower, snorkeling starfish, snowplow cleanup, snowplow pile, snowplow winter, soccer game, south american zoo, starfish sea world, starts shopping, steamed artichoke, stethoscope doctor, strainer pasta, strainer tea, syringe doctor, table with food, tape player, tiger circus, tiger pet, using a can opener, using power drill, waffle iron breakfast, wild lion savana, wildlife preserve animals, wiping dishes, wombat petting zoo, zebra savana, zoo feeding, zoo in australia

## Appendix D Hierarchy of questions for full image annotation

The following is a hierarchy of questions manually constructed for crowdsourcing full annotation of images with the presence or absence of 200 object detection categories in ILSVRC2013 and ILSVRC2014. All questions are of the form “is there a ... in the image?” Questions marked with • are asked on every image. If the answer to a question is determined to be “no” then the answer to all descendant questions is assumed to be “no”. The 200 numbered leaf nodes correspond to the 200 object detection categories.

The goal in the hierarchy construction is to save cost (by asking as few questions as possible on every image) while avoiding any ambiguity in questions which would lead to false negatives during annotation. This hierarchy is not tree-structured; some questions have multiple parents.

### Hierarchy of questions:

- first aid/ medical items
  - (1) stethoscope
  - (2) syringe
  - (3) neck brace
  - (4) crutch
  - (5) stretcher
  - (6) band aid: an adhesive bandage to cover small cuts or blisters
- musical instruments
  - (7) accordion (a portable box-shaped free-reed instrument; the reeds are made to vibrate by air from the bellows controlled by the player)
  - (8) piano, pianoforte, forte-piano
  - percussion instruments: chimes, maracas, drums, etc
    - (9) chime: a percussion instrument consisting of a set of tuned bells that are struck with a hammer; used as an orchestral instrument
    - (10) maraca
    - (11) drum
  - stringed instrument
    - (12) banjo, the body of a banjo is round. please do not confuse with guitar
    - (13) cello: a large stringed instrument; seated player holds it upright while playing
    - (14) violin: bowed stringed instrument that has four strings, a hollow body, an unfretted fingerboard and is played with a bow. please do not confuse with cello, which is held upright while playing
    - (15) harp
    - (16) guitar, please do not confuse with banjo. the body of a banjo is round
  - wind instrument: a musical instrument in which the sound is produced by an enclosed column of air that is moved by the breath (such as trumpet, french horn, harmonica, flute, etc)
    - (17) trumpet: a brass musical instrument with a narrow tube and a flared bell, which is played by means of valves. often has 3 keys on top
    - (18) french horn: a brass musical instrument consisting of a conical tube that is coiled into a spiral, with a flared bell at the end
    - (19) trombone: a brass instrument consisting of a long tube whose length can be varied by a u-shaped slide
    - (20) harmonica

1000个ILSVRC类别的准确率是20.8%。20个PASCAL类别的平均CPL为8.7%，这与ILSVRC中562个最困难类别的CPL相同。

*Clutter*. 直观上，即使是很小的物体在纯色背景上也容易被定位。为了量化背景杂乱程度，我们采用了（Alex e等人，2012年）提出的物体性度量方法——这是一种类别无关的物体检测器，用于评估图像中某个窗口包含连贯物体（任意类别）而非背景（天空、水、草地）的可能性。对于每张包含目标物体实例的图像 $m$ ，其目标实例位于位置 $B_1^m, B_2^m, \dots$ ，我们使用公开可用的物体性软件按窗口包含任意通用物体的概率降序采样1000个窗口 $W_1^m, W_2^m, \dots, W_{1000}^m$ 。设 $\text{OBJ}(m)$ 为在定位到目标类别实例前采样的类物体窗口数量，即 $\text{OBJ}(m) = \min\{k : \max_i \text{IOU}(W_k^m, B_i^m) \geq 0.5\}$ 。对于一个包含 $M$ 张图像类别，我们计算每张图像中此类窗口的平均数量，并定义为

$$\text{CLUTTER} = \log_2\left(\frac{1}{M} \sum_m \text{OBJ}(m)\right) \quad (7)$$

一个类别的杂乱程度越高，根据通用线索定位物体的难度就越大。如果某个物体无法在前1000个窗口中被定位（在ILSVRC中平均每类别有1%的图像属于这种情况，在PASCAL中则为5%），我们设定 $\text{OBJ}(m) = 1001$ 。超过95%的物体能通过这些窗口实现定位，这一事实表明物体性线索已相当强，因此那些平均需要大量窗口的物体将极难检测：例如ILSVRC中的乒乓球（杂乱度9.57，平均需758个窗口）、篮球（杂乱度9.21）、冰球（杂乱度9.17）。PASCAL数据集中最困难的物体是瓶子，其杂乱度得分为8.47。ILSVRC的平均杂乱度得分为3.59。ILSVRC中最困难的250个物体类别子集，其类别数量多一个数量级，且平均杂乱度（5.90）与PASCAL数据集相同。

## Appendix C Manually curated queries for obtaining object detection scene images

在3.3.2节中，我们讨论了用于收集目标检测图像的三种查询类型：(1) 单一目标类别名称或其同义词；(2) 一对目标类别名称；(3) 手动查询，通常针对一个或多个数据不足的目标类别。此处我们提供129条人工整理查询的列表：

下午茶, 蚂蚁搭桥, 狢猱赛跑, 狢猱庭院, 艺术家工作室, 听诊, 婴儿房, 班卓琴乐团, 班卓琴排练, 班卓琴表演, 卡利丰耳机与媒体播放器套装, 骆驼甜点, 骆驼游客, 木匠钻孔, 木工, 野生蜈蚣, 咖啡店, 大陆早餐烤面包机, 大陆早餐华夫饼, 扭扭行走, 沙漠蝎子, 小餐馆, 餐厅, 餐桌, 晚餐, 友善蜻蜓, 蜻蜓孩童, 蜻蜓池塘, 野生蜻蜓, 吹干头发, 哑铃弯举, 风扇吹风, 快餐, 快餐店, 劈柴, 流感疫苗, 金鱼水族馆, 金鱼缸, 高尔夫球场上的高尔夫球车, 健身房哑铃, 仓鼠喝水, 口琴乐团, 口琴排练, 口琴表演, 竖琴合奏, 竖琴乐团, 竖琴排练, 竖琴表演, 可爱刺猬, 刺猬地板, 隐藏刺猬, 河马鸟, 友善河马, 家装DIY电钻, 骑马, 酒店咖啡机, 酒店咖啡壶, 酒店华夫饼机, 水母潜水, 水母浮潜, 厨房, 厨房台面咖啡机, 厨房台面烤面包机, 小厨房, 喂食考拉, 考拉树, 瓢虫花, 瓢虫庭院, 自助洗衣店, 狮子斑马友善, 午餐, 邮递员, 做早餐, 做华夫饼, 墨西哥食物, 摩托车比赛, 办公室, 办公室风扇, 负鼠在树枝上, 管弦乐团, 熊猫玩耍, 熊猫树, 比萨店, 石榴树, 豪猪爬树, 电钻木匠, 钱包店, 小熊猫树, 骑马比赛, 骑摩托滑板车, 学校用品, 潜水海星, 海狮海滩, 海獭, 海胆栖息地, 购买学校用品, 坐在风扇前, 臭鼬和猫, 臭鼬公园, 野生臭鼬, 臭鼬庭院, 蜗牛花, 浮潜海星, 扫雪车清理, 扫雪车雪堆, 扫雪车冬季, 足球比赛, 南美动物园, 海星海洋世界, 开始购物, 燕洋鲷, 听诊器医生, 滤网意面, 滤网茶, 注射器医生, 摆满食物的桌子, 磁带播放器, 老虎马戏团, 宠物老虎, 使用开罐器, 使用电钻, 华夫饼机早餐, 野生狮子草原, 野生动物保护区动物, 擦盘子, 袋熊动物园, 斑马草原, 动物园喂食, 澳大利亚动物园

## Appendix D Hierarchy of questions for full image annotation

以下是为众包对图像进行完整标注而手动构建的问题层级结构，这些图像涉及ILSVRC2013和ILSVRC2014中200个物体检测类别的存在与否。所有问题均采用“图像中是否有……？”的形式。标有{v\*}的问题

● 每张图像都会被问到这些问题。如果某个问题的答案被确定为“否”，那么其所有后续问题的答案也被假定为“否”。200个带编号的叶节点对应着200个物体检测类别。

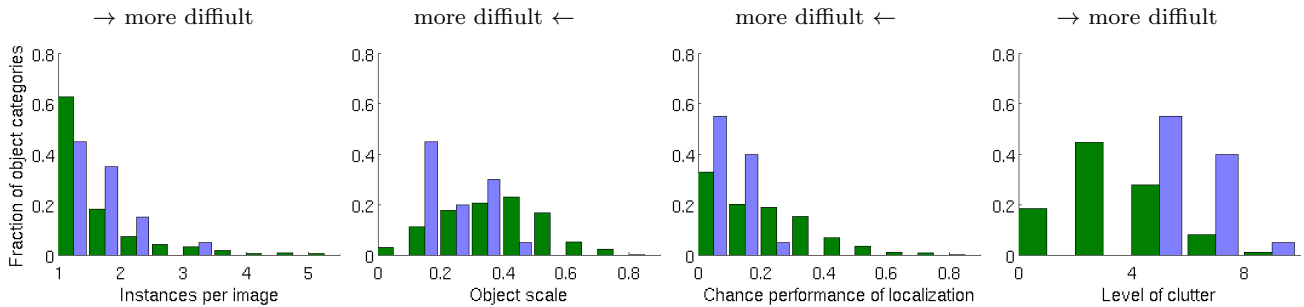
层次结构构建的目标是节省成本（通过对每张图像尽可能少地提问），同时避免问题中的任何歧义，以免在标注过程中导致假阴性。该层次结构并非树状结构；部分问题具有多个父节点。

### Hierarchy of questions:

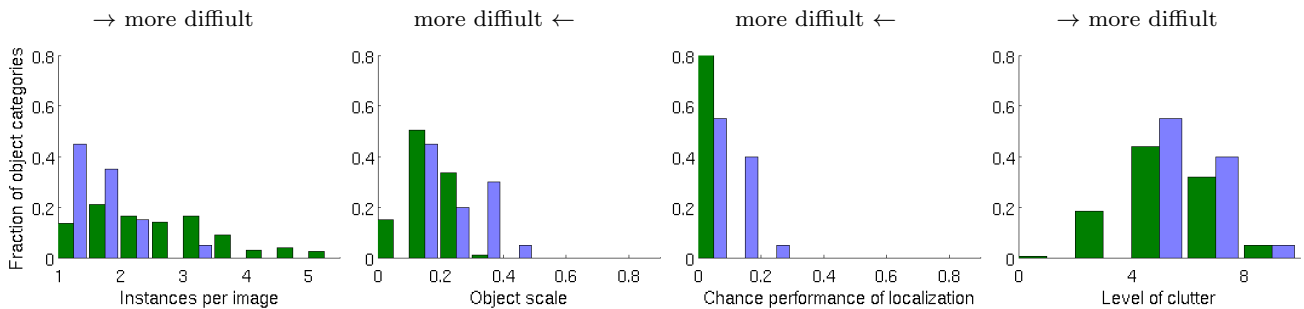
● 急救/医疗物品 ○ (1) 听诊器 ○ (2) 注射器 ○ (3) 颈托 ○ (4) 拐杖 ○ (5) 担架 ○ (6) 创可贴：用于覆盖小伤口或水疱的粘性绷带

● 乐器 ○ (7) 手风琴（一种便携式盒状自由簧乐器；通过演奏者控制风箱的气流使簧片振动）○ (8) 钢琴 ○ 打击乐器：编钟、沙锤、鼓等 ○ (9) 编钟：一种由一组调音钟组成的打击乐器，用槌敲击演奏；常用作管弦乐器 ○ (10) 沙锤 ○ (11) 鼓 ○ 弦乐器 ○ (12) 班卓琴，琴身为圆形。请注意勿与吉他混淆 ○ (13) 大提琴：一种大型弦乐器；演奏者采用坐姿将其直立放置演奏 ○ (14) 小提琴：用琴弓演奏的四弦弓弦乐器，具有空心琴身和五品指板。请注意勿与演奏时需直立放置的大提琴混淆 ○ (15) 竖琴 ○ (16) 吉他，请注意勿与班卓琴混淆。班卓琴的琴身为圆形 ○ 管乐器：通过呼吸驱动封闭空气柱发声的乐器（如小号、圆号、口琴、长笛等）○ (17) 小号：一种铜管乐器，具有细管和喇叭口，通过活塞按键演奏。顶部通常设有三个按键 ○ (18) 圆号：一种铜管乐器，由锥形管卷成螺旋状，末端带有喇叭口 ○ (19) 长号：一种铜管乐器，由长管构成，通过U形滑管改变管长 ○ (20) 口琴

**1000 classes of ILSVRC2012-2014 single-object localization (dark green)  
versus 20 classes of PASCAL 2012 (light blue)**



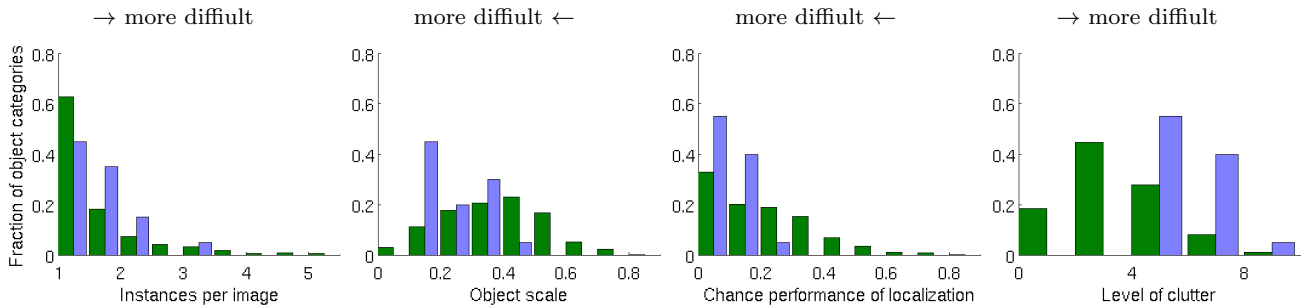
**200 hardest classes of ILSVRC2012-2014 single-object localization (dark green)  
versus 20 classes of PASCAL 2012 (light blue)**



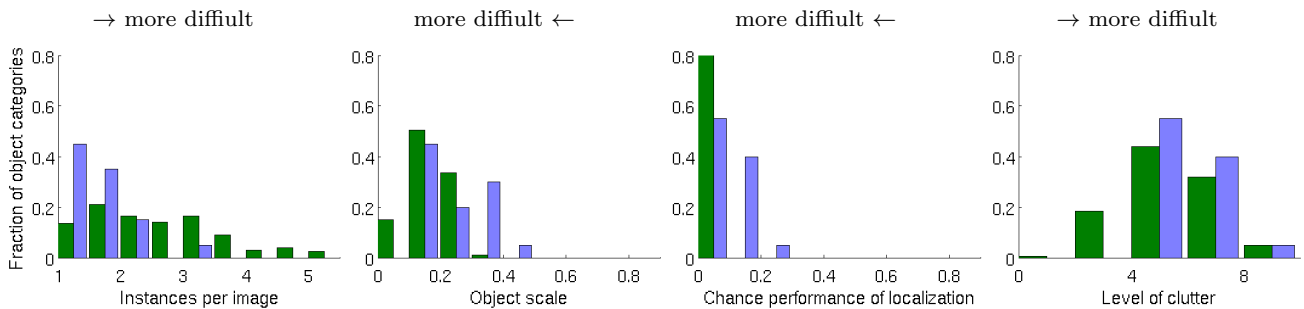
**Fig. 16** Distribution of various measures of localization difficulty on the ILSVRC2012-2014 single-object localization (dark green) and PASCAL VOC 2012 (light blue) validation sets. Object scale is fraction of image area occupied by an average object instance. Chance performance of localization and level of clutter are defined in Appendix B. The plots on top contain the full ILSVRC validation set with 1000 classes; the plots on the bottom contain 200 ILSVRC classes with the lowest chance performance of localization. All plots contain all 20 classes of PASCAL VOC.

- (21) flute: a high-pitched musical instrument that looks like a straight tube and is usually played sideways (please do not confuse with oboes, which have a distinctive straw-like mouth piece and a slightly flared end)
- (22) oboe: a slender musical instrument roughly 65cm long with metal keys, a distinctive straw-like mouthpiece and often a slightly flared end (please do not confuse with flutes)
- (23) saxophone: a musical instrument consisting of a brass conical tube, often with a u-bend at the end
- food: something you can eat or drink (includes growing fruit, vegetables and mushrooms, but does not include living animals)
  - food with bread or crust: pretzel, bagel, pizza, hotdog, hamburgers, etc
  - (24) pretzel
  - (25) bagel, beigel
  - (26) pizza, pizza pie
  - (27) hotdog, hot dog, red hot
  - (28) hamburger, beefburger, burger
  - (29) guacamole
  - (30) burrito
  - (31) popsicle (ice cream or water ice on a small wooden stick)
  - fruit
    - (32) fig
    - (33) pineapple, ananas
    - (34) banana
    - (35) pomegranate
    - (36) apple
    - (37) strawberry
    - (38) orange
    - (39) lemon
  - vegetables
    - (40) cucumber, cuke
    - (41) artichoke, globe artichoke
    - (42) bell pepper
    - (43) head cabbage
  - (44) mushroom
- items that run on electricity (plugged in or using batteries); including clocks, microphones, traffic lights, computers, etc
  - (45) remote control, remote
  - electronics that blow air
    - (46) hair dryer, blow dryer
    - (47) electric fan: a device for creating a current of air by movement of a surface or surfaces (please do not consider hair dryers)
  - electronics that can play music or amplify sound
    - (48) tape player
    - (49) iPod
  - (50) microphone, mike
  - computer and computer peripherals: mouse, laptop, printer, keyboard, etc
  - (51) computer mouse
  - (52) laptop, laptop computer
  - (53) printer (please do not consider typewriters to be printers)
  - (54) computer keyboard
  - (55) lamp
  - electric cooking appliance (an appliance which generates heat to cook food or boil water)
    - (56) microwave, microwave oven
    - (57) toaster
    - (58) waffle iron
    - (59) coffee maker: a kitchen appliance used for brewing coffee automatically
  - (60) vacuum, vacuum cleaner
  - (61) dishwasher, dish washer, dishwashing machine
  - (62) washer, washing machine: an electric appliance for washing clothes
  - (63) traffic light, traffic signal, stoplight
  - (64) tv or monitor: an electronic device that represents information in visual form
  - (65) digital clock: a clock that displays the time of day digitally
  - kitchen items: tools, utensils and appliances usually found in the kitchen
    - electric cooking appliance (an appliance which generates heat to cook food or boil water)
      - (56) microwave, microwave oven
      - (57) toaster
      - (58) waffle iron
      - (59) coffee maker: a kitchen appliance used for brewing coffee automatically
    - (61) dishwasher, dish washer, dishwashing machine
    - (66) stove
    - things used to open cans/bottles: can opener or corkscrew
      - (67) can opener (tin opener)
      - (68) corkscrew
    - (69) cocktail shaker
    - non-electric item commonly found in the kitchen: pot, pan, utensil, bowl, etc
    - (70) strainer
    - (71) frying pan (skillet)
    - (72) bowl: a dish for serving food that is round, open at the top, and has no handles (please do not confuse with a cup, which usually has a handle and is used for serving drinks)
    - (73) salt or pepper shaker: a shaker with a perforated top for sprinkling salt or pepper
    - (74) plate rack
    - (75) spatula: a turner with a narrow flexible blade
    - (76) ladle: a spoon-shaped vessel with a long handle; frequently used to transfer liquids from one container to another
    - (77) refrigerator, icebox
  - furniture (including benches)

**1000 classes of ILSVRC2012-2014 single-object localization (dark green)  
versus 20 classes of PASCAL 2012 (light blue)**



**200 hardest classes of ILSVRC2012-2014 single-object localization (dark green)  
versus 20 classes of PASCAL 2012 (light blue)**



**Fig. 16** ILSVRC2012-2014单目标定位（深绿色）与PASCAL VOC 2012（浅蓝色）验证集上各类定位难度指标的分布情况。物体尺度指单个物体实例平均所占图像面积比例。定位的随机性能与背景杂乱程度的定义详见附录B。上方图表包含含1000个类别的完整ILSVRC验证集；下方图表包含ILSVRC中定位随机性能最低的200个类别。所有图表均包含PASCAL VOC的全部20个类别。

○ (21) 长笛：一种高音乐器，外形呈直管状，通常横向吹奏（请勿与双簧管混淆，双簧管具有独特的稻草状吹嘴和略微外扩的管口）○ (22) 双簧管：一种细长的乐器，长约65厘米，带有金属按键，具有独特的稻草状吹嘴，管口常略微外扩（请勿与长笛混淆）○ (23) 萨克斯管：一种由黄铜锥形管构成的乐器，末端常带有U形弯管

○ (52) 笔记本电脑，便携式电脑 ○ (53) 打印机（请勿将打字机视为打印机）○ (54) 计算机键盘 ○ (55) 灯具 ○ 电炊具（一种产生热量以烹饪食物或烧水的器具）○ (56) 微波炉 ○ (57) 烤面包机 ○ (58) 华夫饼机 ○ (59) 咖啡机：一种用于自动冲泡咖啡的厨房电器 ○ (60) 吸尘器 ○ (61) 洗碗机 ○ (62) 洗衣机：一种用于洗涤衣物的电器 ○ (63) 交通信号灯 ○ (64) 电视或显示器：一种以视觉形式呈现信息的电子设备 ○ (65) 数字时钟：一种以数字形式显示时间的时钟

● 食物：可供食用或饮用的物品（包括种植的水果、蔬菜和蘑菇，但不包括活体动物）○ 带面包或外皮的食物：椒盐卷饼、百吉饼、披萨、热狗、汉堡等 ○ (24) 椒盐卷饼 ○ (25) 百吉饼 ○ (26) 披萨 ○ (27) 热狗 ○ (28) 汉堡 ○ (29) 鳄梨酱 ○ (30) 墨西哥卷饼 ○ (31) 冰棒（插在小木棍上的冰淇淋或雪糕）○ 水果 ○ (32) 无花果 ○ (33) 菠萝 ○ (34) 香蕉 ○ (35) 石榴 ○ (36) 苹果 ○ (37) 草莓 ○ (38) 橙子 ○ (39) 柠檬 ○ 蔬菜 ○ (40) 黄瓜 ○ (41) 洋葱 ○ (42) 甜椒 ○ (43) 卷心菜 ○ (44) 蘑菇

● 厨房用品：通常在厨房中发现的工具、器皿和电器 ○ 电烹饪器具（一种产生热量以烹饪食物或烧水的器具）○ (56) 微波炉 ○ (57) 烤面包机 ○ (58) 华夫饼机 ○ (59) 咖啡机：一种用于自动冲泡咖啡的厨房电器 ○ (61) 洗碗机 ○ (66) 炉灶 ○ 用于打开罐头/瓶子的物品：开罐器或开瓶器 ○ (67) 开罐器 ○ (68) 开瓶器 ○ (69) 鸡尾酒摇酒器 ○ 厨房中常见的非电器物品：锅、平底锅、餐具、碗等 ○ (70) 滤网 ○ (71) 煎锅 ○ (72) 碗：一种用于盛放食物的圆形、顶部开口且无把手的器皿（请勿与杯子混淆，杯子通常有把手，用于盛放饮料）○ (73) 盐或胡椒粉：一种顶部有孔用于撒盐或胡椒粉的瓶子 ○ (74) 盘架 ○ (75) 锅铲：一种带有窄而灵活刀片的翻面工具 ○ (76) 长柄勺：一种长柄勺形容器；常用于将液体从一个容器转移到另一个容器 ○ (77) 冰箱

● 使用电力的物品（插电或使用电池）：包括时钟、麦克风、交通信号灯、计算机等 ○ (45) 遥控器，遥控 ○ 吹风的电子产品 ○ (46) 吹风机，电吹风 ○ (47) 电风扇：通过一个或多个表面的运动产生气流的设备（请勿考虑吹风机）○ 可以播放音乐或放大声音的电子产品 ○ (48) 磁带播放器 ○ (49) iPod ○ (50) 麦克风，话筒 ○ 计算机及计算机外围设备：鼠标、笔记本电脑、打印机、键盘等 ○ (51) 电脑鼠标

● 家具（包括长凳）

- (78) bookshelf: a shelf on which to keep books
- (79) baby bed: small bed for babies, enclosed by sides to prevent baby from falling
- (80) filing cabinet: office furniture consisting of a container for keeping papers in order
- (81) bench (a long seat for several people, typically made of wood or stone)
- (82) chair: a raised piece of furniture for one person to sit on; please do not confuse with benches or sofas, which are made for more people
- (83) sofa, couch: upholstered seat for more than one person; please do not confuse with benches (which are made of wood or stone) or with chairs (which are for just one person)
- (84) table
- clothing, article of clothing: a covering designed to be worn on a person's body
  - (85) diaper: Garment consisting of a folded cloth drawn up between the legs and fastened at the waist; worn by infants to catch excrement
  - swimming attire: clothes used for swimming or bathing (swim suits, swim trunks, bathing caps)
    - (86) swimming trunks: swimsuit worn by men while swimming
    - (87) bathing cap, swimming cap: a cap worn to keep hair dry while swimming or showering
    - (88) maillot: a woman's one-piece bathing suit
  - necktie: a man's formal article of clothing worn around the neck (including bow ties)
    - (89) bow tie: a man's tie that ties in a bow
    - (90) tie: a long piece of cloth worn for decorative purposes around the neck or shoulders, resting under the shirt collar and knotted at the throat (NOT a bow tie)
  - headdress, headgear: clothing for the head (hats, helmets, bathing caps, etc)
    - (87) bathing cap, swimming cap: a cap worn to keep hair dry while swimming or showering
    - (91) hat with a wide brim
    - (92) helmet: protective headgear made of hard material to resist blows
  - (93) miniskirt, mini: a very short skirt
  - (94) brassiere, bra: an undergarment worn by women to support their breasts
  - (95) sunglasses
- living organism (other than people): dogs, snakes, fish, insects, sea urchins, starfish, etc.
  - living organism which can fly
    - (96) bee
    - (97) dragonfly
    - (98) ladybug
    - (99) butterfly
    - (100) bird
  - living organism which cannot fly (please don't include humans)
  - living organism with 2 or 4 legs (please don't include humans):
    - mammals (but please do not include humans)
      - feline (cat-like) animal: cat, tiger or lion
        - (101) domestic cat
        - (102) tiger
        - (103) lion
      - canine (dog-like animal): dog, hyena, fox or wolf
        - (104) dog, domestic dog, canis familiaris
        - (105) fox: wild carnivorous mammal with pointed muzzle and ears and a bushy tail (please do not confuse with dogs)
    - animals with hooves: camels, elephants, hippos, pigs, sheep, etc
      - (106) elephant
      - (107) hippopotamus, hippo
      - (108) camel
      - (109) swine: pig or boar
      - (110) sheep: woolly animal, males have large spiraling horns (please do not confuse with antelope which have long legs)
      - (111) cattle: cows or oxen (domestic bovine animals)
      - (112) zebra
      - (113) horse
      - (114) antelope: a graceful animal with long legs and horns directed upward and backward
    - (115) squirrel
    - (116) hamster: short-tailed burrowing rodent with large cheek pouches
    - (117) otter
    - (118) monkey
    - (119) koala bear
    - (120) bear (other than pandas)
    - (121) skunk (mammal known for its ability to spray a liquid with a strong odor; they may have a single thick stripe across back and tail, two thinner stripes, or a series of white spots and broken stripes)
      - (122) rabbit
      - (123) giant panda: an animal characterized by its distinct black and white markings
      - (124) red panda: Reddish-brown Old World raccoon-like carnivore
    - (125) frog, toad
    - (126) lizard: please do not confuse with snake (lizards have legs)
    - (127) turtle
    - (128) armadillo
    - (129) porcupine, hedgehog
  - living organism with 6 or more legs: lobster, scorpion, insects, etc.
    - (130) lobster: large marine crustaceans with long bodies and muscular tails; three of their five pairs of legs have claws
    - (131) scorpion
    - (132) centipede: an arthropod having a flattened body of 15 to 173 segments each with a pair of legs, the foremost pair being modified as prehensors
    - (133) tick (a small creature with 4 pairs of legs which lives on the blood of mammals and birds)
    - (134) isopod: a small crustacean with seven pairs of legs adapted for crawling
    - (135) ant
  - living organism without legs: fish, snake, seal, etc. (please don't include plants)
    - living organism that lives in water: seal, whale, fish, sea cucumber, etc.
      - (136) jellyfish
      - (137) starfish, sea star
      - (138) seal
      - (139) whale
      - (140) ray: a marine animal with a horizontally flattened body and enlarged winglike pectoral fins with gills on the underside
      - (141) goldfish: small golden or orange-red fishes
    - living organism that slides on land: worm, snail, snake
      - (142) snail
      - (143) snake: please do not confuse with lizard (snakes do not have legs)
- vehicle: any object used to move people or objects from place to place
  - a vehicle with wheels
    - (144) golfcart, golf cart
    - (145) snowplow: a vehicle used to push snow from roads
    - (146) motorcycle (or moped)
    - (147) car, automobile (not a golf cart or a bus)
    - (148) bus: a vehicle carrying many passengers; used for public transport
    - (149) train
    - (150) cart: a heavy open wagon usually having two wheels and drawn by an animal
    - (151) bicycle, bike: a two wheeled vehicle moved by foot pedals
    - (152) unicycle, monocycle
  - a vehicle without wheels (snowmobile, sleighs)
    - (153) snowmobile: tracked vehicle for travel on snow
    - (154) watercraft (such as ship or boat): a craft designed for water transportation
    - (155) airplane: an aircraft powered by propellers or jets
- cosmetics: toiletry designed to beautify the body
  - (156) face powder
  - (157) perfume, essence (usually comes in a smaller bottle than hair spray)
  - (158) hair spray
  - (159) cream, ointment, lotion
  - (160) lipstick, lip rouge
- carpentry items: items used in carpentry, including nails, hammers, axes, screwdrivers, drills, chain saws, etc
  - (161) chain saw, chainsaw
  - (162) nail: pin-shaped with a head on one end and a point on the other
  - (163) axe: a sharp tool often used to cut trees/ logs
  - (164) hammer: a blunt hand tool used to drive nails in or break things apart (please do not confuse with axe, which is sharp)
  - (165) screwdriver
  - (166) power drill: a power tool for drilling holes into hard materials
- school supplies: rulers, erasers, pencil sharpeners, pencil boxes, binders
  - (167) ruler, rule: measuring stick consisting of a strip of wood or metal or plastic with a straight edge that is used for drawing straight lines and measuring lengths
  - (168) rubber eraser, rubber, pencil eraser
  - (169) pencil sharpener
  - (170) pencil box, pencil case
  - (171) binder, ring-binder
- sports items: items used to play sports or in the gym (such as skis, raquets, gymnastics bars, bows, punching bags, balls)
  - (172) bow: weapon for shooting arrows, composed of a curved piece of resilient wood with a taut cord to propel the arrow
  - (173) puck, hockey puck: vulcanized rubber disk 3 inches in diameter that is used instead of a ball in ice hockey
  - (174) ski
  - (175) racket, racquet
  - gymnastic equipment: parallel bars, high beam, etc
    - (176) balance beam: a horizontal bar used for gymnastics which is raised from the floor and wide enough to walk on
    - (177) horizontal bar, high bar: used for gymnastics; gymnasts grip it with their hands (please do not confuse with balance beam, which is wide enough to walk on)
  - ball
    - (178) golf ball
    - (179) baseball
    - (180) basketball
    - (181) croquet ball
    - (182) soccer ball
    - (183) ping-pong ball
    - (184) rugby ball
    - (185) volleyball
    - (186) tennis ball
  - (187) punching bag, punch bag, punching ball, punchball
  - (188) dumbbell: An exercising weight; two spheres connected by a short bar that serves as a handle
- liquid container: vessels which commonly contain liquids such as bottles, cans, etc.
  - (189) pitcher: a vessel with a handle and a spout for pouring
  - (190) beaker: a flatbottomed jar made of glass or plastic; used for chemistry
  - (191) milk can
  - (192) soap dispenser
  - (193) wine bottle
  - (194) water bottle
  - (195) cup or mug (usually with a handle and usually cylindrical)
- bag
  - (196) backpack: a bag carried by a strap on your back or shoulder
  - (197) purse: a small bag for carrying money
  - (198) plastic bag
- (199) person
- (200) flower pot: a container in which plants are cultivated

## Appendix E Modification to bounding box system for object detection

The bounding box annotation system described in Section 3.2.1 is used for annotating images for both the single-object localization dataset and the object detection dataset. However, two additional manual post-processing are needed to ensure accuracy in the object detection scenario:



- (78) 书架：用于存放书籍的架子 ◦ (79) 婴儿床：供婴儿使用的小床，四周有围栏防止婴儿跌落 ◦ (80) 文件柜：用于有序存放文件的办公家具 ◦ (81) 长凳（供多人使用的长型座位，通常由木材或石头制成） ◦ (82) 椅子：供单人使用的有座面的家具；请注意不要与供多人使用的长凳或沙发混淆 ◦ (83) 沙发：供多人使用的软垫座椅；请注意不要与（由木材或石头制成的）长凳或（仅供单人使用的）椅子混淆 ◦ (84) 桌子
- 衣物，服装：设计用于穿戴在人体上的覆盖物 ◦ (85) 尿布：由折叠布料制成，穿过双腿并在腰部固定的服装；婴儿穿着以承接排泄物 ◦ 泳装：用于游泳或沐浴的衣物（泳衣、泳裤、泳帽） ◦ (86) 泳裤：男性游泳时穿着的泳装 ◦ (87) 泳帽：游泳或淋浴时佩戴以保持头发干燥的帽子 ◦ (88) 连体泳衣：女性的一件式泳装 ◦ 领带：男性穿着的正式颈部服饰（包括领结） ◦ (89) 领结：系成蝴蝶结形状的男性领饰 ◦ (90) 领带：用于装饰目的的长条布料，佩戴于颈部或肩部，置于衬衫领下并在喉部打结（非指领结） ◦ 头饰，头具：头部穿戴物（帽子、头盔、泳帽等） ◦ (87) 泳帽：游泳或淋浴时佩戴以保持头发干燥的帽子 ◦ (91) 宽檐帽 ◦ (92) 头盔：由硬质材料制成的防护头具，用于抵御击打 ◦ (93) 迷你裙：极短的裙子 ◦ (94) 胸罩：女性穿着的内衣，用于支撑胸部 ◦ (95) 太阳镜
- 交通工具：用于将人或物体从一地移动到另一地的任何物体 ◦ 带轮子的车辆 ◦ (144) 高尔夫球车 ◦ (145) 扫雪机：用于清除道路积雪的车辆 ◦ (146) 摩托车（或助力车） ◦ (147) 汽车（非高尔夫球车或巴士） ◦ (148) 巴士：运载多名乘客的车辆；用于公共交通 ◦ (149) 火车 ◦ (150) 畜力车：通常有两个轮子、由动物牵引的敞篷重型货车 ◦ (151) 自行车：通过脚踏板驱动的两轮车辆 ◦ (152) 独轮车 ◦ 无轮车辆（雪地摩托、雪橇） ◦ (153) 雪地摩托：用于雪地行驶的履带式车辆 ◦ (154) 水上交通工具（如轮船或小船）：设计用于水上运输的载具 ◦ (155) 飞机：由螺旋桨或喷气发动机驱动的航空器
- 化妆品：用于美化身体的盥洗用品 ◦ (156) 粉饼 ◦ (157) 香水、香精（通常比发胶瓶更小） ◦ (158) 发胶 ◦ (159) 乳霜、药膏、乳液 ◦ (160) 口红、唇彩
- 木工用品：木工中使用的物品，包括钉子、锤子、斧头、螺丝刀、钻头、链锯等 ◦ (161) 链锯 ◦ (162) 钉子：一端有头、另一端呈尖状的针形物 ◦ (163) 斧头：常用于砍伐树木/木材的锋利工具 ◦ (164) 锤子：用于敲入钉子或砸碎物体的钝头手工具（请注意勿与锋利的斧头混淆） ◦ (165) 螺丝刀 ◦ (166) 电钻：用于在硬质材料上钻孔的电动工具
- 文具用品：尺子、橡皮、卷笔刀、铅笔盒、活页夹 ◦ (167) 尺子，规则：由一条木制、金属或塑料直边条组成的测量工具，用于画直线和测量长度 ◦ (168) 橡皮擦，橡胶，铅笔擦 ◦ (169) 卷笔刀 ◦ (170) 铅笔盒，笔袋 ◦ (171) 活页夹，环扣文件夹
- 体育用品：用于体育运动或健身的物品（如滑雪板、球拍、体操杠、弓、沙袋、球） ◦ (172) 弓：射箭武器，由弹性木材弯曲制成，配以绷紧的弦以推动箭矢 ◦ (173) 冰球：硫化橡胶制成的圆盘，直径3英寸，在冰球运动中替代球使用 ◦ (174) 滑雪板 ◦ (175) 球拍 ◦ 体操器械：双杠、平衡木等 ◦ (176) 平衡木：体操用的水平横杆，离地架起且宽度足以行走 ◦ (177) 单杠：体操器械；运动员用手抓握（请勿与可行走的平衡木混淆） ◦ 球 ◦ (178) 高尔夫球 ◦ (179) 棒球 ◦ (180) 篮球 ◦ (181) 网球 ◦ (182) 足球 ◦ (183) 乒乓球 ◦ (184) 橄榄球 ◦ (185) 排球 ◦ (186) 网球 ◦ (187) 沙袋 ◦ (188) 哑铃：锻炼用重物；由短杆连接两个球体作为手柄
- 液体容器：通常盛装液体的器皿，如瓶子、罐子等。 ◦ (189) 水罐：带手柄和倒水口的容器 ◦ (190) 烧杯：由玻璃或塑料制成的平底广口杯；用于化学实验 ◦ (191) 牛奶罐 ◦ (192) 皂液器 ◦ (193) 酒瓶 ◦ (194) 水瓶 ◦ (195) 杯子或马克杯（通常带手柄且多为圆柱形）
- 包 ◦ (196) 背包：用背带背在背上或肩上的包 ◦ (197) 钱包：装钱用的小包 ◦ (198) 塑料袋
- (199) 人
- (200) 花盆：用于栽培植物的容器

## Appendix E Modification to bounding box system for object detection

第3.2.1节中描述的边界框标注系统，既用于单目标定位数据集也用于目标检测数据集的图像标注。然而，在目标检测场景中，为确保准确性还需要进行两项额外的人工后处理：

*Ambiguous objects.* The first common source of error was that workers were not able to accurately differentiate some object classes during annotation. Some commonly confused labels were seal and sea otter, backpack and purse, banjo and guitar, violin and cello, brass instruments (trumpet, trombone, french horn and brass), flute and oboe, ladle and spatula. Despite our best efforts (providing positive and negative example images in the annotation task, adding text explanations to alert the user to the distinction between these categories) these errors persisted.

In the single-object localization setting, this problem was not as prominent for two reasons. First, the way the data was collected imposed a strong prior on the object class which was present. Second, since only one object category needed to be annotated per image, ambiguous images could be discarded: for example, if workers couldn't agree on whether or not a trumpet was in fact present, this image could simply be removed. In contrast, for the object detection setting consensus had to be reached for all target categories on all images.

To fix this problem, once bounding box annotations were collected we manually looked through all cases where the bounding boxes for two different object classes had significant overlap with each other (about 3% of the collected boxes). About a quarter of these boxes were found to correspond to incorrect objects and were removed. Crowdsourcing this post-processing step (with very stringent accuracy constraints) would be possible but it occurred in few enough cases that it was faster (and more accurate) to do this in-house.

*Duplicate annotations.* The second common source of error were duplicate bounding boxes drawn on the same object instance. Despite instructions not to draw more than one bounding box around the same object instance and constraints in the annotation UI enforcing at least a 5 pixel difference between different bounding boxes, these errors persisted. One reason was that sometimes the initial bounding box was not perfect and subsequent labelers drew a slightly improved alternative.

This type of error was also present in the single-object localization scenario but was not a major cause for concern. A duplicate bounding box is a slightly perturbed but still correct positive example, and single-object localization is only concerned with correctly localizing one object instance. For the detection task algorithms are evaluated on the ability to localize *every* object instance, and penalized for duplicate detections, so it is imperative that these labeling errors are corrected (even if they only appear in about 0.6% of cases).

Approximately 1% of bounding boxes were found to have significant overlap of more than 50% with an-

other bounding box of the same object class. We again manually verified all of these cases in-house. In approximately 40% of the cases the two bounding boxes correctly corresponded to different people in a crowd, to stacked plates, or to musical instruments nearby in an orchestra. In the other 60% of cases one of the boxes was randomly removed.

These verification steps complete the annotation procedure of bounding boxes around every instance of every object class in validation, test and a subset of training images for the detection task.

*Training set annotation.* With the optimized algorithm of Section 3.3.3 we fully annotated the validation and test sets. However, annotating *all* training images with all target object classes was still a budget challenge. Positive training images taken from the single-object localization dataset already had bounding box annotations of all instances of one object class on each image. We extended the existing annotations to the detection dataset by making two modifications. First, we corrected any bounding box omissions resulting from merging fine-grained categories: i.e., if an image belonged to the "dalmatian" category and all instances of "dalmatian" were annotated with bounding boxes for single-object localization, we ensured that all remaining "dog" instances are also annotated for the object detection task. Second, we collected significantly more training data for the person class because the existing annotation set was not diverse enough to be representative (the only people categories in the single-object localization task are scuba diver, groom, and ballplayer). To compensate, we additionally annotated people in a large fraction of the existing training set images.

## Appendix F Competition protocol

*Competition format.* At the beginning of the competition period each year we release the new training/validation/test images, training/validation annotations, and competition specification for the year. We then specify a deadline for submission, usually approximately 4 months after the release of data. Teams are asked to upload a text file of their predicted annotations on test images by this deadline to a provided server. We then evaluate all submissions and release the results.

For every task we released code that takes a text file of automatically generated image annotations and compares it with the ground truth annotations to return a quantitative measure of algorithm accuracy. Teams can use this code to evaluate their performance on the validation data.

*Ambiguous objects.* 第一个常见的错误来源是工人在标注过程中无法准确区分某些物体类别。一些常见的混淆标签包括海豹和海獭、背包和钱包、班卓琴和吉他、小提琴和大提琴、铜管乐器（小号、长号、圆号及其他铜管乐器）、长笛和双簧管、长柄勺和锅铲。尽管我们尽了最大努力（在标注任务中提供正反例图片，添加文字说明以提醒用户注意这些类别之间的区别），这些错误仍然持续存在。

在单目标定位设置中，这个问题并不那么突出，原因有二。首先，数据收集方式对图像中存在的物体类别施加了强先验。其次，由于每张图像只需标注一个物体类别，存在歧义的图像可直接被剔除：例如，如果标注人员无法就图像中是否确实存在小号达成一致，这张图像便可直接移除。相比之下，在目标检测设置中，必须对所有图像中的所有目标类别达成标注共识。

为了解决这个问题，一旦收集到边界框标注，我们便手动检查了所有两个不同物体类别的边界框存在显著重叠的情况（约占收集框的3%）。其中约四分之一的框被发现对应错误的物体，因此被移除。虽然可以通过众包完成这一后处理步骤（需设定极其严格的准确度要求），但由于此类情况较少，内部处理反而更快捷（且更准确）。

*Duplicate annotations.* 第二个常见的错误来源是在同一物体实例上绘制了重复的边界框。尽管已明确指示不要在同一物体实例周围绘制多个边界框，且标注界面强制要求不同边界框之间至少存在5像素的差异，这类错误仍然存在。原因之一是初始边界框有时不够精确，后续标注者会绘制一个略微优化的替代框。

这类错误在单目标定位场景中同样存在，但并非主要关注点。重复的边界框是轻微扰动但仍正确的正例，而单目标定位仅关注正确定位单个目标实例。对于检测任务，算法需评估其定位 *every* 个目标实例的能力，并对重复检测进行惩罚，因此必须修正这些标注错误（即使它们仅出现在约0.6%的情况下）。

大约1%的边界框被发现与另一个边界框存在超过50%的显著重叠。

同一物体类别的其他边界框。我们再次在内部手动核验了所有这些情况。大约40%的情况下，两个边界框正确对应了人群中不同的个体、堆叠的盘子，或是管弦乐队中相邻的乐器。在其余60%的情况下，其中一个边界框被随机移除。

这些验证步骤完成了检测任务中验证集、测试集及部分训练集图像内每个对象类别的每个实例周围边界框的标注流程。

*Training set annotation.* 利用第3.3.3节的优化算法，我们完成了验证集和测试集的完整标注。然而，为 *all* 张训练图像标注所有目标物体类别仍存在预算限制。从单物体定位数据集中提取的正样本训练图像已包含每张图像中单个物体类别的所有实例边界框标注。我们通过两项调整将现有标注扩展至检测数据集：首先，修正因合并细粒度类别导致的边界框遗漏——例如，若某图像属于“达尔马提亚犬”类别，且单物体定位任务中已标注所有“达尔马提亚犬”实例的边界框，我们会确保为物体检测任务补标所有剩余的“狗”类实例。其次，针对“人物”类别收集了更多训练数据，因为现有标注集的多样性不足以形成代表性样本（单物体定位任务中仅包含潜水员、新郎和球员三类人物）。为此，我们额外对现有训练集图像中的大量人物实例进行了标注。

## Appendix F Competition protocol

*Competition format.* 每年竞赛期开始时，我们会发布新的训练/验证/测试图像、训练/验证标注以及当年的竞赛规范。随后我们会设定提交截止日期，通常在数据发布后约4个月左右。参赛团队需在此截止日期前将测试图像的预测标注文本文件上传至指定服务器。我们将评估所有提交结果并公布成绩。

对于每个任务，我们都发布了代码，该代码接收一个自动生成的图像标注文本文件，并将其与真实标注进行比较，以返回算法准确性的量化指标。团队可以使用此代码在验证数据上评估他们的表现。

As described in (Everingham et al., 2014), there are three options for measuring performance on test data: (i) Release test images and annotations, and allow participants to assess performance themselves; (ii) Release test images but not test annotations – participants submit results and organizers assess performance; (iii) Neither test images nor annotations are released – participants submit software and organizers run it on new data and assess performance. In line with the PASCAL VOC choice, we opted for option (ii). Option (i) allows too much leeway in overfitting to the test data; option (iii) is infeasible, especially given the scale of our test set (40K-100K images).

We released ILSVRC2010 test annotations for the image classification task, but all other test annotations have remained hidden to discourage fine-tuning results on the test data.

*Evaluation protocol after the challenge.* After the challenge period we set up an automatic evaluation server that researchers can use throughout the year to continue evaluating their algorithms against the ground truth test annotations. We limit teams to 2 submissions per week to discourage parameter tuning on the test data, and in practice we have never had a problem with researchers abusing the system.

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如(Everingham等人, 2014年)所述, 衡量测试数据性能有三种选择: (i) 发布测试图像和标注, 允许参与者自行评估性能; (ii) 发布测试图像但不发布测试标注——参与者提交结果, 组织者评估性能; (iii) 既不发布测试图像也不发布标注——参与者提交软件, 组织者在新数据上运行并评估性能。遵循PASCAL VOC的选择, 我们采用了选项(ii)。选项(i)在过拟合测试数据方面给予过多自由; 选项(iii)不可行, 特别是考虑到我们测试集的规模(4万至10万张图像)。

我们发布了ILSVRC2010图像分类任务的测试标注, 但所有其他测试标注仍处于隐藏状态, 以防止在测试数据上进行微调优化结果。

*Evaluation protocol after the challenge.* 在挑战期结束后, 我们建立了一个自动评估服务器, 研究人员全年都可以使用该服务器, 继续根据真实测试标注来评估他们的算法。我们限制每个团队每周只能提交2次, 以防止在测试数据上进行参数调优, 实际上我们从未遇到过研究人员滥用系统的问题。

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